

Pure-Point Spectrum for the Two-Dimensional Anderson Model at Every Positive Disorder

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Abstract

For each fixed positive disorder strength, we prove that the nearest-neighbor Anderson operator on the square lattice with independent uniform site potentials almost surely has pure-point spectral type throughout its spectrum. This resolves the pure-point assertion of the two-dimensional Anderson localization conjecture for the uniform single-site law.

1 Introduction

Let $h > 0$ and let $(v_x)_{x \in \mathbb{Z}^2}$ be independent random variables, each uniform on $[-h, h]$. We consider the bounded self-adjoint operator

$$(H_v u)(x) = \sum_{|y-x|_1=1} u(y) + v_x u(x) \quad \text{on } \ell^2(\mathbb{Z}^2). \quad (1.1)$$

Thus $v_x = h\omega_x$ in the usual notation with ω_x uniform on $[-1, 1]$. Pure-point spectral type means that the closed span of the eigenvectors is the entire Hilbert space; it does not mean that every point of the spectrum is an eigenvalue.

The planar spectral localization problem asks whether this pure-point conclusion holds throughout the spectrum for every nonzero width of the uniform distribution [14, Example 2.1 and Problem 2]. We prove that it does.

Theorem 1.1. *For each fixed $h > 0$, almost surely H_v has pure-point spectral type, admits a complete orthonormal eigenbasis, and has spectrum*

$$\sigma(H_v) = [-4 - h, 4 + h].$$

Its eigenvalues are dense in this interval. The probability-one event in this statement is allowed to depend on h .

The conclusion includes arbitrarily weak nonzero disorder and the entire spectrum. It resolves the pure-point assertion of the two-dimensional Anderson localization conjecture for a uniform single-site distribution, as formulated in [14, Problem 2]. An arbitrary nondegenerate uniform interval reduces to the centered interval above by subtracting its midpoint from the operator. We do not assert one probability-one event for all disorder strengths, a bound uniform in energy, or a dynamical-localization theorem.

Context and scope

Anderson introduced random site energies as a mechanism for suppressing quantum transport, motivated by spin diffusion and impurity conduction [5]. His discussion includes a flat single-site distribution and nearest-neighbor hopping, although the original physical argument is not a modern proof of the spectral assertion considered here. The conductance-scaling theory of Abrahams, Anderson, Licciardello and Ramakrishnan predicts the absence of true metallic behavior in two dimensions [1]. That physical prediction is distinct from an almost-sure spectral-type theorem. Simon’s formulation isolates the latter question; his diffusion problem is listed separately [14].

Rigorous multidimensional localization theory developed several complementary mechanisms. Fröhlich and Spencer’s multiscale analysis gave Green-function decay and vanishing Kubo diffusion in large-disorder or appropriate extreme-energy regimes [12]. Their 1983 conclusions were explicitly distinguished from a complete pure-point decomposition. Fröhlich, Martinelli, Scoppola and Spencer subsequently established pure-point spectrum with exponentially decaying eigenfunctions for independent identically distributed diagonal potentials with a bounded density in sufficiently large-disorder and suitable tail-energy regimes [11]. Delyon, Lévy and Souillard also obtained multidimensional spectral localization in large-disorder or large-energy regimes [9]. The passage from fixed-energy Green-function information to a complete spectral decomposition is a substantive step, not an automatic consequence of decay outside an exceptional set of energies.

Aizenman and Molchanov developed the fractional-moment approach, obtaining localization at large disorder and at extreme energies under suitable hypotheses on the conditional single-site distributions, including absolute continuity and regularity [3]. This method supplies important ancestry for our use of fractional Green-function moments, but its established regimes do not by themselves yield the all-disorder, all-energy assertion above. Spectral-edge results also have a different scope. Ding and Smart proved localization near an edge for the two-dimensional Bernoulli model and observed that their proof extends to bounded nontrivial iid laws [10]; Hurtado extended edge-localization and unique-continuation methods to independent, not necessarily identically distributed uniformly bounded potentials with uniformly positive variance [13]. The latter localization theorems remain restricted to a bottom interval, even though their distributional assumptions are broader.

Weak-disorder transport on long finite scales poses another, compatible question. Black, Drogin and Hernández obtain quantitative time-averaged diffusion, resolvent and finite-volume eigenfunction results for Gaussian iid potentials on disorder-dependent scales [7]. These weak-coupling limits neither imply nor contradict pure-point spectral type at every fixed positive disorder for the uniform law. The present assertion concerns that fixed-disorder spectral type; it does not assert exponential eigenfunction decay or dynamical localization.

The final spectral step follows the rank-one mechanism of Simon and Wolff [15]. Its input is a suitable inverse-square spectral integral, equivalently a square-summable resolvent column, for almost every configuration and energy. Averaging over the independent site value transfers this information to the random spectral measure. We prove the required averaging and spectral readout explicitly; the main work is to obtain their fixed-energy input under the true uniform product law at arbitrary disorder.

The proof mechanism

Fix $h > 0$ and an interior real energy E . In Section 3, we introduce absorbing diagonal terms in finite tori and their associated input and output channels, called ports. Squared scattering amplitudes define a Dirichlet energy on port data: if S_{ja} is the scattering amplitude from port a to port j and

f is real data on the ports, the energy is

$$\mathcal{Q}_S(f) = \frac{1}{2} \sum_{a,j} |S_{ja}|^2 |f_j - f_a|^2.$$

A Cauchy-distributed real closing value would remove a port with a favorable averaged energy comparison. The bounded physical distribution requires a truncated version and a relative estimate for its error.

The first main ingredient is a family of rare resonant cells (Sections 4 to 6). Their conditional law is embedded in the original product measure by an acceptance likelihood; the true potentials remain independent uniforms before conditioning. The construction gives a weighted boundary impedance estimate uniform in arbitrary passive exterior data, including weakly coupled missing boundary directions. It also gives a transmission lower bound strong enough to make the closing error small relative to the transmission itself. These estimates, together with an upper envelope over adaptive interior configurations, are useful beyond a single prescribed environment.

The second ingredient is a comparison along a hierarchy of increasingly sparse ports (Section 7). A parent-cell alternative controls the apparent singularity at the end of each thinning interpolation. Local generic observation, followed by path estimates, supplies screening at every fixed starting stage (Section 8). For a fixed starting level n , let d_n be the infinite-volume limit of the expected sum of squared port-to-port distances, weighted by $|S_{ja}|^2$ and divided by twice the torus area. These quantities measure the scattering energy of affine coordinate data. The torus side tends to infinity with n fixed to define d_n ; only afterwards does the thinning comparison give a finite limit d as n tends to infinity.

The dimension enters decisively in forcing $d = 0$ (Section 9). An adaptive modification of affine data first makes a local affine first-variation trace small in expectation. Separately, a logarithmic voltage profile separates retained interior ports from an annulus of very weak probes at arbitrarily small two-dimensional capacity cost. If $d > 0$, an analytic test of the scattering matrix has a nonzero value at the center of the disk. Support-preserving shifts of the true potentials and the thinning comparison make its boundary moments arbitrarily small, contradicting submean. Dimension-free weak integral bounds prevent the number of ports from spoiling this argument.

Finally, vanishing affine energy is upgraded to a fixed-real-energy fractional-moment estimate for the true finite-torus Green function (Section 10). Buffered probe barriers, conditional estimates under transmission weights, and resampling of disjoint regions close a contraction uniform over the allowed masks. The infinite-volume conclusion then follows from the spectral theorem, Fubini, and rank-one averaging (Section 11). The property to which Fubini is applied belongs to the true operator, so no common auxiliary construction measurable in energy is needed.

The proof keeps these roles separate: the rare laws prove local comparisons, fixed-stage screening controls limits, two-dimensional capacity makes the limit vanish, and spectral averaging converts fixed-energy information to spectral type.

2 Finite volumes and conventions

Throughout the local and multiscale arguments, fix

$$h > 0, \quad -4 - h < E < 4 + h. \quad (2.1)$$

All constants and auxiliary choices may depend on this pair. Their dependence on the varying geometric scales, exterior data and masks is specified in each assertion. Uniformity in E is neither assumed nor needed for Theorem 1.1.

For integer $l \geq 0$, put $b_l = 3^l$. The global finite volume is the square torus

$$\mathbb{T}_N^2 = (\mathbb{Z}/b_N\mathbb{Z})^2,$$

with the periodic nearest-neighbor adjacency and independent uniform potentials on its sites. Its area is b_N^2 . Distances are shortest coordinate distances on this torus, using the maximum norm unless a different norm is displayed. We write

$$\Lambda_r(x) = \{y : |y - x|_\infty \leq r\}.$$

All local constructions are used with the torus sufficiently large to contain the required neighborhoods without ambiguity. Dirichlet restrictions mean deletion of the hopping bonds leaving the indicated physical site set.

The physical disorder is always the product law of the true variables v_x . Auxiliary selectors, acceptance uniforms, offsets and thinning marks will be introduced explicitly. Conditioning on a rare event can change the law within one cell; it never changes this unconditional physical law. An expectation refers to the sampling and conditioning specified in its statement.

Site and port spaces are complexified finite-dimensional Euclidean spaces. A superscript $\top$ denotes transpose, not adjoint; a superscript $*$ denotes adjoint. Bases used for coordinate projections and their subspaces are real orthonormal bases. We use

$$\operatorname{Im} D = \frac{D - D^*}{2i}, \quad \operatorname{Re} D = \frac{D + D^*}{2}.$$

Unless otherwise indicated, a norm is the Euclidean or induced operator norm. Hilbert–Schmidt and trace norms carry subscripts HS and 1. Coordinate projections may be identified with their ranges. Symbols used for local blocks are reset when the local matrix is defined.

A real reference potential may replace a true potential only at a site carrying an open cell port. Closing that port restores the true value. The matrix $H^\#$ denotes the real symmetric operator with the current reference values. Artificial probes, introduced in the next section, do not alter real reference values. Several ports may share a physical site; their columns are kept distinct.

In bounds for a cell of size L , the notation $\operatorname{poly}(L)$ means a quantity bounded by $C(1 + L)^C$ for fixed constants. The constants may depend on the previously fixed cell-design parameters, including the corner-box width exponent, but not on the exterior matrix, exterior volume, exterior source, or allowed probe mask. The ordering of design parameters is part of each construction. Expressions $\exp(L^{O(1)})$ have the corresponding interpretation. A statement requiring sufficiently large L is applied only after its fixed design parameters have been chosen.

We will repeatedly distinguish the following orders of choice. The disorder and energy are fixed first. Local design constants are fixed before the cell scale is enlarged. For global limits a starting level n is fixed before observation and approximation constants are chosen; a later level m can then be increased to exceed all those thresholds, and the torus level N is increased last. No rate uniform in n is implicit in this convention.

3 Scattering matrices and analytic estimates

All matrices in this section are finite dimensional. A port at a physical site x , of strength $\sigma > 0$, is represented by the column $\sqrt{\sigma} e_x$. In the matrix Y of these columns, rows index physical sites and columns index ports; several distinct ports may be attached to the same site. For a real symmetric matrix $H^\#$ and a real energy E , set

$$\mathcal{L} = H^\# - E - iYY^\top, \quad G = \mathcal{L}^{-1}, \quad S = \mathbf{1} + 2iY^\top GY. \quad (3.1)$$

The entry S_{ja} is the scattering amplitude from input port a to output port j . The physical matrix $H^\#$ may differ from the true Hamiltonian by real diagonal changes at sites carrying ports. We call these changes *reference potentials*. They may depend on the true potentials.

Absorption bounds the response between ports even when the full inverse G is large. Closing ports removes their absorptions and changes real diagonal values; we express this operation as a rational family of scattering matrices and compare its energies after averaging the closing values.

3.1 Invertibility, absorption, and Schur complements

The inverses in Equation (3.1) exist almost surely under any absolutely continuous joint law of the true site potentials. Indeed, if $\mathcal{L}u = 0$, taking the imaginary part of $u^* \mathcal{L}u$ gives $Y^\top u = 0$. Thus u vanishes at every port site, so restoring the true diagonal entries leaves its equation unchanged: $(H - E)u = 0$. For a fixed finite site set, $\det(H - E)$ is a polynomial in its true diagonal variables whose coefficient of their full product is one. It is not identically zero. A nonzero polynomial has a Lebesgue-null zero set, as follows by induction on the number of variables and one-variable slicing. This proves the assertion even when the port set and reference potentials depend on those variables. The same argument applies to deterministic restrictions to physical sites. We always discard the resulting null exceptional sets when forming such inverses.

A matrix D is *passive* if $\text{Im } D \leq 0$, with the Hermitian imaginary part fixed in Section 2. The absorption identities below apply to any real rectangular Y , including real coordinate or subspace inclusions.

Lemma 3.1 (Absorption identities). *Suppose $\mathcal{L}$ is invertible and $W = -\text{Im } \mathcal{L} \geq YY^\top$. Then*

$$\text{Im } G = G^* W G = G W G^*, \quad (3.2)$$

$$\|Y^\top G Y a\|^2 \leq \text{Im}(a^* Y^\top G Y a), \quad \|Y^\top G Y\| \leq 1. \quad (3.3)$$

For the physical system in Equation (3.1), S is symmetric and unitary. If $\mathcal{L}$ is complex symmetric and passive, each defined Schur complement is complex symmetric and passive. An absorption lower bound supported on retained coordinates persists in the Schur complement.

Proof. The inverse difference identities give

$$G - G^* = G^*(\mathcal{L}^* - \mathcal{L})G = G(\mathcal{L}^* - \mathcal{L})G^*,$$

which proves Equation (3.2). Consequently, with $K = Y^\top G Y$,

$$\|K a\|^2 \leq a^* Y^\top G^* W G Y a = \text{Im}(a^* K a) \leq \|a\| \|K a\|.$$

This proves Equation (3.3). In the physical case $W = YY^\top$, we have $\text{Im } K = K^* K$ and hence

$$(\mathbf{1} + 2iK)^*(\mathbf{1} + 2iK) = \mathbf{1} - 4\text{Im } K + 4K^* K = \mathbf{1}.$$

Since the matrix is square, it is unitary. Symmetry follows from $\mathcal{L}^\top = \mathcal{L}$ and the fact that Y is real.

For the Schur assertion, write a symmetric matrix in blocks as

$$D = \begin{pmatrix} D_{11} & C \\ C^\top & D_{22} \end{pmatrix}, \quad D_{\text{eff}} = D_{11} - C D_{22}^{-1} C^\top.$$

For a vector v on the retained coordinates, extend it to $\tilde{v} = (v, -D_{22}^{-1}C^\top v)$. Then

$$D\tilde{v} = (D_{\text{eff}}v, 0), \quad \tilde{v}^* D\tilde{v} = v^* D_{\text{eff}}v.$$

Taking imaginary parts proves passivity. If $-\text{Im } D \geq \text{diag}(W_0, 0)$, the same identity gives $-\text{Im } D_{\text{eff}} \geq W_0$. Symmetry is immediate from the displayed Schur formula. $\square$

We will also use the elementary kernel consequence of passivity: if $Du = 0$ and D is passive, then $\text{Im } D u = \text{Re } D u = 0$. Indeed, the nonnegative matrix $-\text{Im } D$ has zero quadratic form on u , so it annihilates u ; the assertion about the real part follows from $Du = 0$.

3.2 Closing ports and comparing energies

Symmetry and unitarity make $|S_{ja}|^2$ symmetric nonnegative weights with unit row sums. For real scalar or real Euclidean-vector data f on the ports, their Dirichlet energy and its single-port contribution are

$$\mathcal{Q}_S(f) = \frac{1}{2} \sum_{a,j} |S_{ja}|^2 |f_j - f_a|^2, \quad \mathcal{M}_a(f) = \sum_j |S_{ja}|^2 |f_j - f_a|^2. \quad (3.4)$$

For scalar data, writing $F = \text{diag}(f)$, unitarity gives

$$\mathcal{Q}_S(f) = \text{tr}(F^2 - F S F S^*). \quad (3.5)$$

For vector data, one sums this identity over real components.

Lemma 3.2 (Common-phase closure). *Partition the ports into retained ports and a set C to be closed, and write*

$$S = \begin{pmatrix} A & T \\ T^\top & B \end{pmatrix}.$$

Close every port in C and add the real shift $w\sigma_j$ to the physical diagonal for each closed port j of strength σ_j . Where the resulting inverse exists, its scattering matrix on the retained ports is

$$S_z = A + Tz(\mathbf{1} - zB)^{-1}T^\top, \quad z = \frac{w+i}{w-i}. \quad (3.6)$$

The right-hand side is analytic for $|z| < 1$.

Proof. If Y_C consists of the closed port columns, the site matrix becomes $\mathcal{L}_w = \mathcal{L} + (w+i)Y_C Y_C^\top$. Woodbury inversion and $Y_C^\top G Y_C = (B - \mathbf{1})/(2i)$ give the result, using

$$\frac{z}{1-z} = -\frac{w+i}{2i}, \quad \mathbf{1} + (w+i)Y_C^\top G Y_C = \frac{\mathbf{1} - zB}{1-z}.$$

These are rational identities, so the formulas extend across any removable singularities. The determinant of $\mathcal{L}_w$ is a polynomial in w which is nonzero at $w = -i$; hence only finitely many closing values can make the site matrix singular. Finally B is a compression of a unitary matrix and is therefore contractive, which makes $\mathbf{1} - zB$ invertible in the open disk. $\square$

Lemma 3.3 (Conditional single-port comparison). *Consider closing a single port a . Set*

$$u = S_{\hat{a},a}, \quad b = S_{aa}, \quad \gamma = \frac{z}{1 - zb},$$

where $\hat{a}$ means that port a is omitted. Let $\mathcal{F}$ contain the open system and the real data f , before the closing value is sampled. Assume that its conditional law assigns no mass to the finite exceptional set in Lemma 3.2. If $|b| < 1$, put $\mu = \mathbb{E}[\gamma \mid \mathcal{F}]$. Then

$$\mathbb{E}[\mathcal{Q}_{S_z}(f_{\hat{a}}) \mid \mathcal{F}] \leq \mathcal{Q}_S(f) + 2|\mu| \mathcal{M}_a(f). \quad (3.7)$$

If $|b| = 1$, the same assertion holds with zero error, and we use the convention $\mu = 0$.

Proof. First take scalar data and subtract f_a from every datum. This does not change either energy, and it makes the removed datum zero. Let $F = \text{diag}(f_{\hat{a}})$ after this subtraction and $A = S_{\hat{a},\hat{a}}$. By Equation (3.5), expansion of $S_z = A + \gamma uu^\top$ yields the exact identity

$$\mathcal{Q}_{S_z}(f_{\hat{a}}) - \mathcal{Q}_S(f) = -2 \text{Re}(\gamma u^\top F A^* F u) - |\gamma|^2 (u^* F u)^2.$$

The last square is real and nonnegative. Moreover,

$$\left| u^\top F A^* F u \right| \leq \|F u\|^2 = \mathcal{M}_a(f),$$

because A is contractive. Conditional expectation therefore proves Equation (3.7). For vector data the same argument is applied componentwise. If $|b| = 1$, unitarity gives $u = 0$, so closing the port leaves the scattering matrix on the other ports unchanged. $\square$

The use of the conditional mean μ , rather than $\mathbb{E}[|\gamma| \mid \mathcal{F}]$, is important. In later applications the open system and the data must remain measurable without exposing the hidden true value at the port to be closed.

A standard Cauchy law for w makes this conditional mean vanish. To restore a potential within a bounded interval, we truncate that law and estimate the resulting error.

Lemma 3.4 (A cutoff-Cauchy closing law). *Fix a smooth function $\chi : \mathbb{R} \rightarrow [0, 1]$ equal to one near zero and supported in a fixed sufficiently small interval about zero. For a port of strength σ , give its hidden real shift δ the law*

$$\frac{1}{Z_\sigma} \chi(\delta) \frac{\sigma \, d\delta}{\pi(\delta^2 + \sigma^2)}, \quad Z_\sigma = \int_{\mathbb{R}} \chi(\delta) \frac{\sigma \, d\delta}{\pi(\delta^2 + \sigma^2)}. \quad (3.8)$$

Use $w = \delta/\sigma$ in Lemma 3.2. Conditional on the open system, let $g = \sigma G_{xx}$, where x is the site of the port. For all sufficiently small σ , the mean in Lemma 3.3 satisfies

$$|\mu| \leq C, \quad |\mu| \leq C\sqrt{\sigma} \quad \text{if } |g| > \sqrt{\sigma}. \quad (3.9)$$

The constants depend on χ , not on the exterior system.

Proof. The case $|S_{aa}| = 1$ uses the zero-error convention. Otherwise put $b = S_{aa} = 1 + 2ig$, so $|b| < 1$. Under the uncut Cauchy law,

$$z = \frac{\delta + i\sigma}{\delta - i\sigma} = e^{it}, \quad \delta = \sigma \cot(t/2),$$

is Haar distributed on the circle. Its mean of $z/(1-bz) = \sum_{k \geq 0} b^k z^{k+1}$ is zero. Define the deleted angular density

$$d_\sigma(t) = 1 - \chi(\sigma \cot(t/2)).$$

It is bounded by one, supported on an arc of length $O(\sigma)$ about $t = 0$, and Lipschitz with constant $O(\sigma^{-1})$. To see the last claim, derivatives can occur only when δ lies in a fixed compact set separated from zero; there $|t|$ is comparable to σ and $|d\delta/dt| = O(\sigma^{-1})$. Near $t = 0$ the function is constant. Thus $Z_\sigma = 1 - O(\sigma)$ and

$$\mu = -Z_\sigma^{-1} \int_{\mathbb{T}} d_\sigma(t) \frac{e^{it}}{1 - be^{it}} dm_{\mathbb{T}}(t), \quad (3.10)$$

where $dm_{\mathbb{T}}(t) = dt/(2\pi)$.

On the support of d_σ ,

$$|1 - be^{it}| \geq |1 - b| - |b| |1 - e^{it}| \geq 2|g| - C\sigma.$$

If $|g| > \sqrt{\sigma}$, this is at least $|g|$ for small enough σ . The integral in Equation (3.10) is then bounded by $C\sigma/|g| \leq C\sqrt{\sigma}$.

For the uniform bound, $|b| \leq 1/2$ is immediate. If $1/2 < |b| < 1$, use

$$\frac{z}{1 - bz} = b^{-1} \left(\frac{1}{1 - bz} - 1 \right).$$

Write $b = \rho e^{i\beta}$ and use

$$\frac{1}{1 - \rho e^{it}} = \frac{1}{2} (1 + P_\rho(t) + iQ_\rho(t)),$$

where

$$P_\rho(t) = \frac{1 - \rho^2}{1 - 2\rho \cos t + \rho^2}, \quad Q_\rho(t) = \frac{2\rho \sin t}{1 - 2\rho \cos t + \rho^2}.$$

The Poisson contribution is bounded since $P_\rho \geq 0$ has integral one. For the conjugate contribution, if the singular center $-\beta$ is farther than a sufficiently large multiple of σ from the support arc, use $|Q_\rho(t)| \leq C/|t|$ in angular distance to bound the integral absolutely. Otherwise, after recentering at $-\beta$, the density is supported in $|t| \leq C'\sigma$ and has Lipschitz constant C/σ . Oddness gives

$$\left| \int Q_\rho(t) d_\sigma(t - \beta) dm_{\mathbb{T}}(t) \right| \leq C \int_0^{C'\sigma} \frac{1}{t} \frac{t}{\sigma} dt \leq C.$$

The constant part and normalization in Equation (3.10) are harmless. This proves both bounds. $\square$

3.3 Dimension-independent analytic bounds

The common-phase closure formula contains the resolvent $(\mathbf{1} - zB)^{-1}$. We need weak- L^1 bounds for its sandwiched and trace expressions, with no dimensional factor beyond the displayed norms. For a nonnegative random variable X on the circle, write

$$\|X\|_{L^{1,\infty}} = \sup_{t>0} t m_{\mathbb{T}}\{X > t\}.$$

The Hilbert–Schmidt estimate below has the same dimension-independent scale as the dissipative-resolvent bound in [2, Lemma 3.1 and Appendix C]. We prove the disk estimates directly using a Hilbert-space-valued Calderón–Zygmund decomposition [6, Theorem 1].

Lemma 3.5 (Weak- L^1 resolvent estimate). *Let B be a contraction and let U, V be compatible deterministic matrices. At the almost-everywhere boundary values of the rational function in question,*

$$\left\| \left\| U(\mathbf{1} - zB)^{-1}V \right\|_{\text{HS}} \right\|_{L^{1,\infty}} \leq C \|U\|_{\text{HS}} \|V\|_{\text{HS}}. \quad (3.11)$$

For any compatible square matrix D ,

$$\left\| \left\| \text{tr}(D(\mathbf{1} - zB)^{-1}) \right\|_{L^{1,\infty}} \right\| \leq C \|D\|_1. \quad (3.12)$$

Here and below $\|D\|_1$ denotes the trace norm. In particular, every moment of order $0 < p < 1$ is bounded by C_p times the corresponding right-hand scale to the power p .

Proof. We first prove a weak estimate for the conjugate Poisson kernel and then represent the matrix resolvent by a positive matrix-valued measure. Let ν be a finite measure with values in a Hilbert space, and let m_0 be its total variation. Uniformly in $0 \leq \rho < 1$,

$$\left\| \left\| Q_\rho * \nu \right\|_{L^{1,\infty}} \right\| \leq C m_0, \quad \int_{\mathbb{T}} \|P_\rho * \nu\| \, dm_{\mathbb{T}} \leq m_0. \quad (3.13)$$

The Poisson assertion follows from positivity and Fubini. If $m_0 = 0$, both assertions are immediate. Otherwise, for the conjugate assertion, identify the circle with a normalized interval of length one and fix a level $t \geq 2m_0$. Choose maximal dyadic intervals I for which $|\nu|(I) > t|I|$, where $|\nu|$ denotes total variation. They are disjoint and have total length at most m_0/t . Their dyadic parents are not selected, so $|\nu|(I) \leq 2t|I|$.

Replace ν on each selected interval by its vector average. The resulting good measure has a density g with

$$\|g\|_{L^\infty} \leq 2t, \quad \|g\|_{L^1} \leq m_0, \quad \|g\|_{L^2}^2 \leq 2tm_0.$$

These assertions include a possibly singular original measure. The variation measure restricted to the complement of the selected intervals is bounded by $t|J|$ on every dyadic interval J : if J lies inside a selected interval the restriction vanishes there, and otherwise J cannot have variation greater than $t|J|$. Dyadic coverings extend this domination to Borel sets, proving absolute continuity and the density bound. The Fourier multipliers of Q_ρ have modulus at most one, and Hilbert-space Parseval therefore gives $\|Q_\rho * g\|_{L^2}^2 \leq 2tm_0$.

The bad pieces

$$\nu_I = \nu|_I - \frac{\nu(I)}{|I|} \mathbf{1}_I \, dm_{\mathbb{T}}$$

have zero total mass and total variations summing to at most $2m_0$. Interpret a concentric triple interval of length at least one as the whole circle. Outside the triple interval $3I$, cancellation and the uniform kernel derivative bound

$$|Q'_\rho(u)| \leq \frac{C}{\text{dist}(u, 0)^2}$$

give

$$\int_{\mathbb{T} \setminus 3I} \|Q_\rho * \nu_I\| \, dm_{\mathbb{T}} \leq C |\nu_I|(I).$$

Indeed, subtract the kernel evaluated at the center of I , bound its difference by $C|I|/\text{dist}(u, I)^2$, and integrate away from $3I$. For completeness, in angular coordinates and $\rho > 0$,

$$|Q'_\rho(u)| \leq \frac{2\rho}{|1 - \rho e^{iu}|^2} \leq \frac{1}{2 \sin^2(u/2)} \leq \frac{C}{\text{dist}(u, 2\pi\mathbb{Z})^2};$$

the case $\rho = 0$ is trivial. Rescaling the circle only changes the absolute constant, so the bound is uniform as ρ tends to one. The union of the triple intervals has length at most $3m_0/t$. Chebyshev applied to the good L^2 bound and the bad L^1 bound outside this union gives $m_{\mathbb{T}}\{\|Q_\rho * \nu\| > Ct\} \leq Cm_0/t$. Levels below $2m_0$ are trivial. This proves Equation (3.13).

Now define the analytic matrix function

$$F_1(z) = 2(\mathbf{1} - zB)^{-1} - \mathbf{1}.$$

It satisfies $F_1(0) = \mathbf{1}$ and

$$\operatorname{Re} F_1(z) = (\mathbf{1} - \bar{z}B^*)^{-1}(\mathbf{1} - |z|^2 B^*B)(\mathbf{1} - zB)^{-1} \geq 0.$$

For $r < r' < 1$, the positive matrix-valued measure

$$dM_{r'}(t) = \operatorname{Re} F_1(r'e^{it}) \, dm_{\mathbb{T}}(t)$$

has total mass $\mathbf{1}$. Fourier series show that on the circle of radius r , F_1 is its Poisson integral plus i times its conjugate Poisson integral, with kernel radius r/r' . There is no additional constant imaginary part because $F_1(0) = \mathbf{1}$.

The measure $UM_{r'}V$, considered in the Hilbert space of matrices with Hilbert–Schmidt norm, has variation at most $\|U\|_{\text{HS}} \|V\|_{\text{HS}}$. To verify this for a finite measurable partition, write M_j for its positive matrix masses. Then

$$\begin{aligned} \sum_j \|UM_jV\|_{\text{HS}} &\leq \sum_j \|UM_j^{1/2}\|_{\text{HS}} \|M_j^{1/2}V\|_{\text{HS}} \\ &\leq \left(\sum_j \|UM_j^{1/2}\|_{\text{HS}}^2 \right)^{1/2} \left(\sum_j \|M_j^{1/2}V\|_{\text{HS}}^2 \right)^{1/2} = \|U\|_{\text{HS}} \|V\|_{\text{HS}}. \end{aligned}$$

Taking the supremum over partitions proves the claim. Applying Equation (3.13) proves the required weak estimate for UF_1V . The added constant UV in $U(\mathbf{1} - zB)^{-1}V = (UF_1(z)V + UV)/2$ obeys the same scale bound.

For the trace assertion, factor $D = XY$ by singular values so that $\|X\|_{\text{HS}} \|Y\|_{\text{HS}} = \|D\|_1$. For any positive partition masses M_j of total mass $\mathbf{1}$,

$$\sum_j |\operatorname{tr}(DM_j)| \leq \sum_j \|YM_j^{1/2}\|_{\text{HS}} \|M_j^{1/2}X\|_{\text{HS}} \leq \|D\|_1.$$

Thus the scalar measure $\operatorname{tr}(DM_{r'})$ has variation at most the trace norm of D , and the same argument applies.

All estimates are uniform on inner circles. Rationality gives radial limits outside finitely many boundary points. Applying Fatou to level sets, first at a slightly smaller level and then increasing it, gives Equations (3.11) and (3.12) on the boundary. Finally, a tail bound $\mathbb{P}\{X > t\} \leq \min(1, A/t)$ gives

$$\mathbb{E}X^p = p \int_0^\infty t^{p-1} \mathbb{P}\{X > t\} \, dt \leq C_p A^p, \quad 0 < p < 1.$$

□

We record a useful extension of the same proof. If a rational analytic $d \times d$ matrix function F has nonnegative Hermitian real part in the disk, its representing positive measures have total mass $\operatorname{Re} F(0)$. Their Hilbert–Schmidt variation is at most $\operatorname{tr}(\operatorname{Re} F(0)) \leq d \|F(0)\|$. Restoring the constant $i \operatorname{Im} F(0)$, the preceding argument gives

$$\left\| \left\| F(e^{it}) \right\| \right\|_{L^{1,\infty}} \leq Cd \|F(0)\|. \quad (3.14)$$

We need only the resulting polynomial dependence on dimension and the norm at the center.

3.4 Determinant phases and artificial absorptions

To average inverse traces over a real diagonal shift, we integrate the phase of the determinant. The estimate must also allow artificial absorptions that vary with that shift.

Lemma 3.6 (Phase integration). *Let $D(s)$ be a continuously differentiable $d \times d$ matrix on a bounded real interval, with $-\operatorname{Im} D(s) > 0$. Its determinant has a continuous phase Φ such that*

$$-d\pi < \Phi(s) < 0, \quad \Phi'(s) = \operatorname{Im} \operatorname{tr}(D'(s)D(s)^{-1}). \quad (3.15)$$

Suppose, in particular, that

$$D'(s) = P_J - i\tau \operatorname{diag}_J(a'_i(s)), \quad 0 \leq \tau \leq 1, \quad |a'_i(s)|^2 \leq C a_i(s),$$

where P_J is the coordinate projection onto J , and that $-\operatorname{Im} D(s) \geq \tau \operatorname{diag}_J(a_i(s))$. With $G = D^{-1}$,

$$\Phi'(s) \geq \frac{1}{2} \operatorname{tr}_J \operatorname{Im} G(s) - C' d. \quad (3.16)$$

Consequently, on an interval $[a, b]$,

$$\int_a^b \operatorname{tr}_J \operatorname{Im} D(s)^{-1} ds \leq C'' d(1 + b - a). \quad (3.17)$$

The integral bound remains valid for passive matrices at their almost-everywhere existing inverses, by strict regularization.

Proof. Every eigenvalue of a strictly passive matrix lies in the open lower half-plane: test the imaginary part of the quadratic form on an eigenvector. Use the branch of the logarithm whose arguments there lie in $(-\pi, 0)$. The sum of the eigenvalue arguments, counted with algebraic multiplicity, defines a continuous phase of the determinant in $(-d\pi, 0)$. The logarithmic derivative of the determinant gives Equation (3.15); no differentiability of individual eigenvalues is needed.

The absorption identity gives, for every $i \in J$,

$$(\operatorname{Im} G)_{ii} \geq \tau a_i |G_{ii}|^2.$$

Thus the derivative contribution of an artificial diagonal satisfies

$$\tau |a'_i G_{ii}| \leq C^{1/2} \sqrt{\tau} \sqrt{(\operatorname{Im} G)_{ii}}.$$

This is also valid when $a_i = 0$, since then $a'_i = 0$. Taking imaginary parts of the trace derivative and applying Young's inequality yields

$$\Phi' \geq \sum_{i \in J} (\operatorname{Im} G)_{ii} - C^{1/2} \sqrt{\tau} \sum_{i \in J} \sqrt{(\operatorname{Im} G)_{ii}} \geq \frac{1}{2} \operatorname{tr}_J \operatorname{Im} G - C' d.$$

Integration, followed by $\Phi(b) - \Phi(a) \leq d\pi$, proves Equation (3.17). For merely passive D , apply the same argument to $D - i\zeta \mathbf{1}$, $\zeta > 0$. The bounds are independent of ζ ; where D^{-1} exists, the inverses converge as ζ tends to zero, and Fatou proves the stated extension. $\square$

Two variants will be used without changing this argument. If the real derivative is a Hermitian matrix $K(s) \geq cP_J$ for a fixed $c > 0$, its contribution is $\operatorname{tr}(K \operatorname{Im} G) \geq c \operatorname{tr}_J \operatorname{Im} G$. Thus Equation (3.17) holds with constants also depending on c . Additional real diagonal derivatives can instead be retained explicitly in Equation (3.15) and estimated using available absorption bounds. This is how the calibrated root derivative will be handled.

3.5 The artificial probe profile

Separate from the cell ports, we allow at most one artificial probe at each site. A prescribed *mask* chooses among absence, strength τ , and strength $\tau\vartheta(v_i)$, where v_i is always the true site potential, including when the real reference value differs from it. Masks do not depend on the potentials. They may be fixed after conditioning on spatial offsets or other independent prescriptions. Zero strength means absence.

Here $\vartheta : [-h, h] \rightarrow [0, 1]$ is fixed, of class $C^{1,1}$, polynomial on finitely many pieces, positive on $(-h, h)$, and identically one on a plateau containing $[-h/2, h/2]$ and the compact interior potential settings of the cell construction. Such a profile can be chosen after those compact settings are fixed and before the later scale and width parameters are selected. To construct it, choose $a < h$ large enough that the required compact set lies in $[-a, a]$, and use $3t^2 - 2t^3$ on each endpoint transition, with $t = (v + h)/(h - a)$ on $[-h, -a]$ and $t = (h - v)/(h - a)$ on $[a, h]$; put $\vartheta = 1$ on $[-a, a]$. The endpoint values are zero, the derivatives match at the joins, and

$$|\vartheta'(v)|^2 \leq C\vartheta(v), \quad \vartheta(v) \asymp \text{dist}(v, \{-h, h\})^2 \quad \text{near the endpoints.} \quad (3.18)$$

Constants may depend on the fixed profile. In particular, ϑ^{-q} is integrable on $[-h, h]$ for every $q < 1/2$.

For estimates at a single cell of size L , artificial strengths will obey $0 \leq \tau \leq e^{-c_*L}$ with a sufficiently large fixed $c_* > 0$. The screening estimates will also allow $0 \leq \tau \leq 1$. All spatial masks remain arbitrary subject to the independence convention above.

4 The resonant cell law

Throughout this section, $h > 0$ and $E \in (-4 - h, 4 + h)$ are fixed. All choices may depend on these two parameters. A polynomial bound in L means $C(1 + L)^C$, with constants independent of the exterior system, its dimension, and the artificial-probe mask. The construction below specifies an auxiliary law on one cell. Its incorporation into the true product law is deferred to [Proposition 7.1](#).

4.1 Clipping the corners

Fix an integer margin $b \geq 3$. In a tile of odd side $b_l = 3^l$, use coordinates centered at its central site and put

$$L = \frac{b_l - 1}{2} - b, \quad V = \{-L, \dots, L\}^2 \setminus \{(\pm L, \pm L)\}, \quad J = V \setminus \{0\}. \quad (4.1)$$

Only sufficiently large l are used. Let

$$B = \{x \notin V : \text{dist}_1(x, V) = 1\}, \quad p_x = \mathbf{1}_{\{|x|_1=1\}} \quad (x \in J).$$

Thus p is the adjacency column from the root 0 to J . Let $M : \mathbb{R}^B \rightarrow \mathbb{R}^J$ be the adjacency matrix from B to J ; when used on V , it has an additional zero row at the root. Define

$$Z = \{x \in J : M_{x,\cdot} \neq 0\}, \quad Z^+ = \{x \in J : \text{dist}_1(x, Z) \leq 1\}, \quad P = \text{proj}_{(\ker M)^\perp}. \quad (4.2)$$

We identify a projection with its range when specifying a matrix domain. The cell and its corner coupling are shown in [Figure 1](#).

Lemma 4.1 (The clipped boundary). *The set Z consists of the four sides of the square with their corners removed. Every site of Z has an inward nearest neighbor in $J \setminus Z$. The restriction $M : P \rightarrow \mathbb{R}^Z$ is an isomorphism whose singular values lie in $[1, \sqrt{3}]$. In particular,*

$$\|M\| \leq \sqrt{3}, \quad \|Mw\| \geq \|w\| \quad (w \in P).$$

There is an orthogonal decomposition $P = P_0 + Q$: P_0 is the coordinate projection onto the single side sites of B , and Q is the direct sum of four two-dimensional corner planes.

Proof. At the positive corner the relevant sites of B , in their fixed order, are

$$e_1 = (L+1, L-1), \quad e_2 = (L-1, L+1), \quad e_3 = (L, L).$$

Here and below the same symbols also denote the corresponding coordinate vectors. The two rows indexed by $(L, L-1)$ and $(L-1, L)$ give the block

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}. \quad (4.3)$$

Its row space is $\text{span}\{e_1 + e_3, e_2 + e_3\}$. We use the orthonormal basis

$$n = \frac{(-1, -1, 1)}{\sqrt{3}}, \quad r_1 = \frac{(1, -1, 0)}{\sqrt{2}}, \quad r_2 = \frac{(1, 1, 2)}{\sqrt{6}}. \quad (4.4)$$

Here n spans the kernel, and r_1, r_2 span the corner plane. The two nonzero singular values are 1 and $\sqrt{3}$. Use the reflected geometric configuration and the same ordered basis coefficients at the other three corners. Every remaining nonzero row of M contains one entry, equal to 1, in a distinct single side coordinate. This proves the asserted decomposition and singular-value bounds. In fact $|B| = 8L$, $|Z| = 8L - 4$, and $\dim P_0 = 8L - 12$.

For example, an inward neighbor of $(L, y) \in Z$ is $(L-1, y)$, which is not in Z ; the other sides follow by reflection. These choices remain valid next to a clipped corner. All connections from V to its complement pass through B , and $V \cup B$ lies inside the tile because of the fixed margin b . $\square$

4.2 Calibrating the root with a decaying profile

We choose the root reference value so that the Dirichlet cell has a mode at energy E . For arbitrary real values $v|_J$, write

$$A = \text{Adj}_J + \text{diag}_J(v - E).$$

Whenever A is invertible, define

$$c(v) = E + p^\top A^{-1} p. \quad (4.5)$$

The Dirichlet matrix on V , with root potential $c(v)$, then satisfies

$$\begin{pmatrix} c(v) - E & p^\top \\ p & A \end{pmatrix} \begin{pmatrix} 1 \\ -A^{-1} p \end{pmatrix} = 0. \quad (4.6)$$

Thus invertibility of the punctured matrix A permits resonance of the full Dirichlet cell. We must choose the interior law so that A^{-1} is controlled, $c(v)$ stays inside the physical potential interval, and this root-normalized mode is small at the boundary. The transmission estimates will later control how it couples to an arbitrary passive exterior.

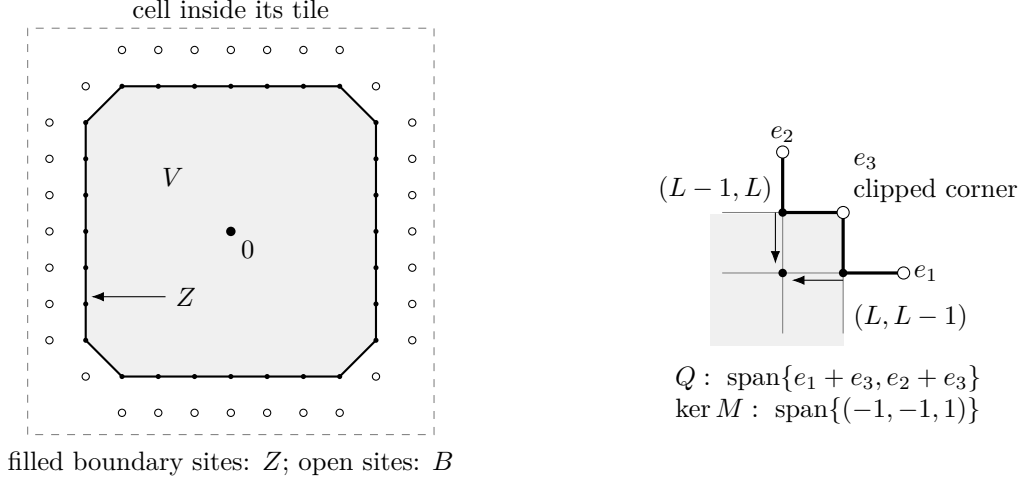


Figure 1: The clipped cell and a magnified positive corner. The two boundary rows couple to a triple of exterior coordinates, leaving exactly one missing direction. The inward arrows reach a site outside Z ; this geometric fact is used in the transmission argument.

We first construct a comparison profile ψ , normalized by $\psi_0 = 1$, and potentials v^0 satisfying its local equation. The profile is defined also on B , where it is small but nonzero, so it need not be a Dirichlet eigenvector. A deformation of this profile will give a polynomial inverse bound for the punctured matrix at v^0 . These two properties will control the exact calibrated mode after the interior values are randomly perturbed.

Choose a rational multiple θ of π and a real phase ϕ such that

$$|E - 4 \cos \theta| < h, \quad m_\theta := \min_{k \in \mathbb{Z}} |\cos(\theta k + \phi)| > 0. \quad (4.7)$$

The first choice is possible because the energy is in $(-4 - h, 4 + h)$ and rational multiples of π are dense. The sequence of cosine values is periodic, so the second condition excludes only finitely many phases modulo 2π . Put

$$d = \frac{h - |E - 4 \cos \theta|}{8} > 0.$$

Define the real sequence f on all integers by

$$\begin{aligned} f_0 &= 0, \\ f_k + f_{k+1} &= \text{sign}(\cos(\theta k + \phi) \cos(\theta(k+1) + \phi)). \end{aligned} \quad (4.8)$$

Recursion in either direction gives $|f_k| \leq |k|$. For small fixed $\nu, c_u > 0$, and $|u| \leq c_u/L$, set

$$\begin{aligned} \psi_x(u) &= e^{-\nu|x|_\infty} \frac{\cos(\theta(x_1 + x_2) + \phi)}{\cos \phi} e^{u f_{x_1 + x_2}} \quad (x \in V \cup B), \\ v_x^0(u) &= E - \sum_{y \sim x} \frac{\psi_y(u)}{\psi_x(u)} \quad (x \in V). \end{aligned} \quad (4.9)$$

Every neighbor in the last sum belongs to $V \cup B$ and $\psi_0 = 1$. Uniformly over the allowed parameters,

$$c e^{-\nu|x|_\infty} \leq |\psi_x(u)| \leq C e^{-\nu|x|_\infty}. \quad (4.10)$$

Indeed $|uf_{x_1+x_2}|$ is bounded by a fixed multiple of c_u and the periodic carrier is bounded away from zero. For neighbors x, y the additional ratio factors differ from 1 by $O(\nu + c_u)$: their radial exponents differ by at most ν , and $|u(f_{y_1+y_2} - f_{x_1+x_2})| \leq Cc_u$. At $\nu = u = 0$, the cosine addition formula gives $v_x^0 = E - 4 \cos \theta$. We therefore fix ν, c_u small enough, once and for all, that

$$|v_x^0(u) - (E - 4 \cos \theta)| \leq d, \quad |v_x^0(u)| \leq h - 7d. \quad (4.11)$$

Lemma 4.2 (Choice of the profile parameter). *There is a fixed exponent C_2 such that, for every sufficiently large L , a deterministic $u_L \in [-c_u/L, c_u/L]$ can be chosen for which the real symmetric Dirichlet matrix*

$$A^0 = \text{Adj}_J + \text{diag}_J(v^0 - E)$$

satisfies $\|(A^0)^{-1}\| \leq L^{C_2}$. The bounds in Equations (4.10) and (4.11) hold for this choice, with constants independent of L .

Proof. During the proof retain u as a variable. Let $\Psi(u) = \text{diag}(\psi_x(u) : x \in J)$, $W(u) = \Psi(u)^2$, and $Q(u) = -\Psi(u)A^0(u)\Psi(u)$. Extend a test vector α by zero at 0 and on B . Counting each unoriented edge meeting J once gives

$$\alpha^* Q(u) \alpha = \sum_{\{x,y\}:\{x,y\} \cap J \neq \emptyset} \psi_x(u) \psi_y(u) |\alpha_x - \alpha_y|^2. \quad (4.12)$$

The sum here contains only edges with the other endpoint in $J \cup \{0\} \cup B$. Its weights may have either sign. Nevertheless, Equation (4.8) gives the exact identity

$$\frac{d}{du}(\psi_x(u) \psi_y(u)) = |\psi_x(u) \psi_y(u)| \quad (x \sim y).$$

Thus $Q'(u)$ is a positive weighted Dirichlet form.

For each $x \in J$, choose a nearest-neighbor path to 0 by decreasing coordinates in absolute value. It lies in V , has at most $2L$ edges, and its max radius never exceeds $|x|_\infty$. Every edge on the path has weight $|\psi_y \psi_z| \geq c|\psi_x|^2$, by Equation (4.10). Weighted Cauchy–Schwarz along the path gives

$$|\psi_x|^2 |\alpha_x|^2 \leq CL \sum_{\{y,z\} \text{ on the path}} |\psi_y \psi_z| |\alpha_y - \alpha_z|^2 \leq CL \alpha^* Q'(u) \alpha.$$

Summing over at most CL^2 sites shows

$$Q'(u) \geq cL^{-3}W(u) \geq L^{-C_1}W(u) \quad (4.13)$$

for some fixed C_1 , after increasing the lower size threshold. One may take $C_1 = 4$. Moreover, all $W(u)$ in the allowed interval are comparable as quadratic forms, with a fixed comparison constant, and

$$-CLW(u) \leq W'(u) \leq CLW(u).$$

Choose $C_2 > C_1 + 3$ and put $t = L^{-C_2}$. For sufficiently large L ,

$$(Q \pm tW)' \geq \frac{1}{2}L^{-C_1}W.$$

More explicitly, from comparability and Equation (4.13) give, whenever $u_2 - u_1 > C'tL^{C_1}$ with a sufficiently large fixed C' ,

$$Q(u_2) - tW(u_2) > Q(u_1) + tW(u_1). \quad (4.14)$$

Let $\lambda_k(u)$ be the k th eigenvalue, in increasing order, of $-A^0(u)$. Congruence by the invertible real diagonal matrix $\Psi(u)$ preserves the numbers of negative, zero, and positive eigenvalues. Consequently, if $\lambda_k(u_1) \geq -t$, the k th eigenvalue of $Q(u_1) + tW(u_1)$ is nonnegative. The strict inequality in Equation (4.14) then makes the k th eigenvalue of $Q(u_2) - tW(u_2)$ positive, so $\lambda_k(u_2) > t$.

It follows that each set $\{u : |\lambda_k(u)| \leq t\}$ has diameter at most $C'tL^{C_1}$. No monotonicity assertion about the individual $\lambda_k(u)$ is needed. The union of these sets, over $|J| = O(L^2)$ indices, has length at most $CL^{C_1+2-C_2} = o(L^{-1})$, whereas the available interval has length $2c_u/L$. Choose u_L outside this union. For definiteness, one may choose the first rational point in a fixed enumeration that belongs to the open complement. Then $\text{dist}(0, \text{spec } A^0) > L^{-C_2}$, proving the inverse bound. $\square$

From now on $u = u_L$ is fixed and suppressed in the notation. Set

$$\epsilon = e^{-\nu L}, \quad \sigma = \epsilon^2. \quad (4.15)$$

In particular, $|\psi_x|$ is comparable to ϵ on B and on Z^+ , with constants independent of L . We have chosen the port strength σ as the square of this designed boundary-profile scale. The exact calibrated mode will satisfy the polynomially lossy comparison in Equation (4.23).

4.3 The branch sampler

We first describe the boundary conditionings used by the sampler. For each corner triple use the fixed basis in Equation (4.4) and put

$$D_j = \text{diag} \left(\frac{(r_j)_i}{n_i} : 1 \leq i \leq 3 \right) \quad (j = 1, 2). \quad (4.16)$$

These matrices satisfy

$$D_j n = r_j, \quad \text{tr } D_j = 0, \quad n^\top D_j n = 0.$$

Thus a common shift changes the real diagonal in the missing direction n , while a shift along D_j changes its coupling to r_j without changing that diagonal. The three diagonal directions I, D_1, D_2 are linearly independent: multiplying a vanishing linear combination by n gives a vanishing linear combination of the orthonormal vectors n, r_1, r_2 . Thus centered coordinates (s, t_1, t_2) specify physical potentials by

$$\text{diag}(v) = \text{diag}(\bar{v}) + sI + t_1 D_1 + t_2 D_2. \quad (4.17)$$

The map has a fixed nonzero Jacobian. In fact its three columns, as vectors of diagonal entries, are orthogonal and have length $\sqrt{3}$. Use this same coordinate convention at all levels, including at exactly coincident triples.

A prescribed corner box is the image under Equation (4.17) of a product of three centered intervals. Fix an exponent κ . The boxes allowed below have all three side lengths at least $c(1+L)^{-\kappa}$ and at most C , lie at distance at least $c(1+L)^{-\kappa}$ from the boundary of $(-h, h)^3$, and keep each physical coordinate in a single polynomial piece of the fixed artificial profile ϑ . Constants may depend on κ ; in the hierarchy this exponent is fixed by K_0 . Uniform measure in these box coordinates is also uniform measure in the physical parallelepiped. Because ϑ is positive in the interior and comparable to squared distance from the support endpoints, every nonzero artificial profile $a_i \in \{1, \vartheta\}$ on a permitted box obeys

$$a_i(v_i) \geq c(1+L)^{-2\kappa}. \quad (4.18)$$

Increasing the exponent absorbs any fixed comparison constants.

Choose $D_1 > C_2 + 2$. A shell-depth parameter D_2 will be fixed after the polynomial thresholds in the boundary and zero-jitter transmission estimates have been fixed. Finally choose D_3 sufficiently large depending on D_1, D_2 and these thresholds. On J sample

$$v_x = v_x^0 + r \mathbf{1}_{Z^+}(x) + \zeta_x, \quad r \sim \text{Unif}[-L^{-D_1}, L^{-D_1}], \quad (4.19)$$

where the ζ_x are independent centered uniforms of half-width

$$w_x = \begin{cases} L^{-D_3}, & L - |x|_\infty \leq D_2 \log L + 5, \\ \epsilon L^{-D_3}, & L - |x|_\infty > D_2 \log L + 5. \end{cases} \quad (4.20)$$

They are independent of r and all boundary sampling.

For these interior draws, use the matrix A and the calibrated value $c(v)$ from Equation (4.5). At the root, put an open cell port a of strength σ , and use $c(v)$ as its real reference potential. To define the hidden true potential, fix a smooth cutoff $\chi : \mathbb{R} \rightarrow [0, 1]$ with $\chi = 1$ on $[-d, d]$ and $\text{supp } \chi \subset [-2d, 2d]$, and sample

$$\begin{aligned} v_0 = c(v) + \delta, \quad \mathbb{P}(\delta \in dt \mid v|_J) &= \frac{1}{Z_\sigma} \chi(t) \frac{\sigma}{\pi(t^2 + \sigma^2)} dt, \\ Z_\sigma &= \int_{\mathbb{R}} \chi(t) \frac{\sigma}{\pi(t^2 + \sigma^2)} dt. \end{aligned} \quad (4.21)$$

This is exactly the cutoff Cauchy law used in Lemma 3.4. The open reference $c(v)$ does not depend on the hidden draw δ .

The boundary law is as follows. On each corner triple sample uniformly in its prescribed box, independently between the four triples. On every single side site x sample

$$v_x = \xi_x + s_0, \quad \xi_x \sim \text{Unif}[-h/4, h/4], \quad s_0 \sim \text{Unif}[-h/4, h/4], \quad (4.22)$$

using independent ξ_x and one common s_0 . These draws are independent of the triples and the interior draws. Sites of the tile outside $V \cup B$ retain their independent uniform $[-h, h]$ laws, independently of all draws just described. Apart from the common shifts r, s_0 and the dependence of the root center $c(v)$ on $v|_J$, all sampling is independent.

Proposition 4.3 (Properties of the branch law). *For sufficiently large L , the above choices have the following properties, uniformly over permitted corner boxes:*

(i) *Every interior draw has*

$$\|A^{-1}\| \leq 2L^{C_2}, \quad \|-A^{-1}p - \psi|_J\| \leq CL^{C_2+1}\epsilon. \quad (4.23)$$

The exponents in these bounds are independent of D_2 , provided D_3 is subsequently chosen sufficiently large.

(ii) *All true potentials on V lie in a fixed compact subinterval of $(-h, h)$. The root center satisfies*

$$|c(v) - v_0^0| \leq CL^{C_2+1}\epsilon. \quad (4.24)$$

The plateau of ϑ can therefore be fixed to contain every true value on V and every single side value in B , for all the cells used. The artificial absorptions on these sites are constant as r , the jitters, the common side shift, and the hidden root draw vary.

(iii) Relative to the true product law on the tile conditioned on these same four corner boxes, the branch has a density $\mathcal{L}$ satisfying

$$0 \leq \mathcal{L} \leq \exp(CL^3 + CL^2 \log L), \quad \int \mathcal{L} \, d\mathbb{P}_{\text{true}, \text{boxes}} = 1. \quad (4.25)$$

The corner-box conditional laws and the laws outside $V \cup B$ are unchanged.

Artificial probes may have any prescribed mask of absent, constant, or profile-dependent strengths, as in [Section 3.5](#). For subsequent cell estimates their common scale is restricted to $0 \leq \tau \leq e^{-c_* L}$, with a fixed $c_* > 2\nu$.

Proof. The perturbation from A^0 is diagonal and has norm at most $L^{-D_1} + L^{-D_3}$. Taking $D_3 > D_1$, [Lemma 4.2](#) and a Neumann series give the first bound in [Equation \(4.23\)](#). The profile equation on J gives the exact residual identity

$$A^0 \psi|_J + p = -M\psi|_B.$$

The right-hand side has norm at most $C\sqrt{L}\epsilon$. On Z^+ , $|\psi_x| \leq C\epsilon$, so the additional common-shift residual has norm at most $CL\epsilon L^{-D_1}$. On the jitter shell,

$$|\psi_x| \leq C\epsilon L^{\nu D_2},$$

where the extra depth 5 has been absorbed in C . Its jitter residual is bounded by $CL\epsilon L^{\nu D_2 - D_3}$. Off that shell, the half-width already has a factor ϵ , and $|\psi_x| \leq C$, giving an upper bound $CL\epsilon L^{-D_3}$. Thus $D_3 > \nu D_2 + 2$ ensures

$$\|A\psi|_J + p\| \leq CL\epsilon.$$

Multiplication by A^{-1} proves the second bound in [Equation \(4.23\)](#). Notice that the resulting exponent $C_2 + 1$ was fixed before D_2 .

Since $\psi_0 = 1$, the profile equation at 0 reads $v_0^0 = E - p^\top \psi|_J$. Subtracting this from the definition of $c(v)$ proves [Equation \(4.24\)](#). Increase the minimum cell size so that all interior perturbations and the last calibration error are at most d . By [Equation \(4.11\)](#), $|v_x| \leq h - 6d$ on J , $|c(v)| \leq h - 6d$, and $|v_0| \leq h - 4d$ after the hidden draw. The side values lie in $[-h/2, h/2]$. Choose the plateau to contain these compact intervals. This choice uses only the original design margin d , and is therefore available before the large width and shell exponents are fixed. Each present artificial profile on these sites is then identically 1, proving the constancy assertion, also at the root.

For the density estimate, first condition on r and s_0 . Each jitter has density at most $(2w_x)^{-1}$ in its physical coordinate, hence likelihood at most h/w_x relative to a uniform $[-h, h]$ coordinate. For small σ , $Z_\sigma \geq 1/2$, so the conditional root density is at most C/σ . Its likelihood relative to the true uniform coordinate is at most C_h/σ . Each single side coordinate has likelihood at most 4. Multiplying these bounds before averaging the common shifts gives

$$\log \mathcal{L} \leq C|J| + D_3|J| \log L + \nu L \#\{x \in J : w_x = \epsilon L^{-D_3}\} + 2\nu L + C|B| \leq CL^3 + CL^2 \log L.$$

An upper bound of this form also holds when some individual displayed logarithms are negative. The four corner factors cancel against the identical box conditionings in the reference law. The outside factors cancel as well. All sampler laws are normalized, so integration gives the last assertion in [Equation \(4.25\)](#). $\square$

We record explicitly the order of choices used later. The carrier, d, ν, c_u, C_1, C_2 , the compact cutoff and plateau, and the corner-box exponent are fixed first. Next fix D_1 . The polynomial threshold in [Proposition 5.1](#) and the regular inverse thresholds in the zero-jitter part of [Proposition 6.1](#) can then be fixed without D_2, D_3 . At depth $L - |x|_\infty = \lfloor D_2 \log L \rfloor$, [Equation \(4.10\)](#) gives

$$\frac{|\psi_x|}{\epsilon} \geq cL^{\nu D_2}.$$

Choose D_2 large enough that this dominates the already fixed polynomial errors and reflected-field bounds in that argument. Finally choose D_3 to satisfy both the residual requirements above and the inverse and path-entry stability requirements of [Proposition 6.1](#). Enlarging D_3 changes neither the exponents in [Equation \(4.23\)](#) nor the earlier zero-jitter bounds. All parameters are now fixed; only after that do we increase the lower cell-size threshold. This order prevents the depth choice from depending on a polynomial exponent that itself grows with the depth.

5 A weighted projected impedance estimate

We retain the boundary geometry and the conditional boundary law of [Lemma 4.1](#) and [Proposition 4.3](#). In this section the interior of the cell does not enter the calculation. Its only relevant feature is that it couples to the boundary space through the projection P . We write

$$P = P_0 + Q, \quad Q = Q_1 + Q_2 + Q_3 + Q_4,$$

where P_0 projects onto the single side sites and Q_a projects onto the two-dimensional plane of the a th corner triple. The four complementary directions in the triples are denoted by n_a . In the prescribed ordering of each triple, $n_a = (-1, -1, 1)/\sqrt{3}$. Thus $\text{rank } Q = 8$, whereas the dimension of the whole boundary space is $O(L)$.

All norms below use the fixed real orthonormal bases of these spaces. Constants may depend on the fixed disorder, profile, and corner-box parameters. They will not depend on the exterior matrices, their norms, the source vector, or the artificial-probe mask. Until an exponent H is selected below, every expression denoted by $\text{poly}(L)$ means a bound $C(1+L)^C$ whose constants are independent of H . A final polynomial bound is allowed to depend on that selected, fixed exponent.

Proposition 5.1 (Weighted projected impedance). *Let T be any fixed complex symmetric matrix on the boundary space with $-\text{Im } T \geq 0$, and let k be any fixed vector there. For each boundary site prescribe a_i to be one of 0, 1, and ϑ , with the prescription independent of the boundary variables. Set*

$$\mathcal{T} = T + \text{diag}_B(v_i - i\tau a_i(v_i)), \quad \rho = \left\| P(\mathcal{T} - iP)^{-1}k \right\|^2. \quad (5.1)$$

Fix $c_ > 0$ and suppose $0 \leq \tau \leq e^{-c_* L}$. There are constants $c > 0$, $C < \infty$, and a sufficiently large lower threshold for L such that*

$$\mathbb{E}_B \left[\rho \mathbf{1}_{\{\|P\mathcal{T}^{-1}P\| \leq C(1+L)^C\}} \right] \geq c \mathbb{E}_B \rho. \quad (5.2)$$

Here the expectation uses the boundary law conditional on its four corner boxes. The constants are uniform in all the exterior data just described. Moreover, changing only the single side variables to any other bounded allowed settings, including their artificial absorptions, changes ρ by at most fixed multiplicative factors, pointwise wherever the inverses exist.

The weight in Equation (5.1) matters: a small unweighted exceptional set could contain nearly all the response of a weakly coupled exterior mode. We first control the inverse on the corner planes Q while leaving unit absorption on the single side coordinates P_0 . The corner estimates must retain a fixed fraction of the response weight. We then average the common real side shift to bound the whole projected inverse $PT^{-1}P$ with the auxiliary P_0 absorption removed.

5.1 Elementary reductions and polynomial estimates

We first record the comparison that permits bounded changes of the absorbed coordinates.

Lemma 5.2 (Comparison of absorbed responses). *Suppose D_1, D_2 are invertible matrices with $-\text{Im } D_j \geq P$. If $D_2 - D_1 = P\Delta P$ and $\|\Delta\| \leq K$, then, for every k ,*

$$(1 + K)^{-1} \|PD_1^{-1}k\| \leq \|PD_2^{-1}k\| \leq (1 + K) \|PD_1^{-1}k\|.$$

Proof. The inverse difference identity gives

$$PD_2^{-1}k = PD_1^{-1}k - PD_2^{-1}P\Delta PD_1^{-1}k.$$

By Lemma 3.1, $\|PD_2^{-1}P\| \leq 1$. This proves the upper bound; interchanging the indices proves the lower bound. $\square$

The inverses used in this section exist almost everywhere in the stated boundary law. To see this without any nonsingularity assumption on T , recall that a kernel vector of a passive matrix is annihilated by both its Hermitian real and imaginary parts. Absorption on P restricts such a vector to the four missing directions. On that space the Hermitian real part has the form of a fixed real symmetric matrix plus the independent common corner shifts on its diagonal. Its determinant is a nonzero polynomial, with the product of those shifts as its leading monomial. The same reasoning applies after one missing direction is deleted or after bounded changes in plane blocks. With absorption only on P_0 , a possible kernel lies on the full corner triples. The four common shifts act there as scalar shifts on their respective three-dimensional blocks; the determinant again has a nonzero leading monomial. Finally, $\mathcal{T}$ itself has physical real diagonal variables with a joint density, so its Hermitian real part is nonsingular almost everywhere. These observations also justify almost-everywhere slicing below.

The response weights are integrable. For example, writing $G = (\mathcal{T} - iP)^{-1}$, the Ward identity implies

$$\|PGk\|^2 \leq k^*(\text{Im } G)k \leq \|k\|^2 \text{tr Im } G.$$

The joint law of the physical boundary diagonals has a bounded density on a compact set; the bound here may depend on L . Slice that set along a common real shift of all the physical diagonals. The phase estimate Lemma 3.6, including its artificial-diagonal derivative term, bounds the integral of $\text{tr Im } G$ along each slice. Integration over the transverse coordinates proves finiteness. One may add $-i\zeta\mathbf{1}$ first and pass to the limit by Fatou. In particular, after some boundary variables are frozen, all integrals below are finite for almost every such choice.

We shall use two elementary facts about polynomials. Their degree bounds are fixed throughout this section.

Lemma 5.3 (Polynomial norms and products). *On a rectangular box in a fixed number of real variables, let p be a polynomial of degree at most d with values in a finite-dimensional Hilbert space. There is a constant depending only on d and the number of variables such that*

$$\sup \|p\|^2 \leq C \text{ avg } \|p\|^2. \quad (5.3)$$

If the length of a coordinate interval is at least ℓ , then

$$\sup \|\partial_j p\| \leq C\ell^{-1} \sup \|p\|. \quad (5.4)$$

If q is a scalar polynomial of degree at most d , then

$$\sup \|pq\| \geq c \sup \|p\| \sup |q|. \quad (5.5)$$

The constants do not depend on the dimension of the target Hilbert space or the side lengths of the box.

Proof. Affinely rescale the box to a unit cube. For scalar polynomials of bounded degree, the asserted norm comparisons and derivative estimate follow from equivalence of norms on a fixed finite-dimensional vector space. For Equation (5.3), choose a point where $\|p\|$ is maximal and a unit linear functional attaining that norm at that point; apply the scalar estimate to its composition with p . The derivative estimate follows in the same way by projecting at a maximizing derivative.

For the scalar product estimate, normalize the two supremum norms to one. The two unit spheres of bounded-degree scalar polynomials are compact. The supremum norm of their product has a positive minimum: otherwise two nonzero limiting polynomials would have identically zero product. Projecting the vector polynomial at one of its maximizing points reduces Equation (5.5) to this scalar result. $\square$

The following one-dimensional observation will prevent an extremely narrow response peak from defeating a short-exception estimate.

Lemma 5.4 (Resolution of a weighted slice). *Let I be an interval of length at least ℓ , let N be a Hilbert-space-valued polynomial of fixed degree on I , and let z be a continuously differentiable complex function satisfying*

$$|z(s)| \geq c_0 a, \quad |z'(s)| \leq C_0, \quad a > 0.$$

Then

$$\sup_{s \in I} \frac{\|N(s)\|^2}{|z(s)|^2} \leq \frac{C}{\min(\ell, a)} \int_I \frac{\|N(s)\|^2}{|z(s)|^2} ds.$$

Here C depends only on c_0, C_0 and the degree bound.

Proof. Take a maximizing point s_0 and an interval $I_0 \subset I$ containing it, of length a sufficiently small fixed multiple of $\min(\ell, a)$. The derivative bound gives $|z(s)| \leq 2|z(s_0)|$ on I_0 . Applying Equation (5.3) on I_0 yields

$$\int_{I_0} \frac{\|N(s)\|^2}{|z(s)|^2} ds \geq \frac{c|I_0| \|N(s_0)\|^2}{|z(s_0)|^2}.$$

This proves the claim, including when s_0 is an endpoint. $\square$

5.2 Eight corner tests

Remove the common shift from the single side variables and omit their artificial absorptions. Denote the resulting boundary matrix by $\mathcal{T}_0$, and set

$$w_0 = \left\| P(\mathcal{T}_0 - iP)^{-1} k \right\|^2.$$

The independent single side uniforms are still present. By [Lemma 5.2](#), w_0 and ρ are pointwise comparable by fixed factors. We now hold those independent side uniforms fixed.

Choose an exponent H , to be fixed after the polynomial losses below, and put

$$\eta = L^{-H}, \quad G_\eta = (\mathcal{T}_0 - iP_0 - i\eta Q)^{-1}.$$

For each vector b' in the eight prescribed real orthonormal corner-plane bases, define

$$\mathcal{E}_{b'} = \left\{ \eta \operatorname{Im}(b'^\top G_\eta b') \geq \frac{1}{40} \right\}. \quad (5.6)$$

Outside their union,

$$\|Q(\mathcal{T}_0 - iP_0)^{-1}Q\| \leq \eta^{-1}. \quad (5.7)$$

Indeed, if $\|QG_\eta Q\| < (2\eta)^{-1}$, inverse difference and a Neumann series give the bound in [Equation \(5.7\)](#). Hence failure of that bound forces $\|QG_\eta Q\| \geq (2\eta)^{-1}$. Ward on Q then gives

$$\eta \operatorname{tr}(Q \operatorname{Im} G_\eta Q) \geq \eta^2 \|QG_\eta Q\|_{\text{HS}}^2 \geq \frac{1}{4}.$$

If all eight tests fail to occur, the left side is less than $8/40 < 1/4$, a contradiction.

Fix one test and condition on all boundary variables outside its corner triple. Use centered box coordinates (s, t_1, t_2) there, where s is the common shift and t_j is the shift whose diagonal matrix maps the missing direction n to the plane basis vector r_j . In particular, the t_j shift has zero n, n entry. Write $t = (t_1, t_2)$. Every side length of the box is at least $\ell \geq (1+L)^{-C}$ and at most a fixed constant. All probabilities and averages on this box are normalized uniform ones.

There is first an unweighted bound

$$\mathbb{P}_{s,t}(\mathcal{E}_{b'}) \leq \operatorname{poly}(L)\eta. \quad (5.8)$$

For fixed t , apply [Lemma 3.6](#) while varying s . Its real derivative is the identity on the tested triple. The artificial derivatives satisfy the profile condition and are controlled by their actual absorption, so the integral of the triple partial trace of $\operatorname{Im} G_\eta$ is polynomially bounded. Since b' is a unit vector in that triple, its quadratic form is bounded by this trace. Divide by the interval length, then average over t , to obtain [Equation \(5.8\)](#). Neither the phase range nor this polynomial loss depends on H or on the exterior norm.

We must strengthen [Equation \(5.8\)](#) to a bound for w_0 -mass.

5.3 A fixed remainder at one corner

The exceptional test still uses the actual inverse G_η . To measure its weight, we replace w_0 by a comparable response whose remainder after deleting the tested missing direction n is fixed. In $\mathcal{T}_0 - iP$, freeze the real variations of the tested triple within its plane-plane block at the box center. Replace the artificial absorption in that plane-plane block by τ times the plane identity. Retain every fixed term from T and retain all actual mixed terms and the missing-direction diagonal. Call this modified matrix $\tilde{D}$.

The modification is bounded and supported on the tested plane. It also preserves absorption at least P plus the original artificial absorption of the tested triple. In fact the change of artificial absorption is

$$\tau Q_a(\mathbf{1} - \operatorname{diag}(a_i(v_i)))Q_a \geq 0,$$

because $0 \leq a_i \leq 1$. Consequently

$$C^{-1}w_0 \leq W := \|P\tilde{D}^{-1}k\|^2 \leq Cw_0. \quad (5.9)$$

The constant is fixed by [Lemma 5.2](#).

Delete the tested missing direction n , and denote the inverse of the remaining block of $\tilde{D}$ by G_a . This inverse is independent of (s, t) . The off-diagonal column from n and its Schur denominator have the form

$$p_1(s, t) = p_{10} + t_1 r_1 + t_2 r_2 - i\tau \sum_{i \text{ in the triple}} a_i(v_i) n_i Q e_i, \quad (5.10)$$

$$s - e(s, t) = s - e_{\text{fix}} - p_1^\top G_a p_1 - i\tau \sum_{i \text{ in the triple}} a_i(v_i) n_i^2. \quad (5.11)$$

The fixed vector p_{10} may have components outside the tested triple; no bound on it is assumed. All variable parts of p_1 lie in the tested plane. The constants in [Equation \(5.11\)](#) include the fixed diagonal at n ; there is no real t -dependent contribution to that diagonal.

Write $k_n^\wedge$ for k with its n coordinate deleted, and set

$$a = PG_a k_n^\wedge, \quad d_1 = PG_a p_1, \quad \beta = p_1^\top G_a k_n^\wedge - k_n. \quad (5.12)$$

The block inverse formula yields

$$W = \left\| a + \frac{d_1 \beta}{s - e} \right\|^2 = \frac{\|N_1\|^2}{|s - e|^2}, \quad N_1 = a(s - e) + d_1 \beta. \quad (5.13)$$

The norm of d_1 measures the response on the absorbed coordinates induced by unit amplitude in the missing direction. The scalar Schur denominator $s - e$ controls how sharply W can concentrate. It need not agree with the corresponding denominator in the actual test G_η . Each profile lies on one fixed polynomial piece throughout the corner box. Hence p_1, d_1, β, e, N_1 are polynomials in (s, t) of bounded degree, independently of all exterior data. Their coefficients need not be bounded.

Lemma 5.5 (Corner Schur estimates). *Uniformly throughout the box,*

$$\text{Im } e \geq \|d_1\|^2, \quad |\partial_s e| \leq C\tau(1 + \|d_1\|), \quad (5.14)$$

$$\|\partial_s d_1\| \leq C\tau, \quad |\partial_s \beta| \leq C\tau \|a\|, \quad (5.15)$$

$$\|\partial_{t_j} d_1\| \leq C, \quad |\partial_{t_j} e| \leq C(1 + \|d_1\|). \quad (5.16)$$

If at least one artificial probe is present in the tested triple and $\tau > 0$, then also

$$\text{Im } e \geq \frac{\tau}{\text{poly}(L)}. \quad (5.17)$$

For every fixed t ,

$$\int \frac{\|d_1(s, t)\|^2}{|s - e(s, t)|^2} ds \leq \int \text{Im} \frac{1}{s - e(s, t)} ds \leq \text{poly}(L). \quad (5.18)$$

Proof. Apply absorption to the Schur extension $(-G_a p_1, 1)$ at n . Its quadratic form under $\tilde{D}$ is the scalar $s - e$, and its absorbed P -coordinates have norm $\|d_1\|$. This proves the first inequality. Every derivative of p_1 is supported in the tested plane. Its s derivative has size $O(\tau)$, whereas its t_j derivative is $r_j + O(\tau)$. Since $\|PG_a P\| \leq 1$, these facts give the bounds on d_1 . Differentiating

Equations (5.11) and (5.12), using transpose symmetry, gives the other derivative bounds: for example,

$$\partial_s e = 2(\partial_s p_1)^\top G_a p_1 + i\tau \sum_i \partial_s(a_i(v_i))n_i^2.$$

The projection of $G_a p_1$ onto the support of $\partial_s p_1$ is controlled by d_1 . Thus arbitrarily large fixed components of p_{10} cause no loss.

For Equation (5.17), an active profile in the triple is at least an inverse polynomial, by the box clearance from the support endpoints. If $\|d_1\| \geq 1/(2\sqrt{3})$, the first bound already supplies a fixed positive lower bound. Otherwise the corresponding physical coordinate of $(-G_a p_1, 1)$ has modulus at least $1/(2\sqrt{3})$, since $|n_i| = 1/\sqrt{3}$. Its artificial absorption supplies $\tau/\text{poly}(L)$.

For the integral bound, let $y = \text{Im } e$ and $V = \text{Im}(s - e)^{-1} = y/|s - e|^2$. The first inequality of Equation (5.18) follows from $y \geq \|d_1\|^2$. If e is independent of s , ordinary scalar shift integration gives $\int V \leq \pi$; real poles are null sets. Otherwise $\tau > 0$ and a probe is present. Combining Equations (5.14) and (5.17),

$$\frac{|\partial_s e|}{\sqrt{y}} \leq C\tau \left(\frac{1}{\sqrt{y}} + 1 \right) \leq \text{poly}(L).$$

The argument of $s - e$ lies in $(-\pi, 0)$, and its derivative is

$$\frac{d}{ds} \arg(s - e) = V - \text{Im} \frac{\partial_s e}{s - e}.$$

The error is bounded by $\text{poly}(L)\sqrt{V}$. Young's inequality absorbs half the integral of V ; the remaining constant and the phase variation are polynomially bounded on the bounded interval. This proves Equation (5.18). $\square$

5.4 Weighted exceptions when the coupling is appreciable

Put

$$D = \max_{s,t} \|d_1(s, t)\|, \quad \alpha_0 = \frac{1}{100}.$$

All maxima are taken over the closed box, which does not affect its conditional law. The bound $\text{Im } e \geq \|d_1\|^2$ ties the width of the response to this coupling. When $D \geq \eta^{\alpha_0}$, we can isolate the few transverse parameters where the coupling becomes small and use the unweighted test bound elsewhere. The case where D is small throughout the box will instead require comparing the length of each exceptional slice with its response width.

We first treat $D \geq \eta^{\alpha_0}$.

Suppose initially that $|s - e| \geq 1$ throughout the box. The maximum and minimum of this denominator are then comparable by a fixed factor. Indeed, Equations (5.15) and (5.16) show that d_1 varies by at most a fixed constant. If D is large, then $\text{Im } e \geq cD^2$ throughout, whereas the variation of $s - e$ is at most $C(1 + D)$. If D is bounded, the same variation bound and the assumed lower bound one suffice. By Equations (5.3) and (5.8),

$$\mathbb{E}_{s,t}[W \mathbf{1}_{\mathcal{E}_{b'}}] \leq \text{poly}(L)\eta \mathbb{E}_{s,t} W. \quad (5.19)$$

It remains to consider the case where $|s - e| < 1$ somewhere. At that point $\|d_1\| < 1$; the derivative bounds then imply that D is bounded throughout the box. Since s is bounded, the same argument shows that e and $s - e$ are bounded as well. Set

$$M_1 = \max \|N_1\|, \quad B_1 = \max |\beta|.$$

The bounded denominator and Equation (5.3) give

$$\mathbb{E}_{s,t}W \geq cM_1^2. \quad (5.20)$$

Furthermore,

$$\|a\| + DB_1 \leq \text{poly}(L)M_1. \quad (5.21)$$

Here is the required quantitative argument. By the product estimate,

$$DB_1 \leq C \max \|d_1\beta\| \leq C(M_1 + \|a\|).$$

Differentiate $N_1 = a(s - e) + d_1\beta$ in s and use Equations (5.4), (5.14) and (5.15). As D is bounded, this gives

$$\|a\| \leq C\ell^{-1}M_1 + C\tau(\|a\| + B_1).$$

Substitute the preceding estimate for B_1 . Since $D \geq \eta^{\alpha_0}$, the coefficient $C\tau(1 + D^{-1})$ is less than one half for sufficiently large L after H is fixed. Absorption proves Equation (5.21). Its polynomial exponent is independent of H .

Choose $\Delta = \eta^{1/8}$ and let

$$\mathcal{A} = \{t : \min_s \|d_1(s, t)\| < \Delta\}.$$

Then

$$\mathbb{P}_t(\mathcal{A}) \leq C \frac{\Delta}{D}. \quad (5.22)$$

To prove this, Equation (5.10) and Ward show that $d_1(s, t)$ is within $C\tau$ of an affine function of t , uniformly in s . Rescale the two t intervals to unit intervals. If $\mathcal{A}$ is nonempty, the affine function has a value of norm at most $\Delta + C\tau$, and its maximum norm is at least $D - C\tau$. Project it by a real unit linear functional attaining that latter norm. Because $\Delta/D \rightarrow 0$ and $\tau \ll \Delta$, one of the two scalar slopes has magnitude at least cD . Slicing in that coordinate bounds the fraction where the scalar magnitude is at most $\Delta + C\tau$ by $C\Delta/D$. This includes $\mathcal{A}$.

For each $t \in \mathcal{A}$, Equations (5.13) and (5.18) gives

$$\mathbb{E}_sW \leq \text{poly}(L)(\|a\|^2 + B_1^2) \leq \text{poly}(L)(1 + D^{-2})M_1^2.$$

On the complement of $\mathcal{A}$, absorption gives $|s - e| \geq \|d_1\|^2 \geq \Delta^2$, so $W \leq \Delta^{-4}M_1^2$. Using Equation (5.8) there and Equation (5.22) on $\mathcal{A}$, we obtain

$$\mathbb{E}_{s,t}[W\mathbf{1}_{\mathcal{E}_{b'}}] \leq \text{poly}(L) \left(\frac{\Delta}{D^3} + \frac{\eta}{\Delta^4} \right) \mathbb{E}_{s,t}W. \quad (5.23)$$

We used the boundedness of D and Equation (5.20) in the last step. If $M_1 = 0$, all the weighted integrals vanish and the inequality holds directly. In particular, this argument never divides by a possibly zero expected weight. The powers in Equation (5.23) satisfy

$$\frac{\Delta}{D^3} \leq \eta^{1/8-3\alpha_0}, \quad \frac{\eta}{\Delta^4} = \eta^{1/2}, \quad \frac{1}{8} - 3\alpha_0 > 0.$$

5.5 Weighted exceptions when the coupling is small

We now assume $D < \eta^{\alpha_0}$ and write $b' = r_j$ for the tested basis vector. Vary t_j between two points separated by at least half its interval length. From Equations (5.10) and (5.12),

$$a_{b'} := PG_a b', \quad \|a_{b'}\| \leq C\ell^{-1}(D + \tau). \quad (5.24)$$

This can be made as small as a prescribed fixed constant by choosing H sufficiently large, then L sufficiently large. The inference uses variation over the whole t_j interval; it does not assume that the exterior has a lower bound on its coupling.

Next delete n in the actual matrix G_η^{-1} , without freezing its plane block. Let $G_{\eta,\text{rem}}$ denote the remaining inverse, put

$$d_{1,\eta} = PG_{\eta,\text{rem}}p_1,$$

and write $s - e_\eta$ for its Schur denominator at n . Its off-diagonal column is exactly the same p_1 as before. The remaining matrices for G_a and $G_{\eta,\text{rem}}$ differ by a bounded matrix supported on Q . Moreover, $\|QG_{\eta,\text{rem}}Q\| \leq \eta^{-1}$ by Ward. Inverse difference and symmetry therefore give

$$\eta \left| b'^\top G_{\eta,\text{rem}} b' \right| \leq C \|a_{b'}\|, \quad (5.25)$$

$$\left| b'^\top d_{1,\eta} \right| \leq \left| b'^\top d_1 \right| + C \|a_{b'}\| \|Qd_{1,\eta}\|. \quad (5.26)$$

For clarity, the inverse identity used in both estimates is $G_{\eta,\text{rem}} = G_a + G_a \Delta G_{\eta,\text{rem}}$, where $\Delta = Q \Delta Q$ has bounded norm. The first left factor is thus controlled by $PG_a b'$, regardless of the exterior norm.

Take the bound in Equation (5.24) small enough that the first term in the Schur inverse formula cannot by itself cause $\mathcal{E}_{b'}$. On that exception,

$$\frac{\eta \left| b'^\top d_{1,\eta} \right|^2}{|s - e_\eta|} \geq c. \quad (5.27)$$

On the other hand, absorption on the actual Schur extension yields

$$\text{Im } e_\eta \geq \eta \|Qd_{1,\eta}\|^2.$$

Together with Equation (5.27), this implies $\|Qd_{1,\eta}\| \leq C \left| b'^\top d_{1,\eta} \right|$. Insert Equation (5.26) and absorb its small last coefficient. We conclude that, on the exception,

$$\|Qd_{1,\eta}\| \leq C \|d_1\|, \quad |s - e_\eta| \leq C \eta \|d_1\|^2. \quad (5.28)$$

The exceptional set is short on every almost-everywhere s slice. To establish this rigorously, let $D_t = \max_s \|d_1(s, t)\|$. Differentiating the actual Schur formula shows that, on the exception,

$$|\partial_s e_\eta| \leq C \|Qd_{1,\eta}\|^2 + C\tau(1 + \|Qd_{1,\eta}\|) \leq CD^2 + C\tau(1 + D) < \frac{1}{2}.$$

The first term comes from the real derivative of the tested plane block. Its artificial derivative is absorbed in the same bound; the mixed and missing-direction derivatives give the terms containing τ . Thus $\text{Re}(s - e_\eta)$ has derivative at least one half on the exception.

There are only boundedly many components of that exceptional slice away from poles. Indeed, with all other variables fixed, the matrix varies polynomially only on the three coordinates of the tested triple. In a basis containing those coordinates, expansion of the determinant uses variable factors from at most three rows. Its degree, and the degrees of the relevant cofactors, are bounded by a constant depending only on the profile degree, not on the dimension or norm of the exterior. The same assertion holds after deleting n . Clearing the squared absolute value of a denominator converts Equation (5.6) into a real polynomial inequality of bounded degree. Such a set has a bounded number of intervals and isolated points, except for poles, which also have bounded number on an almost-everywhere slice.

On each interval component, Equation (5.28) confines $\text{Re}(s - e_\eta)$ to an interval of length $C\eta D_t^2$, while its derivative is at least one half. Summing over the bounded number of components gives

$$|\{s : (s, t) \in \mathcal{E}_{b'}\}| \leq C\eta D_t^2. \quad (5.29)$$

If $D_t = 0$, Equation (5.28) and the Schur formula rule out the exception off its null set of poles.

For $D_t > 0$, the denominator of the modified weight has a matching minimum scale:

$$\text{Im } e(s, t) \geq cD_t^2 \quad \text{throughout the slice,} \quad |\partial_s(s - e)| \leq C. \quad (5.30)$$

Here is the point requiring the artificial-absorption floor. By Equation (5.15), the oscillation of d_1 on the slice is $O(\tau)$. If its norm is everywhere at least $D_t/2$, absorption proves the first bound immediately. Otherwise $D_t \leq C\tau$. In this latter case an artificial term must be present, since without one d_1 is independent of s . Now Equation (5.17) gives $\text{Im } e \geq \tau / \text{poly}(L) \geq cD_t^2$ for sufficiently large L . The derivative bound follows from Equation (5.14) and $D < \eta^{\alpha_0}$.

The exceptional length $O(\eta D_t^2)$ must be compared with the resolution $\min(\ell, D_t^2)$ supplied by the modified denominator. The real parts of the two Schur denominators need not vanish at the same point. Apply Lemma 5.4 to N_1 and $s - e$, using Equation (5.30). It gives

$$\max_s W(s, t) \leq \frac{C}{\min(\ell, D_t^2)} \int W(s, t) \, ds.$$

Combining with Equation (5.29),

$$\int W(s, t) \mathbf{1}_{\mathcal{E}_{b'}} \, ds \leq \frac{C\eta D_t^2}{\min(\ell, D_t^2)} \int W(s, t) \, ds \leq \text{poly}(L)\eta \int W(s, t) \, ds.$$

The last inequality uses $D_t \leq D < 1$ and the inverse-polynomial lower bound on ℓ . Averaging over t proves

$$\mathbb{E}_{s,t}[W \mathbf{1}_{\mathcal{E}_{b'}}] \leq \text{poly}(L)\eta \mathbb{E}_{s,t} W \quad (5.31)$$

in the small-coupling case. This estimate remains uniform as the coupling vanishes; no positive lower bound for D was used here.

5.6 Restoring the side shift

We now choose the parameters. All polynomial losses in Equations (5.19), (5.23) and (5.31), and the box width exponent, have been fixed independently of H . Choose H large enough that their products with

$$\eta^{1/8-3\alpha_0}, \quad \eta^{1/2}, \quad \eta$$

tend to zero, and that Equation (5.24) is uniformly small in its case. With this H fixed, take L sufficiently large that $\tau \ll \Delta$, $\tau/\eta^{\alpha_0} \ll 1$, and all preceding absorptions are valid. The assumption $\tau \leq e^{-c_* L}$ ensures every one of these requirements for any fixed $c_* > 0$.

For each of the eight tests, condition on the other corner variables, apply the appropriate estimate, and compare W with the same weight w_0 by Equation (5.9). Integration and summation give

$$\mathbb{E}[w_0 \mathbf{1}_{\mathcal{G}}] \geq \frac{3}{4} \mathbb{E} w_0, \quad \mathcal{G} = \bigcap_{b'} \mathcal{E}_{b'}^c, \quad (5.32)$$

for sufficiently large L . This is still conditional on the independent side uniforms. The threshold can be chosen uniformly in those frozen values and in all exterior data. Zero conditional weights cause no difficulty, since every estimate was an inequality between integrals.

Fix data in $\mathcal{G}$, before restoring the common side shift. On the space P consider the rational matrix function

$$F(z) = P(\mathcal{T}_0 + zP_0)^{-1}P, \quad \text{Im } z < 0.$$

It is analytic there and has nonnegative imaginary part. To check analyticity, a kernel at one point of the lower half-plane would be annihilated by P_0 and by $\mathcal{T}_0$, hence would also be a kernel at $z = -i$, where the inverse exists. On $\mathcal{G}$, Equation (5.7) bounds the Q, Q block of $F(-i)$ by η^{-1} . Ward gives a bound one on the P_0, P_0 block and

$$\|P_0(\mathcal{T}_0 - iP_0)^{-1}Q\|^2 \leq \|Q \text{Im}(\mathcal{T}_0 - iP_0)^{-1}Q\| \leq \eta^{-1}.$$

Transpose symmetry gives the same bound on the other mixed block. Thus

$$\|F(-i)\| \leq C\eta^{-1}. \quad (5.33)$$

For completeness, map the unit disk to the lower half-plane by $z = -i(1+w)/(1-w)$. Then $-iF(z(w))$ has positive semidefinite Hermitian real part, and its value at zero obeys Equation (5.33). The positive-real-part weak estimate Equation (3.14) gives a boundary weak- L^1 bound at most $C(\text{rank } P)\eta^{-1}$. One can obtain this bound directly from the positive matrix measure in its Herglotz representation: its total trace mass is at most $(\text{rank } P)\|F(-i)\|$, while the constant imaginary part has the same polynomial bound. The boundary change of variable sends Haar measure to $dx/(\pi(1+x^2))$.

The Cauchy density has a positive lower bound on the fixed interval $[-h/4, h/4]$. Therefore there is a polynomial $R_L = C(1+L)^C$, now allowed to depend on H , such that for at least half the common side shifts x in their prescribed uniform law,

$$\|P(\mathcal{T}_0 + xP_0)^{-1}P\| \leq R_L. \quad (5.34)$$

This conclusion uses only the norm at $-i$, not the norm of T .

Restore the artificial absorptions at the single side sites. Their change has norm at most τ and is supported on P_0 . For a shift satisfying Equation (5.34), the inverse difference identity on P and a Neumann series give

$$\|PT^{-1}P\| \leq 2R_L$$

as soon as $\tau R_L \leq 1/2$. This holds for sufficiently large L . Throughout the common-shift average, with or without these side absorptions, Lemma 5.2 compares the weight to w_0 by fixed factors. Hence at least a fixed fraction of the ρ -weight survives the good shifts for every corner setting in $\mathcal{G}$. Integrate this statement, use Equation (5.32), and finally average over the initially frozen independent side uniforms. Since $\mathbb{E}\rho$ and $\mathbb{E}w_0$ are comparable, this proves Equation (5.2). The pointwise comparison assertion in Proposition 5.1 is already contained in Lemma 5.2. $\square$

5.7 The Schur form used by the interior

The following formulation makes the interface with the transmission calculation explicit.

Corollary 5.6 (Eliminating the missing directions). *Let $N = \mathbf{1} - P$ on the boundary space. Almost everywhere in the boundary law, the N, N block of $\mathcal{T}$ is invertible. Define its Schur matrix and transformed source by*

$$\begin{aligned} \hat{T} &= PTP - PTN(NTN)^{-1}NTP, \\ \hat{k} &= Pk - PTN(NTN)^{-1}Nk. \end{aligned}$$

Then $\hat{T}$ is complex symmetric and passive, and

$$\begin{aligned}\hat{T}^{-1} &= P\mathcal{T}^{-1}P, \\ \hat{k} &= \hat{T}P\mathcal{T}^{-1}k, \\ P(\mathcal{T} - iP)^{-1}k &= (\hat{T} - i\mathbf{1}_P)^{-1}\hat{k}.\end{aligned}$$

In particular $\rho = \left\|(\hat{T} - i\mathbf{1}_P)^{-1}\hat{k}\right\|^2$, and [Equation \(5.2\)](#) controls $\hat{T}^{-1}$ on a set carrying a fixed fraction of this weight.

Proof. On the four-dimensional space N , the Hermitian real part is a fixed real symmetric matrix plus the four common corner shifts on its diagonal; the traceless box directions have zero diagonal entry there. Its determinant is a nonzero polynomial, so the block is invertible almost everywhere by passivity and the density of those shifts. The identities are the block inverse and source elimination formulas. Symmetry and passivity persist under Schur elimination by [Lemma 3.1](#). $\square$

6 Transmission through a cell

Throughout this section, the geometry and branch parameters are those of [Lemmas 4.1](#) and [4.2](#) and [Proposition 4.3](#). Constants may depend on the fixed energy, disorder, profile, corner-box exponents, and parameters already chosen, but not on exterior matrices or source vectors. A polynomial bound means $C(1+L)^C$ with this convention; different occurrences may have different exponents. All assertions concern sufficiently large L after the parameters have been fixed.

6.1 Exterior data and the two branch estimates

Suppose that the root is the only cell port in $V \cup B$, and that the source port lies outside this set. Artificial probes with a prescribed mask may be present. Eliminate the exterior physical sites. Its effect on B is a fixed complex-symmetric passive matrix T , and the source gives a fixed vector k , including its input multiplier. The remaining boundary matrix is

$$\mathcal{T} = T + \text{diag}_B(v_i - i\tau a_i(v_i)).$$

The exterior is held fixed throughout the cell integration, as are the four corner boxes. In particular, no bound on $\|T\|$ or $\|k\|$ is assumed.

Let P also denote its range, equipped with a real orthonormal basis. Eliminating the four missing boundary directions gives

$$\hat{T} = (P\mathcal{T}^{-1}P)^{-1}, \quad \hat{k} = \hat{T}P\mathcal{T}^{-1}k, \quad k_* = (\hat{T} - iP)^{-1}\hat{k}, \quad \rho = \|k_*\|^2. \quad (6.1)$$

These formulas are used at their almost-everywhere inverses. The missing-direction block is generically invertible by [Corollary 5.6](#). Schur complementation preserves symmetry and passivity, and gives the identity

$$k_* = P(\mathcal{T} - iP)^{-1}k.$$

Thus ρ is exactly the weight in [Proposition 5.1](#), and

$$\left\|(\hat{T} - iP)^{-1}\right\| \leq 1. \quad (6.2)$$

Hereafter M is restricted to P , and is extended by a zero row at the root when it is regarded as a map into V .

Let G be the inverse on $V \oplus P$ with the root open, reference potential $c(v)$, and root strength $\sigma = \epsilon^2$. Its internal artificial absorptions are constant during the branch integration: the true values lie on the plateau specified in [Proposition 4.3](#). Set

$$u = \sqrt{\sigma} G e_0, \quad g = \sigma G_{00}, \quad t = \widehat{k}^\top u_P. \quad (6.3)$$

Since k includes the source input multiplier, the root-to-source scattering amplitude satisfies $|S_{\text{source},a}|^2 = 4|t|^2$. The sign convention for the source does not affect this identity. The inverse exists almost everywhere even though $c(v)$ depends on other potentials. Indeed, absorption forces a kernel vector to vanish at the root. Restoring its true real diagonal therefore preserves that vector. After eliminating the fixed boundary data, the real part is a fixed matrix plus the true real diagonals on V ; their absolutely continuous joint law makes its determinant nonzero almost everywhere. This also justifies the almost-everywhere slices used below.

Proposition 6.1 (Branch transmission). *Choose the branch parameters in the order described below and fix $c_* > 2\nu$. For $0 \leq \tau \leq e^{-c_*L}$, uniformly in the fixed exterior data and allowed corner boxes,*

$$\mathbb{E}|t|^2 \geq e^{-C(\log L)^2} \mathbb{E}_B \rho, \quad (6.4)$$

$$\mathbb{E}[|\mu| |t|^2] \leq e^{-cL} \mathbb{E}|t|^2. \quad (6.5)$$

Here μ is the conditional single-root closure coefficient of [Lemmas 3.2 and 3.4](#). The expectation is the branch expectation, and $\mathbb{E}_B$ averages its boundary law. The hidden root value has already been averaged in μ ; averaging it again does not change any other quantity in these formulas.

6.2 The cutoff error and the small-response region

We first derive [Equation \(6.5\)](#) assuming [Equation \(6.4\)](#). Fix the boundary data and all jitters, and vary the common real shift r on Z^+ . Write P_+ for the coordinate projection onto Z^+ , extended by zero on the root and on P . Since $c(v) = E + p^\top A^{-1}p$ and $A' = P_+$ on J , differentiation gives

$$c' = -\left\|P_+ A^{-1}p\right\|^2, \quad g' = -u^\top P_+ u - \sigma^{-1} c' g^2. \quad (6.6)$$

The profile estimate in [Equation \(4.23\)](#), together with $|\psi_i| \leq C\epsilon$ on Z^+ , implies $|c'|/\sigma \leq C(1+L)^C$. Root absorption and [Lemma 3.1](#) imply

$$|g|^2 \leq \text{Im } g \leq 1. \quad (6.7)$$

Let $I_r = [-L^{-D_1}, L^{-D_1}]$. The derivative of a continuous determinant phase of G^{-1} is

$$\Phi'(r) = \text{tr}(P_+ \text{Im } G P_+) + \frac{c'}{\sigma} \text{Im } g.$$

The phase has range at most $\pi \dim(V \oplus P)$ after strict regularization. Integrating, using [Equation \(6.7\)](#), and then removing the regularization as in [Lemma 3.6](#) gives

$$\int_{I_r} \text{tr}(P_+ \text{Im } G P_+) \, dr \leq C(1+L)^C. \quad (6.8)$$

In particular, Ward and symmetry bound the integral of $\|P_+ u\|^2$ by the same polynomial. We need an additional factor when g is small.

Put $D = -\operatorname{Im} G^{-1}$. Complex symmetry makes D real symmetric and gives

$$u - \bar{u} = 2iG^*Du, \quad u^*Du = \operatorname{Im} g.$$

For each coordinate, Cauchy–Schwarz in the form D bounds the square of the corresponding entry of G^*Du by $(\operatorname{Im} G)_{ii} \operatorname{Im} g$. Consequently, with entrywise imaginary part on vectors,

$$\|P_+ \operatorname{Im} u\|^2 \leq (\operatorname{Im} g) \operatorname{tr}(P_+ \operatorname{Im} GP_+). \quad (6.9)$$

Take $q = \sqrt{\sigma} = \epsilon$, and set $\Omega_q = \{r \in I_r : |g(r)| \leq q\}$, excluding null singular sets. [Equations \(6.8\)](#) and [\(6.9\)](#) give

$$\int_{\Omega_q} \|P_+ \operatorname{Im} u\|^2 \, dr \leq C(1+L)^C q.$$

There are only polynomially many component intervals of Ω_q . To see this, A is affine in r , c is a rational function obtained by inverting A , and G is a further matrix inverse in cell dimension $O(L^2)$. Thus numerator and denominator degrees of g are bounded by a fixed polynomial in L . The inequality $|g|^2 \leq q^2$ becomes a real polynomial inequality after clearing the squared denominator. Its intervals and the poles are therefore polynomially many; the values of the exterior coefficients do not affect this degree bound. Integrate the real part of [Equation \(6.6\)](#) on these intervals, first truncating away from any endpoints where necessary. Every endpoint value of $\operatorname{Re} g$ has modulus at most q . Hence

$$\int_{\Omega_q} \operatorname{Re}(u^\top P_+ u) \, dr \leq C(1+L)^C q + C(1+L)^C q^2 |I_r|.$$

Using $\|P_+ u\|^2 = \operatorname{Re}(u^\top P_+ u) + 2\|P_+ \operatorname{Im} u\|^2$, we obtain

$$\int_{\Omega_q} \|P_+ u\|^2 \, dr \leq C(1+L)^C q. \quad (6.10)$$

Normalized uniform integration in r only introduces the polynomial factor $(2L^{-D_1})^{-1}$.

The Z rows of the equation $G^{-1}u = \sqrt{\sigma}e_0$ involve only u_P and $P_+ u$. Their coefficients are uniformly bounded, and M is bounded below on P . The P rows then give, respectively,

$$\|u_P\| \leq C \|P_+ u\|, \quad \hat{T}u_P = -M^\top u_J, \quad \|\hat{T}u_P\| \leq C \|P_+ u\|.$$

It follows from symmetry and [Equation \(6.1\)](#) that

$$|t| = \left| k_*^\top (\hat{T} - iI_P) u_P \right| \leq C\sqrt{\rho} \|P_+ u\|. \quad (6.11)$$

The cutoff bounds in [Lemma 3.4](#) now yield

$$\mathbb{E}[|\mu| |t|^2] \leq Cq\mathbb{E}|t|^2 + C(1+L)^C q\mathbb{E}_B \rho. \quad (6.12)$$

The first term treats $|g| > q$; the second follows from [Equations \(6.10\)](#) and [\(6.11\)](#) on Ω_q . Since $q = e^{-\nu L}$, substitution of [Equation \(6.4\)](#) absorbs the factor $e^{C(\log L)^2}$ and proves [Equation \(6.5\)](#) for some fixed $c > 0$.

6.3 Regular boundary and shift data

We turn to Equation (6.4). By Proposition 5.1, it is enough to work on boundary data satisfying

$$\|\hat{T}^{-1}\| \leq C(1+L)^C, \quad (6.13)$$

since this event carries a fixed positive fraction of $\mathbb{E}_B \rho$. Fix such data, assume $\rho > 0$, and initially set every jitter to zero. Define the boundary Schur matrix

$$\mathcal{B} = \hat{T} - M^\top A^{-1} M.$$

We claim that at least half of the uniform r -interval satisfies

$$\|\mathcal{B}^{-1}\| \leq C(1+L)^C. \quad (6.14)$$

All exponents in this assertion can be fixed before D_2, D_3 .

Indeed,

$$\mathcal{B}' = M^\top A^{-1} P_+ A^{-1} M \geq c I_P.$$

For the last inequality, put $w = A^{-1} M x$. The Z rows of $A w = M x$ have bounded coefficients, use only $P_+ w$, and give $\|M x\| \leq C \|P_+ w\|$. The lower singular-value bound for M gives $\|P_+ w\| \geq c \|x\|$, which proves the assertion. For $\xi > 0$ let $K_\xi = (\mathcal{B} - i\xi I_P)^{-1}$. If $\|\mathcal{B}^{-1}\| > \xi^{-1}$, inverse difference implies $\|K_\xi\| \geq (2\xi)^{-1}$. Ward then implies

$$\xi \operatorname{tr} \operatorname{Im} K_\xi \geq \frac{1}{4}. \quad (6.15)$$

The same conclusion follows at a singular matrix from a null vector. The determinant phase derivative of $\mathcal{B} - i\xi I_P$ is at least $c \operatorname{tr} \operatorname{Im} K_\xi$. Thus phase integration bounds the Lebesgue measure of the failed event in Equation (6.14) by $C\xi \dim P$. Dividing by $2L^{-D_1}$ and choosing ξ to be a sufficiently small inverse polynomial proves the claim. The only dependence on the exterior here is through coefficients of a passive matrix; its norm never enters.

For regular zero-jitter data, omit both the root and artificial absorption in J , and write

$$R_0 = \begin{pmatrix} A & M \\ M^\top & \hat{T} \end{pmatrix}^{-1}, \quad \Gamma = (R_0)_{JJ}, \quad z_0 = (R_0)_{JP} \hat{k}, \quad U_0 = \Gamma p / \epsilon. \quad (6.16)$$

The subscript 0 denotes the absence of internal artificial absorption; the boundary matrix $\hat{T}$ is unchanged. These definitions will also be used after the jitters have been varied. The field z_0 is the response in J to the exterior source. Define its normalized coupling to the root by

$$F_0 = p^\top z_0 / \epsilon.$$

If we adjoin the calibrated open root, still omitting the artificial absorptions on V , block inversion gives the analogue of t for this auxiliary system:

$$-\frac{F_0}{(c(v) - E - i\sigma - p^\top \Gamma p) / \sigma}.$$

The prescribed internal absorptions will be restored below. We first obtain a lower bound for this numerator using the shell jitters, then control the denominator. A diagonal change at a site j changes F_0 through the product $(U_0)_j (z_0)_j$. We will find a source component near the boundary

and a large component of U_0 at logarithmic depth, and use the jitters to propagate the source component to that deeper site.

Block inversion through A and $\mathcal{B}$ gives polynomial bounds on R_0 and Γ . Moreover,

$$\Gamma^{-1} = A - M\hat{T}^{-1}M^\top \quad (6.17)$$

has polynomial norm by Equation (6.13).

The source normalization requires some care because $\hat{k}$ and $\hat{T}$ need not be bounded. Put $C_{\text{ext}} = \hat{T} - iI_P$, $\Delta = \mathcal{B} - C_{\text{ext}} = iI_P - M^\top A^{-1}M$, and $b_0 = \mathcal{B}^{-1}\hat{k}$. The identities

$$b_0 = (I_P - \mathcal{B}^{-1}\Delta)k_*, \quad k_* = (I_P + C_{\text{ext}}^{-1}\Delta)b_0 \quad (6.18)$$

use only polynomially bounded $\Delta, \mathcal{B}^{-1}$ and the norm bound Equation (6.2). Consequently,

$$(1+L)^{-C}\sqrt{\rho} \leq \|b_0\| \leq C(1+L)^C\sqrt{\rho}, \quad \|z_0\| \leq C(1+L)^C\sqrt{\rho}. \quad (6.19)$$

Since $Az_0 = -Mb_0$, the sparse Z rows also imply $\|P_+z_0\| \geq c\|b_0\|$. There is therefore a site $j_0 \in Z^+$ with

$$|(z_0)_{j_0}| \geq (1+L)^{-C}\sqrt{\rho}. \quad (6.20)$$

The other field has the representation

$$U_0 = \frac{A^{-1}p}{\epsilon} + A^{-1}M\mathcal{B}^{-1}\frac{M^\top A^{-1}p}{\epsilon}. \quad (6.21)$$

The profile estimate and $|\psi_i| \leq C\epsilon$ on Z show $\|M^\top A^{-1}p\| \leq C(1+L)^C\epsilon$. Thus the reflected term in Equation (6.21) has polynomial norm, with exponent fixed independently of D_2 . At depth $d = \lfloor D_2 \log L \rfloor$, the profile has

$$\frac{|\psi_i|}{\epsilon} \geq ce^{\nu d} \geq c' L^{\nu D_2}.$$

Choose D_2 large enough that this exceeds both the reflected term and the profile error in units of ϵ , with fixed slack. Every site i at that depth then satisfies

$$|(U_0)_i| \geq 2. \quad (6.22)$$

6.4 A path of resolvent entries

Join j_0 to a site at depth d by a physical nearest-neighbor path of length $O(1 + \log L)$ in J , moving inward by decreasing absolute coordinates. The corner clipping guarantees that the path can avoid Z after its first vertex. For each successive pair y, j , with $j \notin Z$, Equation (6.17) gives

$$(\Gamma^{-1})_{yj} = 1.$$

We next replace this physical edge by a uniformly bounded number of large entries of Γ . This replacement is needed because Γ need not be spatially local.

Write $B_\Gamma = \Gamma^{-1}$, $n_J = |J|$, and $K_\Gamma = \max(1, \|B_\Gamma\|)$. Let

$$S_y = \text{supp}(B_\Gamma e_y), \quad S_j = \text{supp}(B_\Gamma e_j), \quad \delta_\Gamma = (2n_J K_\Gamma^2)^{-1}.$$

Form the undirected graph on $S_y \cup S_j$ with an edge between $a \in S_y$ and $b \in S_j$ whenever $|\Gamma_{ab}| \geq \delta_\Gamma$. The endpoints y, j belong to this vertex set because $(B_\Gamma)_{yj} = 1$. If they were not connected, let F

be the diagonal indicator of the component of y , extended by zero outside the graph. The inverse commutator identity would give

$$\begin{aligned} 1 &= |[F, B_\Gamma]_{yy}| = |(B_\Gamma[F, \Gamma]B_\Gamma)_{yy}| \\ &\leq \delta_\Gamma \|(B_\Gamma)_{y,:}\|_1 \|(B_\Gamma)_{:,j}\|_1 \leq \delta_\Gamma n_J K_\Gamma^2 = \frac{1}{2}. \end{aligned}$$

Indeed every nonzero summand with $F_{aa} \neq F_{bb}$ must cross components, so its Γ entry is below threshold. This contradiction proves connectivity.

Every graph edge meets S_j , which consists of j and at most its four physical neighbors: $j \notin Z$ makes the nonlocal term in Equation (6.17) vanish in its column. Thus S_j is a vertex cover of size at most five. A simple connecting path has at most ten edges, since a simple path with no two successive vertices outside a set of size five has at most eleven vertices. The other support S_y lies in the physical star of y together with Z . All these vertices lie in the shell $L - |i|_\infty \leq D_2 \log L + 5$; the fixed buffer accommodates the physical stars.

Concatenate the paths for the physical edges and erase loops. We obtain distinct sites

$$j_0, j_1, \dots, j_s, \quad s \leq C(1 + \log L), \quad |\Gamma_{j_a j_{a+1}}| \geq (1 + L)^{-C}, \quad (6.23)$$

all carrying shell jitters, with j_s at depth d . Fixed tie rules make all choices measurable functions of the regular zero-jitter data alone.

6.5 Propagation by the independent jitters

Choose D_3 sufficiently large after D_2 . For every allowed simultaneous jitter perturbation, its diagonal operator norm is at most L^{-D_3} . Inverse difference applied to R_0 therefore preserves its polynomial norm, the bounds for Γ^{-1} , and the path-edge lower bounds, with constant-factor slack. In particular, if superscripts denote perturbed and zero-jitter values, then

$$z_0^{\text{new}} - z_0^{\text{old}} = -\Gamma^{\text{new}} \Delta_J z_0^{\text{old}},$$

where Δ_J is the jitter diagonal. Equations (6.19) and (6.20) show that the norm and starting-component bounds also retain constant-factor slack. The terminal bound retains, say, $|(U_0)_{j_s}| \geq 1$ throughout these perturbations. For this last assertion use Equation (6.21) and the uniform profile estimate again: the large total size of U_0 is not controlled by an additive small-perturbation argument.

Condition on all nonpath jitters. Expose the path jitters successively, setting unexposed ones to zero only in the intermediate calculations. Let $w = L^{-D_3}$ be their common half-width. At step $a < s$, changing the diagonal at j_a by $\zeta \in [-w, w]$ gives the exact rank-one formula

$$(z_0^{\text{new}})_{j_{a+1}} = (z_0)_{j_{a+1}} - \frac{\Gamma_{j_{a+1}j_a}(z_0)_{j_a}\zeta}{1 + \Gamma_{j_a j_a}\zeta}. \quad (6.24)$$

Values on the right refer to the state just before this update. The denominator has modulus between $1/2$ and $3/2$ after increasing D_3 if necessary. For $\zeta \in [-w, -w/2]$ and $\zeta' \in [w/2, w]$, the two values on the right differ by

$$|\Gamma_{j_{a+1}j_a}(z_0)_{j_a}| \frac{|\zeta - \zeta'|}{|1 + \Gamma_{j_a j_a}\zeta| |1 + \Gamma_{j_a j_a}\zeta'|} \geq cw(1 + L)^{-C} |(z_0)_{j_a}|.$$

If one of the two intervals contains a value below one quarter of this separation in modulus, every value in the other interval is above that threshold. If it contains no such value, that interval itself is successful. Hence the conditional success probability is at least $1/4$, and success gives

$$|(z_0^{\text{new}})_{j_{a+1}}| \geq (1+L)^{-C} |(z_0)_{j_a}|. \quad (6.25)$$

Constants and the exponent here include the fixed D_3 .

Finally expose the jitter at j_s . Applied to the scalar F_0 , the update is

$$F_0^{\text{new}} = F_0 - \frac{(U_0)_{j_s}(z_0)_{j_s}\zeta}{1 + \Gamma_{j_s j_s}\zeta}.$$

Since $|(U_0)_{j_s}| \geq 1$, the same two-interval argument gives a final success of probability at least $1/4$ and size at least $(1+L)^{-C} |(z_0)_{j_s}|$. Only the current component is used at each step; later jitters need not preserve previously propagated components. There are $O(1 + \log L)$ updates, all at distinct sites and hence using independent uniforms. Combining [Equations \(6.20\)](#) and [\(6.25\)](#), we conclude that

$$\mathbb{P}_{\text{jitters}} \left\{ \left| p^\top z_0 / \epsilon \right| \geq e^{-C(\log L)^2} \sqrt{\rho} \right\} \geq e^{-C(1+\log L)}. \quad (6.26)$$

This holds conditionally on every regular zero-jitter boundary and r datum, and uniformly after conditioning on nonpath jitters.

6.6 Restoring absorption and the root

Reinsert the prescribed artificial absorptions in J . If their diagonal is τA_{art} , with $0 \leq A_{\text{art}} \leq I_J$, the rootless inverse becomes

$$R_\tau = \left(R_0^{-1} - i\tau \begin{pmatrix} A_{\text{art}} & 0 \\ 0 & 0 \end{pmatrix} \right)^{-1}.$$

Write $\Gamma_\tau = (R_\tau)_{JJ}$ and $z_\tau = (R_\tau)_{JP}\hat{k}$. Inverse difference and the polynomial bounds already established give

$$\|\Gamma_\tau - \Gamma\| \leq \tau C(1+L)^C, \quad \|z_\tau - z_0\| \leq \tau C(1+L)^C \sqrt{\rho}. \quad (6.27)$$

Thus the error in $p^\top z_0 / \epsilon$ is at most $\epsilon^{-1} \tau C(1+L)^C \sqrt{\rho}$.

Now restore the open root. Its scalar Schur denominator is

$$d = c(v) - E - i\sigma - i\tau a_0 - p^\top \Gamma_\tau p.$$

For the auxiliary system with internal artificial absorptions omitted, the calibration and block formula for Γ give

$$\frac{c(v) - E - i\sigma - p^\top \Gamma p}{\sigma} = -i - \frac{p^\top A^{-1} M \mathcal{B}^{-1} M^\top A^{-1} p}{\sigma}. \quad (6.28)$$

The boundary profile estimate gives $\|M^\top A^{-1} p\| \leq C(1+L)^C \epsilon$, and hence the second term is polynomially bounded. Reinstating absorption changes the normalized denominator by at most $\sigma^{-1} \tau C(1+L)^C$, including the absorption at the root. Because $c_* > 2\nu$, this is small for large L . We have therefore proved, uniformly on all the data being used,

$$|d/\sigma| \leq C(1+L)^C.$$

Passivity and the open root also give $-\text{Im } d \geq \sigma$, so this denominator is nonzero.

Block inversion now identifies the amplitude, with an irrelevant sign, as

$$t = -\frac{p^\top z_\tau / \epsilon}{d/\sigma}. \quad (6.29)$$

The numerator error in Equation (6.27) is exponentially smaller than the successful lower bound in Equation (6.26). The polynomial denominator and the success probability can therefore be incorporated into $e^{-C(\log L)^2}$. For each good boundary datum, integration over the regular r event, of probability at least $1/2$, and then all jitters gives

$$\mathbb{E}_{r,\text{jitters}} |t|^2 \geq e^{-C(\log L)^2} \rho.$$

When $\rho = 0$, invertibility in Equation (6.1) gives $\hat{k} = 0$, and the inequality is immediate. Finally integrate over the weighted good boundary event from Proposition 5.1. This proves Equation (6.4) and completes the proof of Proposition 6.1.

For clarity, the choices above have no circular dependence. First fix the profile and its plateau, the box-width exponents, and D_1 as in Proposition 4.3. The polynomial in Proposition 5.1, the regular zero-jitter inverse thresholds, and the exponents for the profile residual and reflected field can then be fixed without D_2, D_3 . Choose D_2 to obtain the profile height in Equation (6.22). Choose D_3 afterwards to guarantee the profile estimates and all the perturbation margins used above. The constants in Equation (6.26) may depend on this last choice. Any fixed $c_* > 2\nu$ then makes the absorption errors small for sufficiently large L ; no polynomial threshold is chosen by using the norm of an exterior matrix.

6.7 An envelope for arbitrary interior choices

For comparison between different interior laws we need a bound that does not require the branch on V . Keep the same geometry, boundary data, exterior source, and prescribed artificial mask. Allow any set $D \subset V$ of internal cell ports, at most one per site, with strengths $s_d \in [e^{-C_0 L}, 1]$ and real reference potentials bounded in modulus by a fixed C_0 . All other internal real diagonals are their true values, and there are no cell ports on B . The set D , strengths, and references may depend arbitrarily on the true interior potentials and on auxiliary choices.

Proposition 6.2 (Interior envelope). *For almost every fixed boundary datum there is a nonnegative function $\mathfrak{E}(v_V)$ depending only on the true interior potentials, the fixed exterior and boundary data, and the artificial mask, such that every allowed internal configuration satisfies*

$$\sum_{d \in D} |S_{d,\text{source}}|^2 \leq \mathfrak{E}(v_V), \quad \mathbb{E}_{V,\text{true}} \mathfrak{E} \leq e^{C(1+L)^C} \rho. \quad (6.30)$$

The same integrated bound, with adjusted constants, holds under any density bounded by $e^{C(1+L)^C}$ relative to the true product law on V . The constants are uniform in the exterior data and in all adaptive interior choices.

Proof. First fix a candidate subset D and let E_D denote its coordinate inclusion in $V \oplus P$. Replace all its strengths by 1 and all its real reference potentials by their true values, retaining the artificial probes. Write G'_D for the inverse after replacement and G_D for the inverse of any of the original configurations with this set of sites. The two inverse matrices differ by

$$(G'_D)^{-1} = G_D^{-1} + E_D \Delta_D E_D^\top, \quad \|\Delta_D\| \leq C.$$

Let $s_* = e^{-C_0 L}$. Absorption before replacement gives $\|E_D^\top G_D E_D\| \leq s_*^{-1}$. For any source vector x , inverse difference gives the pointwise identity

$$E_D^\top G_D x = (I_D + E_D^\top G_D E_D \Delta_D) E_D^\top G'_D x.$$

Consequently, including the factor 4 for squared scattering amplitudes and using $s_d \leq 1$,

$$\sum_{d \in D} |S_{d, \text{source}}|^2 \leq e^{C(1+L)} \|E_D^\top (G'_D)_{VP} \hat{k}\|^2. \quad (6.31)$$

In this formula and below, E_D is also used for its inclusion in V when the block dimensions specify this meaning. The replacement matrix depends on D and true potentials only, so the right side is independent of the original reference values and strengths.

All these inverses can be justified on a common full-measure set. For G'_D , eliminate P and use the true diagonal densities on V ; the real-part determinant is a nonzero diagonal polynomial. If an original matrix had a kernel vector, absorption would make it vanish on D . Restoring true reference values and replacing strengths at those sites would leave it a kernel vector of the replacement matrix. Thus the original inverse exists whenever the replacement inverse does, even for potential-dependent choices. There are only finitely many subsets D , so their generic sets can be intersected.

Fix D . Ward and transpose symmetry imply

$$X_D := \|(G'_D)_{VV} E_D\|_{\text{HS}}^2 \leq \text{tr Im}(G'_D)_{VV}. \quad (6.32)$$

We bound the expectation of the trace under the true product law by common-shift phase integration. If $n_V = |V|$, parameterize the cube $[-h, h]^{n_V}$ by $v_{n_V} = t$ and $v_i = t + w_i$ for $i < n_V$. This change has Jacobian 1; the allowed t form an interval of length at most $2h$, and each $w_i \in [-2h, 2h]$. Along this interval every true real diagonal in V shifts at rate 1. The only other derivatives are $-i\tau a'_i(v_i)$, which obey $|a'_i|^2 \leq C a_i$. The absorption argument of [Lemma 3.6](#) therefore gives

$$\Phi'(t) \geq \frac{1}{2} \text{tr Im}(G'_D)_{VV} - C \dim(V \oplus P).$$

The phase range and interval length bound the unnormalized trace integral by $C(1+L)^C$. Integrating over the transverse box and dividing by $(2h)^{n_V}$ costs at most $e^{C(1+L)^C}$; the constants may depend on the fixed h . Strict regularization followed by Fatou handles null singular slices. Thus

$$\mathbb{E}_{V, \text{true}} X_D \leq \mathbb{E}_{V, \text{true}} \text{tr Im}(G'_D)_{VV} \leq e^{C(1+L)^C}, \quad (6.33)$$

uniformly in D and in the exterior matrices.

Let H'_V be the V, V block of $(G'_D)^{-1}$. It has bounded operator norm, since its diagonals are true potentials with unit cell absorption and bounded artificial absorption. The V rows of $(G'_D)^{-1} G'_D = I$ give

$$M(G'_D)_{PV} E_D = E_D - H'_V (G'_D)_{VV} E_D.$$

The lower singular-value bound for M implies the pointwise estimate

$$\|(G'_D)_{PV} E_D\|_{\text{HS}}^2 \leq C(|D| + X_D). \quad (6.34)$$

For the exterior source, use the V, P block of $G'_D (G'_D)^{-1} = I$ and [Equation \(6.1\)](#):

$$\begin{aligned} E_D^\top (G'_D)_{VP} \hat{k} &= E_D^\top (G'_D)_{VP} (\hat{T} - iI_P) k_* \\ &= -E_D^\top (G'_D)_{VV} M k_* - iE_D^\top (G'_D)_{VP} k_*. \end{aligned}$$

Transpose symmetry, the bounded norm of M , and Equation (6.34) now show

$$\left\| E_D^\top (G'_D)_{VP} \hat{k} \right\|^2 \leq C\rho(|D| + X_D). \quad (6.35)$$

Again, this uses no norm bound on $\hat{T}$ or $\hat{k}$. Equations (6.33) and (6.35) give the desired integrated bound for each replacement problem.

Finally define the pointwise envelope

$$\mathfrak{E}(v_V) = e^{C(1+L)} \sum_{D \subset V} \left\| E_D^\top (G'_D)_{VP} \hat{k} \right\|^2,$$

with the empty-set summand zero and any value on the common null exceptional set. There are $2^{|V|} = e^{O(L^2)}$ summands. Equation (6.31) proves pointwise domination for every adaptive choice, and summing the expectation bounds proves Equation (6.30). Multiplication by a bounded density gives the final assertion of the proposition. $\square$

When this envelope is compared with the branch lower bound, the corner-box conditionings must coincide. The single-side laws on B may differ: their settings and artificial absorptions change ρ only by the pointwise bounded factors in Proposition 5.1. This is the precise boundary comparison needed when an adaptive collection of descendant ports is replaced by an open parent root.

7 A hierarchy of probability splits and energy comparisons

Throughout this section the disorder strength $h > 0$ and the interior energy E are fixed. All parameters in the cell construction are fixed as well. Constants may depend on these choices, including the corner-box exponent chosen below, but not on the outer torus, the upper thinning level, or the prescribed artificial-probe mask. We write

$$b_l = 3^l, \quad L_l = \frac{b_l - 1}{2} - b, \quad q_l = p_l = \exp(-b_l^6).$$

Only levels above a sufficiently large fixed l_0 will be used. We increase l_0 whenever necessary to apply all the preceding cell estimates.

7.1 An iid implementation of the rare cell laws

On the torus of side b_N , choose one uniform lattice offset, independently of the true iid potentials. For each $l_0 \leq l \leq N$, partition the torus into squares of side b_l whose centers are congruent to that offset modulo b_l . These partitions are nested. In each tile use the translated geometry and branch law of Lemma 4.1 and Proposition 4.3, with $L = L_l$. The word *ancestor* will always refer to strict containment in this nested family unless the contrary is stated.

We first record the geometric feature that makes conditioning compatible across levels. The corner triples belonging to different tiles and levels are disjoint except when they coincide exactly. At a coincident triple the ordering of its three sites, and the two real plane-basis vectors used for its box coordinates, are the same. In fact, the missing corner and its two companions have fixed displacements from the corresponding two ends of a tile. Passing to an extreme child preserves those positions:

$$b_l + \frac{b_l - 1}{2} - b = \frac{b_{l+1} - 1}{2} - b.$$

Consequently, if an ancestor triple meets a descendant tile, it is one of that descendant's own triples. Also, the set $V \cup B$ of a descendant is contained in that of its ancestor, and every descendant center belongs to the ancestor's V . The latter inclusion will ensure that all descendant reference-potential changes are interior changes when we use the parent-cell envelope.

Put $K_0 = 10^6$. For each distinct corner triple, give its three true potentials a single box label as follows. In each product of polynomial pieces of ϑ lying in $(-h, h)^3$, use the fixed linear box coordinates from [Proposition 4.3](#). Partition the resulting region into maximal dyadic cubes whose concentric doubles lie in the region. Their inverse images are the corner boxes. All labels are defined from the true potentials, but we will expose only specified labels. At exact corner coincidences we use the same label, rather than successively refining it. Null boundaries may be assigned arbitrarily.

Conditional on a box label, the three true variables are uniform on that box: the original density is constant, and the coordinate change is linear with constant nonzero Jacobian. Labels of disjoint triples are independent. If a box has width less than ξ , its points lie within $C\xi$ of the boundary of one of the finitely many piece regions. Indeed, away from those boundary strips a dyadic cube of width comparable to ξ containing the point has its concentric double inside the region, so a maximal cube containing it cannot be smaller. Thus

$$\mathbb{P}(\text{width of a given triple's box} < \xi) \leq C\xi. \quad (7.1)$$

A level- l tile has *valid own labels* when the widths of all four corner boxes are at least $b_l^{-K_0}$. The doubled-box requirement also provides the polynomial clearance from support endpoints and piece boundaries required in the cell estimates.

Assign to every tile U of level l an independent selector $\eta_U \in \{0, 1\}$ of success probability q_l . These selectors are independent of all true potentials. A tile is *eligible* if it is selected and has no selected strict ancestor. Eligible tiles are pairwise disjoint, even when a selected tile later fails its branch test. An eligible tile is *allowed* if its own labels and those of every strict ancestor are valid at their respective levels.

For an allowed tile U , let $\mathcal{L}_U$ be the likelihood of its cell branch law relative to the true product law with its four corner boxes fixed. It changes the law on V and the noncorner single-side coordinates of B , leaves the conditional laws on the four corner triples unchanged, and leaves all remaining sites of the tile unchanged. Its root factor is the conditional density centered at the reference $c(v)$ constructed from the true values on J . Apart from the allowance gate, the likelihood uses only data in U and its own corner boxes. In particular it does not use ancestor labels in any other way. The density bound of [Proposition 4.3](#) gives, uniformly over valid boxes,

$$\mathcal{L}_U \leq \exp(CL_l^3 + CL_l^2 \log(2 + L_l)).$$

After increasing l_0 , we have $p_l \mathcal{L}_U \leq 1$.

Give each tile a further independent uniform variable t_U on $[0, 1]$. Its acceptance indicator is

$$A_U = \mathbf{1}_{\{U \text{ eligible and allowed}\}} \mathbf{1}_{\{t_U \leq p_l \mathcal{L}_U\}}. \quad (7.2)$$

The selectors, acceptance uniforms, and offset are auxiliary randomness; the true potentials are sampled first from their unchanged iid law.

Proposition 7.1 (Conditional sampling). *Fix the offset and all selector marks. Expose the box identities of the eligible tiles and of their ancestors. Conditional on this information, the unexplored variables in disjoint eligible tiles are independent. For an allowed eligible tile U of level l , its conditional acceptance probability is p_l , and conditional on acceptance its site variables have exactly the branch law of [Proposition 4.3](#) with the indicated corner conditioning.*

These conclusions remain valid for U after fixing status decisions or other local choices in disjoint eligible tiles, provided the exposed box identities are retained in the conditioning. In particular,

$$\mathbb{P}(A_U = 1 \mid \text{offset}) \leq q_l p_l. \quad (7.3)$$

The unconditional law of the true potentials remains the iid product law. The same construction can be made prospectively by omitting all selectors and ancestor tests above a specified cutoff level.

Proof. The selector marks determine eligibility without using any potential. Consider an eligible tile U . Any exposed triple that meets U is either one of its own triples or an ancestor triple coinciding with one of them. To see this also for labels exposed on behalf of another eligible tile, use nesting: that other tile is disjoint from U , and an ancestor of it that meets U either contains U or is disjoint from its interior. In the containing case the corner-coincidence property applies. Thus no exposed label imposes an additional condition on the interior of U or on its single-side variables.

The conditional true law in U is consequently precisely the reference law defining $\mathcal{L}_U$: the four corner triples are uniform in their fixed boxes, and all other sites have their product laws. These conditional laws factor across eligible tiles, since their site sets are disjoint and the only repeated labels are identical labels on the same triple. For an allowed tile,

$$\mathbb{E}[\mathcal{L}_U \mid \text{selectors and exposed labels}] = 1.$$

Integrating the independent uniform t_U proves that its acceptance probability is p_l . More generally, for every bounded local test function F ,

$$\mathbb{E}[F \mathbf{1}_{\{A_U=1\}} \mid \text{selectors and exposed labels}] = p_l \mathbb{E}[F \mathcal{L}_U \mid \text{selectors and exposed labels}].$$

After division by p_l , this is exactly the branch expectation. Local status decisions in disjoint eligible tiles multiply the conditional product density by factors on those tiles alone, so do not alter this conclusion.

The event $A_U = 1$ requires its own selector to be one. Conditional on eligibility and all required valid labels, the additional acceptance probability is p_l ; dropping the other restrictions gives Equation (7.3). Finally, acceptance only attaches auxiliary indicators to the initially sampled product field, and never resamples or changes that field. Omitting the higher-level rules leaves the same argument on the truncated hierarchy. $\square$

For a stage n , $l_0 \leq n \leq N$, place active cell ports at precisely the accepted roots of levels at least n . Use their branch strengths and reference potentials. Every site without an active cell port has its true real potential. Because eligible tiles are disjoint, there is at most one cell port at a site. Denote the scattering matrix without artificial probes by $S_{n,N}$.

We may also prescribe at most one artificial probe per site, absent or of strength τ or $\tau \vartheta(v_i)$. Its mask may depend on the offset and on external randomness independent of the potentials and branch decisions. The profile is always evaluated at the true value, including at an active cell root. When comparing stages $n < m$, we assume

$$0 \leq \tau \leq \exp(-c_* b_m). \quad (7.4)$$

This is sufficient for the cell estimate at every removed level and at its immediate parent. In particular all constants below are uniform over these masks. For notational convenience, expectations denoted by $\mathbb{E}_{\text{off}}$ may include any external independent randomness used in the geometry or deterministic data prescriptions.

Lemma 7.2 (The hidden root law). *At an accepted root whose cell port is still open, condition on the selectors, the needed corner-box identities, all branch acceptance indicators, and the other true potentials. Do not condition on the numerical value of the root tile's acceptance uniform. The root's true value then has the form*

$$v_0 = c(v) + \delta,$$

where δ has the centered, cutoff Cauchy law of [Lemma 3.4](#), with the scale of that cell port. The open response, all other active references and strengths, and any artificial probe at this root are independent of this hidden draw under the stated conditioning.

Proof. The root does not belong to any exposed triple. By [Proposition 7.1](#), acceptance tilts its tile's conditional law by the branch likelihood. With all other true values fixed, the only remaining factor is exactly the specified root density centered at $c(v)$; the reference is a function of the values on J . Acceptance indicators of other eligible tiles concern disjoint sites. Indicators of ineligible descendants are identically zero and give no further information about the root.

The open system uses $c(v)$ in place of v_0 . The support of the hidden draw lies on the plateau of ϑ , so its artificial probe, if present, has constant strength there. References and strengths in other active cells depend only on their disjoint tiles. These facts prove the final assertion. Restoring true values at other roots does not change it, since those values also belong to disjoint eligible tiles. Conditioning on the numerical acceptance uniform, in contrast, could truncate the root density and is expressly excluded. $\square$

7.2 Thinning passes and admissible data

Fix $n < m \leq N$. A thinning pass acts on a designated union of level- m tiles, chosen from the offset and any independent geometric randomness. In that region remove all accepted root ports of levels $n, \dots, m-1$, restoring their true real values. Process these levels in increasing order. At a given level interpolate by retaining each eligible accepted root independently with probability θ , with θ decreasing from one to zero. Use fresh independent thinning marks; never recompute branch choices. One may subsequently perform a second pass on the complementary union of level- m tiles. Each such pass is covered separately by the estimates below, with its own starting system. A *prefix* may end between levels or at an intermediate value of θ .

Let f be deterministic finite real scalar or Euclidean-vector site data, possibly depending on the offset. Its restriction supplies data on every port set. Let g be data fixed at the beginning of the pass such that, for a deterministic $c_g < \infty$,

$$|g_i - g_j|^2 \leq c_g^2 |f_i - f_j|^2 \quad \text{for all sites } i, j. \quad (7.5)$$

We permit g to depend on selectors, branch statuses, and the start-of-pass response, and on external independent randomness. The precise admissibility requirement is that, at each root closure to which the energy comparison is applied, g is measurable in the conditioning that preserves that root's hidden law in [Lemma 7.2](#). In particular it must not inspect still hidden true values needed for future closures, and it must not depend on the thinning marks of the pass. This formulation allows response-dependent adjustments while retaining the conditional closure identity.

For a level- l tile U , let $D(U)$ be its immediate parent, and put

$$h_D = \sup_{i,j \in D} |f_i - f_j|.$$

Define the nonnegative charge

$$C_{n,m}(f) = \sum_{l=n}^{m-1} q_l p_l \mathbb{E}_{\text{off}} \sum_{U: \text{level}(U)=l} h_{D(U)}^2. \quad (7.6)$$

The sum includes all tiles of a level, even when the pass acts on only part of the torus; this only enlarges the bound.

Theorem 7.3 (Energy comparison under thinning). *There is a deterministic sequence $\zeta_n \downarrow 0$, after replacing it by a decreasing majorant if necessary, such that for all sufficiently large n , every $n < m \leq N$, every allowed artificial-probe mask, and every pass or prefix just described,*

$$\mathbb{E}Q_{\text{end}}(g) \leq \mathbb{E}Q_{\text{start}}(g) + Cc_g^2 (\zeta_n \mathbb{E}Q_{\text{start}}(f) + \mathcal{C}_{n,m}(f)). \quad (7.7)$$

The constants and ζ_n are independent of m, N , the pass region, the mask satisfying Equation (7.4), and the finite dimensions of the real vector data.

We prove the theorem by first establishing the estimate that controls the interpolation near $\theta = 0$. In a current level- l system, let $\mathcal{A}_l$ be the thinnable roots in the pass region that are still present. For $a \in \mathcal{A}_l$, write $U(a)$ for its level- l tile. Set

$$Z_l(\theta) = \mathbb{E} \sum_{a \in \mathcal{A}_l} \sum_{j: \text{site}(j) \notin D(U(a))} |S_{ja}|^2 |f_j - f_a|^2, \quad (7.8)$$

$$B_l(f) = q_l p_l \mathbb{E}_{\text{off}} \sum_{U: \text{level}(U)=l} h_{D(U)}^2. \quad (7.9)$$

Port data in these formulas are evaluated at their physical sites. Trivially,

$$Z_l(\theta) \leq 2\mathbb{E}Q_{\text{current}}(f). \quad (7.10)$$

Lemma 7.4 (The parent-alternative cap). *There are constants C, C_1 such that, with $K_l = \exp(C(1 + b_{l+1})^{C_1})$, for $0 \leq \theta \leq 1$,*

$$Z_l(\theta) \leq \theta K_l \mathbb{E}Q_{\text{current}}(f) + C\theta B_l(f). \quad (7.11)$$

The estimate holds during either pass and during every level interpolation in that pass.

Proof. First replace the datum f_a by that at its parent center c_D :

$$|f_j - f_a|^2 \leq 2|f_j - f(c_D)|^2 + 2h_D^2.$$

The additional term is bounded using the unit row mass of S and Equation (7.3). Each accepted level- l root survives with probability θ independently, so the total additional expectation is at most $C\theta B_l(f)$. It remains to treat the first term, grouped by parent D .

Fix such a parent in the pass region. The presence of a thinnable accepted child implies that all selector marks on D and above it are zero and that all labels on D and above it are valid. For comparison, however, leave the selector of D itself free. Condition on:

- (i) absence of selected strict ancestors of D ;
- (ii) the identities of the valid labels of D and its strict ancestors;
- (iii) true potentials outside the parent's $V \cup B$;
- (iv) exterior selector marks, auxiliary uniforms, and thinning outcomes determining choices in tiles disjoint from D .

Only this conditioned event can contribute to the upper expression; outside it that expression is zero. In particular, do not expose finer labels, statuses, or true potentials inside the parent.

No eligible tile strictly contains D under this conditioning. Changing D 's selector affects only its descendants. Exterior eligibility is unchanged, and any shared ancestor validity tests use the labels already fixed. If a label on behalf of an exterior tile meets the parent, it is one of the fixed parent triples. Hence the exterior active ports, their strengths and references, and their already restored true values are fixed across the alternatives below. The remaining law on the parent is the true product law with precisely the parent corner-box conditioning.

Fix an exterior source port j whose physical site is outside D . Schur elimination of the fixed exterior produces the same passive matrix T and source vector k at the parent's B in both alternatives. Artificial probes in the exterior are part of these fixed data. All possible descendant centers are in the parent's V , and there are no cell ports on its B . Every active descendant strength is between $\exp(-CL_{l+1})$ and one, and its reference potential is bounded, because its level does not exceed l .

In the alternative contributing child survivors, apply [Proposition 6.2](#) to all possible internal active subsets. The pointwise envelope furnished there depends only on the true potentials and the fixed exterior data. The same pointwise envelope applies to every internal branch choice and thinning outcome, so all internal selector, label, and acceptance restrictions may be dropped in its upper bound. Assign a fresh Bernoulli- θ survival mark to each of the nine level- l children, including those not accepted. A child contribution is possible only if at least one of these marks is one. This event has probability at most 9θ , independently of the envelope. Thus, writing $\mathcal{S}_D$ for the thinnable survivors in D , the conditional upper bound is

$$\mathbb{E} \left[\sum_{a \in \mathcal{S}_D} |S_{ja}|^2 \middle| \text{fixed data} \right] \leq \theta \exp(C(1 + b_{l+1})^{C_1}) \mathbb{E}_{B, \text{true}} \rho. \quad (7.12)$$

Here ρ is the parent's boundary weight used in [Proposition 6.1](#); the expectation includes its fixed corner boxes.

In the comparison alternative, D itself is selected and accepted. It is eligible because its strict ancestor selectors are zero, and allowed because its labels and those above are valid. By [Proposition 7.1](#), this alternative has conditional probability $q_{l+1}p_{l+1}$ and its conditional law is the parent branch law. Its root a_D is still active: the pass is presently at level l , and it has not thinned level $l + 1$. If this is the second pass, the earlier pass did not touch D , since its region and the present region are unions of level- m tiles and $l + 1 \leq m$. Therefore the alternative is an actual member of the current ensemble, not a new system outside that ensemble. The lower estimate in [Proposition 6.1](#) yields

$$\mathbb{E} \left[\mathbf{1}_{\{a_D \text{ active}\}} |S_{ja_D}|^2 \middle| \text{fixed data} \right] \geq cq_{l+1}p_{l+1} \exp(-C(\log L_{l+1})^2) \mathbb{E}_{B, \text{branch}} \rho. \quad (7.13)$$

The harmless normalization of a scattering amplitude is absorbed in c .

The two boundary averages in [Equations \(7.12\) and \(7.13\)](#) have identical corner-box laws. They differ only in the single-side values and their permitted artificial absorptions. The pointwise weight comparison accompanying [Proposition 5.1](#) compares all such bounded settings by a fixed factor. Integrating that comparison shows that the two averages of ρ are comparable, without imposing an exterior norm bound or assuming independence after a branch tilt. Inverting the parent incidence probability costs

$$(q_{l+1}p_{l+1})^{-1} = \exp(2b_{l+1}^6),$$

which can be absorbed in K_l . Consequently, under the specified conditioning,

$$\mathbb{E} \sum_{a \in \mathcal{S}_D} |S_{ja}|^2 \leq \theta K_l \mathbb{E} \left[\mathbf{1}_{\{a_D \text{ active}\}} |S_{ja_D}|^2 \right]. \quad (7.14)$$

Absent exterior source ports contribute zero on both sides. Their status was fixed in the conditioning, so the inequality can be summed over exterior source indices and then integrated.

Multiply by $|f_j - f(c_D)|^2$ and sum over parents. For an active parent this datum is exactly f_{a_D} . Each parent appears only once at this level, and the resulting oriented parent-to-exterior sum is at most $2\mathcal{Q}_{\text{current}}(f)$. Combining it with the oscillation charge proves Equation (7.11), after enlarging K_l by a fixed factor. $\square$

Proof of Theorem 7.3. Fix a level l during the pass. Conditional on the initial data and the true potentials, the expectation over its finitely many thinning marks is a polynomial in θ . Differentiation gives a sum of single-root kept-versus-removed differences, with the other roots retained independently with parameter θ . Equivalently, for $\theta > 0$, sum over the roots currently retained and divide by θ . The data g are frozen throughout this interpolation. All scattering energies involved are integrable: unitarity and Equation (7.5) bound them by finite deterministic pair-data oscillations and the finite number of ports.

For one retained thinnable root a , use the conditioning of Lemma 7.2 and the admissibility of g . Restoring its true potential is exactly the closure operation of Lemma 3.3. Its expected increase is at most

$$2|\mu_a| \mathcal{M}_a(g),$$

where μ_a is the conditional cutoff coefficient. Use Equation (7.5) to dominate the squared differences by those of f , with factor c_g^2 . After this domination, g plays no role in any further conditioning or cell integration.

For ports j in the parent $D(U(a))$, the uniform bound on $|\mu_a|$ from Lemma 3.4 and row mass give a cost at most $Cc_g^2 h_{D(U(a))}^2$ on occurrence of this root. Its incidence is at most $\theta q_l p_l$. After division by θ , the summed local cost is at most $Cc_g^2 B_l(f)$.

For links to ports outside the parent, apply the relative estimate in Proposition 6.1. To justify its use, fix selectors, the needed box identities, the tile's acceptance, exterior true potentials off its $V \cup B$, and choices in the disjoint eligible tiles, as well as the independent thinning outcomes. Its branch law is unchanged by Proposition 7.1. All other active cell ports and their reference-potential changes are exterior, since the accepted tile has no eligible descendants. Accordingly the source and Schur data are fixed in the cell average. For deterministic f this gives the upper bound

$$Cc_g^2 e^{-cL_l} \frac{Z_l(\theta)}{\theta}$$

for the exterior part of the derivative in the direction of removal. Writing $F_l(\theta) = \mathbb{E}\mathcal{Q}_{\text{current}}(f)$, the complete upper derivative bound is therefore

$$Cc_g^2 \left(B_l(f) + e^{-cL_l} \frac{Z_l(\theta)}{\theta} \right). \quad (7.15)$$

Combine Equations (7.10) and (7.11). Increasing K_l to be at least two and using $\min(a, b+c) \leq \min(a, b) + c$ gives

$$\frac{Z_l(\theta)}{\theta} \leq \min\left(\frac{2}{\theta}, K_l\right) F_l(\theta) + CB_l(f).$$

The logarithmic interpolation cost is explicit:

$$\int_0^1 \min\left(\frac{2}{\theta}, K_l\right) d\theta = 2 + 2\log(K_l/2) \leq C(1 + \log K_l).$$

In particular the additive charge is not multiplied by this logarithm. Define

$$a_l = Ce^{-cL_l}(1 + \log K_l), \quad \alpha_n = \sum_{l=n}^{\infty} a_l.$$

The constants in K_l are fixed independently of l, m, N . Since $\log K_l$ is polynomial in $b_{l+1} = 3b_l$ and L_l is proportional to b_l for large levels, α_n is finite and tends to zero. Integrating Equation (7.15) over a whole level, or any prefix of it, costs at most

$$c_g^2 \left(a_l \sup_{0 \leq \theta \leq 1} F_l(\theta) + CB_l(f) \right).$$

At $\theta = 0$ use continuity of the finite thinning polynomial.

Let M be the supremum of $\mathbb{E}\mathcal{Q}(f)$ over all prefixes of the pass, including all intermediate interpolation parameters. First take $g = f$. Summing the preceding estimates up to any prefix gives

$$M \leq \mathbb{E}\mathcal{Q}_{\text{start}}(f) + \alpha_n M + C\mathcal{C}_{n,m}(f).$$

Choose n sufficiently large that $\alpha_n < 1/2$. Then

$$M \leq \frac{\mathbb{E}\mathcal{Q}_{\text{start}}(f) + C\mathcal{C}_{n,m}(f)}{1 - \alpha_n}. \quad (7.16)$$

Now apply the same integrated estimates to the given g and substitute Equation (7.16). The resulting increase is at most

$$c_g^2 \left(\frac{\alpha_n}{1 - \alpha_n} \mathbb{E}\mathcal{Q}_{\text{start}}(f) + \frac{C}{1 - \alpha_n} \mathcal{C}_{n,m}(f) \right).$$

This proves the theorem with a decreasing majorant of $\alpha_n/(1 - \alpha_n)$ as ζ_n . All bounds are uniform over the specified levels and masks. For a complementary pass the same proof starts with that pass's own initial energy; already performed changes in the other region are simply fixed exterior data in each local comparison. $\square$

7.3 The two useful scales of the additive charge

Lemma 7.5 (Charge tails). *Put $r_n = \sum_{l=n}^{\infty} q_l p_l$, so that $r_n \rightarrow 0$.*

(i) *If f has global lattice Lipschitz constant at most K , then*

$$b_N^{-2} \mathcal{C}_{n,m}(f) \leq CK^2 r_n, \quad n < m \leq N.$$

(ii) *Suppose f vanishes outside a square of side at most AR and has Lipschitz constant at most K/R . If $b_m \leq R$ and the square is interpreted in a coordinate neighborhood of the torus, then*

$$\mathcal{C}_{n,m}(f) \leq C_A K^2 r_n.$$

Both assertions hold for offset-dependent locations and prescriptions, and for real Euclidean-vector data.

Proof. There are b_N^2/b_l^2 level- l tiles. The diameter of an immediate parent is at most Cb_{l+1} in the lattice distance. In the first case, $h_D^2 \leq CK^2b_{l+1}^2$. The contribution of this level, divided by area, is at most

$$CK^2q_l p_l \frac{b_{l+1}^2}{b_l^2} \leq CK^2q_l p_l.$$

Sum over l .

In the second case, parents missing the support have zero oscillation. At level l , at most $C_A(1 + R/b_l)^2$ children have a parent meeting the support. Also $h_D^2 \leq CK^2b_{l+1}^2/R^2$. Thus the entire level contributes at most

$$C_A K^2 q_l p_l (1 + R/b_l)^2 \frac{b_{l+1}^2}{R^2} \leq C_A K^2 q_l p_l,$$

because $b_{l+1} \leq b_m \leq R$. Summation proves the second claim. These are deterministic counts conditional on the offset, so averaging does not change the bounds. $\square$

8 Screening and affine energies at a fixed stage

Throughout this section the starting level n is fixed and sufficiently large. Constants bearing a subscript n may depend arbitrarily on n , as well as on the fixed h, E . They do not depend on the torus size once that size is sufficiently large. We use the stage- n construction of [Proposition 7.1](#). We also allow the prescribed artificial probes: at each prescribed site there is at most one such probe, of strength τ or $\tau\vartheta(v_x)$, where $0 \leq \tau \leq 1$. A mask specifies which of these choices is made at each site. Masks, observation grids, and test centers may depend on the hierarchy offset and on external independent randomness, but not on the potentials or on the branch decisions. When conditioning on the offset below, we also fix any such external randomness. Uniform assertions about these prescriptions mean estimates for every individual prescription, not a single event on which all prescriptions work simultaneously.

Write $\Lambda_r(x)$ for the max-norm ball of radius r , with shortest coordinate distances on the torus. A tile of level $l > n$ is *hazardous* if its selector is present, regardless of acceptance, or if one of its own corner labels fails the level- l validity test. The construction gives, conditional on the offset,

$$\mathbb{P}(U \text{ is hazardous} \mid \text{offset}) \leq Cb_l^{-K_0}, \quad K_0 = 10^6. \quad (8.1)$$

Consequently, for a square of radius A , the probability that a hazardous tile of side greater than D meets that square is at most

$$C \sum_{b_l > D} (1 + A/b_l)^2 b_l^{-K_0}. \quad (8.2)$$

All the counting estimates below are used within a small fixed fraction of the torus side, where squares may be unwrapped in the usual way.

8.1 Prospective observations and a generic observation bound

Choose a large power of three Q_1 , and tile the torus by observation cores of side Q_1 . Set $\beta = 1/12$ and require $Q_1^\beta \geq b_n$. For a core centered at z , test candidate level- n roots in $\Lambda_{3Q_1}(z)$ using the prospective rule from [Proposition 7.1](#): retain all local acceptance rules, but omit ancestor selectors and validity tests at sides exceeding Q_1^β . The resulting test for one candidate uses randomness within distance CQ_1^β of that candidate, conditional on the offset.

Each test succeeds with probability at least $cq_n p_n > 0$. To see this, first expose the required selector statuses and the needed corner-box identities. Conditional on the candidate being eligible and all required labels being valid, its acceptance likelihood integrates to one, so its selector and acceptance contribute $q_n p_n$. The probability of the required strict-ancestor selector absences and the own and ancestor validity conditions is bounded below by a positive absolute constant: their failures have a summable union bound along the ancestor chain, made small by the choice of the starting level. This also explains why the local likelihood is not additionally conditioned on finer corner data.

Cover $\Lambda_{2Q_1}(z)$ by mesh squares of side comparable to $Q_1^{1/4}$. In each such square choose candidates separated by a sufficiently large multiple of Q_1^β . There are at least $cQ_1^{2(1/4-\beta)} = cQ_1^{1/3}$ candidates with disjoint randomness neighborhoods, for sufficiently large Q_1 at fixed n . Their success indicators are independent conditional on the offset. The probability that some mesh square contains no successful root is therefore at most

$$CQ_1^{3/2} \exp(-c_n Q_1^{1/3}). \quad (8.3)$$

In particular, with arbitrarily high probability, the prospective observation roots form a $CQ_1^{1/4}$ -net of $\Lambda_{2Q_1}(z)$, also for nearest-neighbor graph distance after changing the absolute constant.

The next argument applies to every possible observation set with this net property. Its equations always use the true potentials.

Lemma 8.1 (Generic local observation). *For sufficiently large Q_1 , almost every choice of the true potentials on $\Lambda_{3Q_1}(z)$ has the following property. For every set $\mathcal{O} \subset \Lambda_{3Q_1}(z)$ that is a $CQ_1^{1/4}$ -net of $\Lambda_{2Q_1}(z)$, there is a finite constant C_{obs} such that*

$$\|u|_{\text{core}}\| \leq C_{\text{obs}} \left(\|(H - E)u|_{\Lambda_{3Q_1}(z)}\| + \|u|_{\mathcal{O}}\| \right) \quad (8.4)$$

for every vector on $\Lambda_{3Q_1+1}(z)$. After Q_1 is fixed, a deterministic C_{obs} can be chosen so that this bound holds for every such set $\mathcal{O}$ outside an event of arbitrarily small probability. The exceptional probability is uniform in the offset and in all sufficiently large tori.

Proof. It suffices to show that a vector annihilated by both constraints on the right of Equation (8.4) vanishes on the core. All these constraints are real, so a counterexample would have a real counterexample. Fix its support pattern $\mathcal{P} \subset \Lambda_{3Q_1+1}(z)$ and an observation-set pattern $\mathcal{O}$, and suppose that $\mathcal{P}$ meets the core. Set $\ell = \lceil C' Q_1^{1/4} \rceil$, with C' large enough for the graph-distance net property.

For any fixed sufficiently small $c_0 > 0$, some integer $r \in [Q_1, 2Q_1]$ satisfies

$$|\mathcal{P} \cap (\Lambda_{r+1}(z) \setminus \Lambda_{r-\ell}(z))| \leq c_0 \ell^{-2} |\mathcal{P} \cap \Lambda_{r-\ell}(z)|. \quad (8.5)$$

Indeed, if this failed at every radius, apply its reverse inequality at radii separated by $\ell + 1$. At each step the support count would grow by a factor greater than $1 + c_0 \ell^{-2}$. The first inner square contains the nonempty core support, and there are at least cQ_1/ℓ steps. The resulting count would be at least $\exp(cQ_1/\ell^3)$, whereas the available neighborhood has only $O(Q_1^2)$ sites. Since Q_1/ℓ^3 grows as a positive multiple of $Q_1^{1/4}$, this is impossible for large Q_1 .

For a radius satisfying Equation (8.5), write

$$d'_0 = |\mathcal{P} \cap \Lambda_{r-\ell}(z)|, \quad p = |\mathcal{P} \cap \Lambda_r(z)|, \quad m = |\mathcal{P} \cap \Lambda_{r+1}(z)|.$$

Every deep support site $x \in \mathcal{P} \cap \Lambda_{r-\ell}(z)$ has an observed zero within graph distance ℓ . Follow a shortest path to such a zero and take its first vertex outside $\mathcal{P}$. This is a missing site in $\Lambda_r(z)$ adjacent to the support. At that site the equation $(H - E)u = 0$ is a nonzero homogeneous linear equation in the m coordinates on $\mathcal{P} \cap \Lambda_{r+1}(z)$; its diagonal term vanishes.

Select a maximal collection of these missing sites with pairwise graph distance greater than two. Their neighbor sets are disjoint, and each row is nonzero, so the corresponding linear equations are independent. By maximality, each deep site is within distance $\ell + 2$ of a selected missing site. A graph ball of this radius has at most $C\ell^2$ sites. Thus the number k of independent equations satisfies

$$k \geq c\ell^{-2}d'_0.$$

On the other hand, $m - p \leq c_0\ell^{-2}d'_0$ by Equation (8.5). Choosing $c_0 < c$ gives

$$m - k < p. \tag{8.6}$$

The equations at the p nonzero sites in $\Lambda_r(z)$ determine their true potentials through

$$v_x = E - \frac{\sum_{y: |y-x|_1=1} u_y}{u_x}.$$

The admissible nonzero coordinates lie in a linear space of dimension at most $m - k$. On the set where the relevant denominators are nonzero, these formulas define a smooth map into $\mathbb{R}^p$. Cover its domain by countably many bounded patches on which the denominators are bounded away from zero. On each patch the map is Lipschitz, and its image is Lebesgue null by Equation (8.6). For completeness, a bounded patch in dimension $a < p$ may be covered by $O(\varepsilon^{-a})$ cubes of side ε ; their Lipschitz images have total p -dimensional covering volume $O(\varepsilon^{p-a})$, which tends to zero. The joint law of the p true potentials has a density, so this support and observation pattern has probability zero. There are only finitely many patterns and admissible radii. Taking their union proves the kernel assertion, including for an observation set chosen from the potentials.

Let A be the map consisting of the residual and observation constraints, and let B be restriction to the core. The assertion $\ker A \subset \ker B$ makes B a well-defined linear map on $\text{im } A$ through A . Finite dimensionality gives a finite operator bound, which is Equation (8.4); complex vectors obey the same bound by complexification. The maximum of these finite bounds over all possible net sets is finite almost surely. Taking a sufficiently high quantile gives a deterministic C_{obs} . The local true product law is unchanged by offset or by passage to a larger torus, proving the last assertion. No deterministic unique-continuation assertion is used. $\square$

8.2 Projected resolvents and shield comparisons

The decay estimate below uses an exponential-conjugation argument of Combes–Thomas type [8]. We first establish the projected inverse bound on the observed coordinates; no bound on the full inverse is required.

Call an observation core *good* if the prospective roots form the specified net, the deterministic observation bound holds, and no hazardous tile of side greater than Q_1^β meets Λ_{4Q_1} about its center. The prospective roots of a good core are actual stage- n ports: all omitted ancestor requirements hold. Moreover, every real reference change in its tested rows occurs at a cell port of side at most Q_1^β . Such a port has strength bounded below by a positive number depending on Q_1 .

We may count only a subset of these good cores. In a partly thinned or Dirichlet-restricted system, we count only cores for which the physical neighborhood $\Lambda_{3Q_1+1}(z)$ remains present, and for which all cell ports, their strengths, and the real reference values in the tested rows $\Lambda_{3Q_1}(z)$

agree with the full stage- n system. We call these tests *intact*. The neighbor layer preserves every hopping term in the true equations; artificial probes may still have any of the allowed masks. Let J_g be the coordinate embedding of the sites in the counted cores, and set

$$\Pi = [J_g \ Y], \quad G = (H^\# - E - iYY^\top)^{-1},$$

where Y includes every current port, whether or not it lies in a counted core.

Proposition 8.2 (Projected screening). *For all the systems just described,*

$$\|\Pi^\top G \Pi\| \leq C_n. \quad (8.7)$$

Give cost one to nearest-neighbor edges whose endpoints both belong to counted good cores, and cost zero to other edges; any of the positive costs may also be dropped. If two source and response sectors of Π have cost distance at least t , their resolvent block has norm at most $C_n \exp(-c_n t)$.

Suppose, further, that two systems agree on an inner region and a surrounding shield, including all ports there, and that counted good tests in the shield are usable in both systems. Count costs only on shared good-good edges in the shield. If every crossing of the shield from the inner region to the exterior costs at least t , then the responses between common inner ports differ by at most $C_n \exp(-c_n t)$ in operator norm. This includes comparison with a Dirichlet restriction buffered beyond the shield.

Proof. Put $u = G \Pi w$, and abbreviate $A = \|J_g^\top u\|$, $B = \|Y^\top u\|$, and $W = \|w\|$. In each tested neighborhood the true residual is

$$(H - E)u = \Pi w + (H - H^\#)u + iYY^\top u.$$

There are at most two ports at a site and their strengths are uniformly bounded, so $\|Y\|$ is uniformly bounded. The real changes are bounded and, within these rows, supported at cell ports with the positive strength lower bound just noted. Thus their contribution is bounded by $C_n B$. The observed values obey the same bound, using the accepted level- n ports. The testing neighborhoods have bounded overlap. Summing the squared observation inequalities gives

$$A \leq C_n(W + B). \quad (8.8)$$

Taking the imaginary part of the inverse equation, or using the Ward identity of [Lemma 3.1](#), yields

$$B^2 \leq (A + B)W \leq C_n(W + B)W.$$

This implies $B \leq C_n W$, and then $A \leq C_n W$ by [Equation \(8.8\)](#). It proves [Equation \(8.7\)](#) without a bound on the full inverse.

Let F be a real function on physical sites with $|F(x) - F(y)|$ at most the assigned cost of every nearest-neighbor edge. Conjugate the site matrix by multiplication by e^{sF} . Diagonal entries do not change, and every changed hopping entry has both endpoints among the counted good sites. Hence

$$e^{sF}(H^\# - E - iYY^\top)e^{-sF} = H^\# - E - iYY^\top + J_g D_s J_g^\top, \quad \|D_s\| \leq C(e^s - 1).$$

For a sufficiently small fixed $s > 0$, depending on n , we have $\|D_s J_g^\top G J_g\| < 1/2$. The inverse difference formula and [Equation \(8.7\)](#) therefore bound the conjugated projected inverse by another

C_n . Assign to each column of Π the value of F at its physical site, so that repeated columns at a site have the same weight. If F_Π is this diagonal matrix, the conclusion is

$$\left\| e^{sF_\Pi} \Pi^\top G \Pi e^{-sF_\Pi} \right\| \leq C_n. \quad (8.9)$$

Choose F to be cost distance from the source sector. Restriction of Equation (8.9) to a response sector at distance t gives the stated exponential attenuation.

For the two-system comparison, use this same distance from the inner region, with costs counted only in the shield. For t larger than an absolute constant, choose a multiplier χ equal to one where $F \leq t/3$, zero where $F \geq 2t/3$, and affine in between. It is supported in the common region, and changes only across counted good-good edges. Cut off the first system's columns sourced at common inner ports by χ . In the second inverse equation the intended source is unchanged, and the only error is a hopping commutator supported on good coordinates. Each edge of that commutator has $F \geq t/3 - 1$ at both endpoints. The weighted bound in the first system makes the error, as a good-coordinate source, at most $C_n e^{-c_n t}$ in operator norm. Correcting it with the second system's projected inverse costs at most another C_n . Taking the common inner-port response proves the comparison. Bounded t is covered by the unweighted projected bounds after enlarging the constant. The same calculation applies to a buffered Dirichlet restriction, because the cutoff vanishes before its boundary. $\square$

8.3 Crossing costs and the wall variant

By Equation (8.3) and Lemma 8.1, the probability of a local test failure can be made arbitrarily small, first by taking Q_1 large and then C_{obs} large. The additional hazard cost for one core is at most

$$C \sum_{b_l > Q_1^\beta} (1 + Q_1/b_l)^2 b_l^{-K_0},$$

which tends to zero as Q_1 grows. Consequently, in any neighborhood of radius a fixed multiple of Q_1 , the probability that some core meeting that neighborhood is bad is arbitrarily small. All these probabilities are conditional on the offset, uniformly in that offset.

Proposition 8.3 (Full-stage path estimate). *There is an absolute $\alpha > 0$ such that, for the full stage- n good-core costs and all sufficiently large r up to a small fixed fraction of the torus side,*

$$\mathbb{P} \left(\begin{array}{c} \text{some nearest-neighbor crossing from } \Lambda_r(x) \text{ to } \Lambda_{2r}(x)^c \\ \text{has cost less than } c_n r^\alpha \end{array} \middle| \text{offset} \right) \leq C_n r^{-50}. \quad (8.10)$$

Only costs of edges with both endpoints in the intervening band need be counted. The constants are uniform in the center and the allowed prescriptions.

Proof. Fix a sufficiently large absolute integer B_1 , and use the integer radii

$$r_0 = 100Q_1, \quad r_j = B_1^j r_0,$$

as long as $r_j \leq b_N/100$. Let p'_j be the worst conditional probability, over offsets, centers, and sufficiently large tori, of a crossing of the r_j -band with cost less than 2^j . Trim a crossing at its two radial boundaries, so that the tested segment lies entirely in the band.

A crossing of the r_{j+1} -band with cost less than 2^{j+1} forces two failures at scale r_j at well-separated centers from a bounded mesh list. Indeed, take successive visits to radii rounded from

$5r_{j+1}/4$, $3r_{j+1}/2$, and $7r_{j+1}/4$, and replace each visited site by a mesh center within distance r_j . For B_1 large enough, the corresponding r_j -annuli lie in the larger band, are pairwise disjoint, and are crossed by subsegments of the original path. Their centers are more than $20r_j$ apart. If two of these crossings each had cost at least 2^j , the original cost would be at least 2^{j+1} . Thus at least two are failures. The number of possible mesh-center pairs is an absolute constant after B_1 has been fixed.

Temporarily truncate the hazard tests used for these smaller annuli at side r_j . Each resulting failure event then depends only on randomness within distance at most $5r_j$ of its center. This includes the local prospective tests, the true observation equations, and the complete corner triples and selectors of every tested tile. The two separated events are independent conditional on the offset. Truncation can only enlarge the good set, so each truncated failure probability is at most p'_j . If no omitted hazardous tile meets the needed neighborhoods, the truncated and full costs coincide. The union bound Equation (8.2) therefore gives

$$p'_{j+1} \leq C(p'_j)^2 + C \sum_{b_l > r_j} (1 + r_{j+1}/b_l)^2 b_l^{-K_0} \leq C(p'_j)^2 + Cr_j^{-K_0}. \quad (8.11)$$

The constants here may depend on the fixed absolute B_1 .

Choose a small absolute $a > 0$ with $Ca \leq B_1^{-50}/2$. The base tests above allow $p'_0 \leq a$. Increase Q_1 , if necessary, so that the hazard term at the base is at most $aB_1^{-50}/2$. Since $K_0 > 50$, induction in Equation (8.11) gives

$$p'_j \leq aB_1^{-50j}.$$

Set $\alpha = \log_{B_1} 2$; thus $2^j = (r_j/r_0)^\alpha$. This proves the assertion at recursion radii. For a general sufficiently large radius r , take the largest r_j with $r_j \leq r/100$. A crossing of the r -band has a translated r_j -band crossing wholly inside it, at one of a bounded mesh list of centers. Here r_j is comparable to r , with absolute comparison constants. The union bound gives Equation (8.10), after changing c_n, C_n . This construction also covers noninteger radii and uses no costs at the outer band boundaries. $\square$

Proposition 8.4 (Screening on retained walls). *Fix an aligned square grid of side L_* , deterministic given the offset. Suppose a system retains the full stage- n data on a wall set containing strips of width δL_* along every grid boundary, where $\delta > 0$ is a fixed absolute constant. Count only good cores whose centers are at least $5Q_1$ inside this wall set, so that all their tests are intact. There is an absolute constant A_w such that, for L_* sufficiently large at fixed n and*

$$A_w L_* \leq r \leq cb_N,$$

every crossing of the r -band has wall cost at least $c'_n r^\alpha$ outside an event of conditional probability at most $C'_n r^{-30}$. Here c'_n, C'_n and the threshold on L_ may depend on n ; A_w depends only on the absolute strip fraction and geometric constants. In particular, the constants in the bound are uniform as L_* grows. As before, only edges within the crossed band need be counted.*

Proof. It suffices to count good cores whose centers are at least $5Q_1$ inside the prescribed boundary strips. These cores are among those allowed in the statement, and dropping the other costs can only decrease crossing costs. This counted geometric region is deterministic conditional on the offset.

Let $r_0^w = KL_*$ for a sufficiently large fixed absolute K . A crossing of its band contains a smaller annulus crossing deep inside one wall strip. To verify this, unwrap coordinates about the large-band center x and stop the path when it first reaches radius approximately $7r_0^w/4$. One of its signed coordinates attains this value. Between signed coordinate distances $5r_0^w/4$ and $3r_0^w/2$ from x , choose

a parallel grid boundary line. Such a line exists because K is large. The path must visit this line, while its other coordinate has modulus at most $7r_0^w/4 + 1$.

Put $\rho = \lfloor c_w L_* \rfloor$ for a small fixed $c_w > 0$. Choose a mesh center within $\rho/10$ of this visit. The entire ρ -annulus about that center lies both inside the original large band and inside the wall strip; choosing c_w sufficiently small and then L_* sufficiently large also leaves the $5Q_1$ buffer for all relevant core tests. By its later visit to radius $7r_0^w/4$, the path has left the small annulus. It therefore provides the claimed crossing. Since r_0^w/ρ is bounded by an absolute constant, only boundedly many mesh centers are needed. Applying [Proposition 8.3](#) at those centers gives a base wall threshold $a_n L_*^\alpha$, with failure probability at most $C_n L_*^{-50}$.

Run the same recursion as in [Equation \(8.11\)](#), now with the deterministic wall restriction at every scale. The three-annulus geometry is unchanged. After truncating hazards, separated events still use disjoint randomness neighborhoods, since the counted strips add no randomness conditional on the offset. At radius $r_j^w = B_1^j K L_*$ the threshold is

$$2^j a_n L_*^\alpha = a_n K^{-\alpha} (r_j^w)^\alpha.$$

The failure recurrence is again a square of the preceding probability plus $O((r_j^w)^{-K_0})$. For a sufficiently large lower threshold on L_* , its base bound $C_n L_*^{-50}$ and this recurrence imply $C'_n (r_j^w)^{-30}$ at all recursion radii: at a new radius $B_1 r$, the two terms are $O_n(r^{-60})$ and $O(r^{-K_0})$, which are bounded by $C'_n (B_1 r)^{-30}$ once the base radius is large. This induction uses constants independent of L_* . Finally use a comparable smaller recursion scale and a bounded mesh list inside a general band, as in [Proposition 8.3](#). Taking $r \geq 100r_0^w$ suffices. This is an absolute multiple of L_* , as claimed. $\square$

8.4 Jump tails and fixed-stage limits

At full stage n , with the possible extra probes, define

$$d_{n,N}(x) = \frac{1}{2} \sum_{a: \text{site}(a)=x} \sum_j |S_{ja}|^2 \text{dist}_2(x, \text{site}(j))^2. \quad (8.12)$$

Here dist_2 is Euclidean distance formed from the shortest coordinate displacements. Let $d_{n,N}^{\leq D}(x)$ and $d_{n,N}^{> D}(x)$ denote the same sum restricted respectively to jump lengths at most D and greater than D .

Proposition 8.5 (Fixed-stage limits). *For each fixed sufficiently large n there is a finite number $d_n \geq 0$ with the following properties. The jump tails satisfy*

$$\sup \mathbb{E} \left[d_{n,N}^{> D}(x) \mid \text{offset} \right] \leq \omega_n(D), \quad \omega_n(D) \rightarrow 0, \quad (8.13)$$

uniformly in the site, sufficiently large tori, prescribed masks, and $0 \leq \tau \leq 1$. For a deterministic site x , with expectation including the uniform offset,

$$\mathbb{E} d_{n,N}(x) \rightarrow d_n \quad (N \rightarrow \infty, \tau \rightarrow 0), \quad (8.14)$$

uniformly in the prescriptions. If A is a growing square, whose position may be chosen from the offset and external independent randomness, then

$$\frac{1}{|A|} \sum_{x \in A} d_{n,N}(x) \rightarrow d_n \quad \text{in mean and in probability.} \quad (8.15)$$

The estimates for this convergence are uniform conditional on the offset, over such square positions and prescriptions, as the square side and the torus size grow and $\tau \rightarrow 0$. Bounded-length jumps capture these spatial averages to any given tolerance with arbitrarily high probability if the jump cutoff is chosen first and sufficiently large.

Proof. At most two ports occur at a site: one accepted cell root and one artificial probe. Thus the source rank in Equation (8.12) is at most two. For large D , use a separating annulus about x with radius comparable to $D/200$. Its cost is at least $c_n D^\alpha$ outside an event of conditional probability at most $C_n D^{-50}$, by Proposition 8.3. The projected attenuation in Proposition 8.2 applies from the source ports at x to all ports at distance greater than D simultaneously. Unitarity bounds that block on the exceptional event. The source-rank bound gives

$$\mathbb{E} \left[\sum_{a: \text{site}(a)=x} \sum_{j: \text{dist}_2(x, \text{site}(j)) > D} |S_{ja}|^2 \middle| \text{offset} \right] \leq C_n (e^{-c_n D^\alpha} + D^{-50}). \quad (8.16)$$

Multiply the bound at dyadic distances $2^q D$ by $C(2^q D)^2$ and sum. The series is convergent and tends to zero with D , proving Equation (8.13). This also controls jumps comparable to the torus side: their separating annulus can still be taken at radius $D/200$, within the allowed small torus fraction. There is no final unestimated shell.

We next construct local approximants, keeping n fixed. Fix a jump cutoff D . First restrict the effective site data to a Dirichlet square of a sufficiently large fixed radius R_{loc} about x . A shield buffered inside this square encloses x and all targets at distance at most D . The shield comparison of Proposition 8.2 and the path estimate give arbitrarily small expected error in the truncated sum as R_{loc} grows. On its exceptional event use unitarity; the truncated sum is bounded by CD^2 . The comparison concerns a block with source rank at most two, so its error does not acquire a factor from the number of remote ports.

Next choose a fixed superscale b_k much larger than R_{loc} and omit all hierarchy rules above that scale when forming the restricted effective data. Use the prospective ancestor convention, the same true potentials, and the same selectors and acceptance uniforms below the cutoff. The restricted effective data agree with the original data unless an omitted hazardous tile meets the restriction. Indeed, absent such a hazard, no omitted selector can suppress a retained root, and every omitted ancestor validity gate is satisfied. The local likelihoods do not otherwise use ancestor-label identities. The probability of disagreement is bounded by

$$C \sum_{b_l > b_k} (1 + R_{\text{loc}}/b_l)^2 b_l^{-K_0},$$

which tends to zero with k . A retained reference value and its likelihood use only their finite tile. The resulting truncated Dirichlet quantity therefore depends on randomness within distance $O(R_{\text{loc}} + b_k)$ of x .

For this fixed finite approximant we may remove artificial probes by sending τ to zero. To keep its channel space fixed, pad it with one artificial channel at every site of the restriction and give absent channels strength zero. Such a column is zero in Y and contributes only an isolated identity entry to S . Its inverse at $\tau = 0$ exists almost surely: a kernel vector of the passive matrix vanishes at cell-port sites, where all real reference changes are supported, and hence would be a kernel vector of the true Dirichlet restriction minus E . The latter has nonzero determinant almost surely by the joint diagonal density. Inverse continuity now gives $S \rightarrow S_{\text{cell}} \oplus 1$ on this fixed channel space. In particular, the cell-port scattering block converges. Mixed entries involving an artificial probe tend to zero, as do off-diagonal entries between artificial probes. Their diagonal entries do not contribute

to Equation (8.12), since their jump length is zero. Bounded convergence applies to the truncated sums, which are uniformly bounded by CD^2 . There are only finitely many local masks, including the choices of constant or profile strength, and finitely many offset patterns modulo the fixed superscale. Taking a maximum over these finitely many choices makes this convergence uniform. This step does not require any uniform lower bound for $\vartheta(v_x)$.

Combining these constructions with the jump-tail bound proves the following useful precise statement. For every $\varepsilon > 0$ there is a bounded no-extra local function Z_x , with fixed dependence radius $\rho_{\text{loc}} = O(R_{\text{loc}} + b_k)$, such that

$$\mathbb{E}[|d_{n,N}(x) - Z_x| \mid \text{offset}] < \varepsilon \quad (8.17)$$

for every sufficiently large N and sufficiently small τ , uniformly in masks, offsets, and sites. The choices of D , R_{loc} , and k are made in that order, before these limits.

Conditional on the offset, the Z_x have finite dependence range: two collections farther apart than $2\rho_{\text{loc}}$ are independent. Their conditional means are periodic at the superscale b_k , and the pattern for a different offset is a translate of the same pattern. Thus their period-averaged mean, say μ_Z , is independent of offset. It is also their mean at any deterministic site after averaging over the uniform offset. For all sufficiently large tori these local laws are identical. By Equation (8.17), the numbers $\mathbb{E}d_{n,N}(x)$ are Cauchy in the joint limit, uniformly in the prescriptions. Their finite limit is d_n , and $|\mu_Z - d_n| \leq \varepsilon$.

For a square A of side a , covering its interior by complete b_k -periods gives

$$\left| \frac{1}{|A|} \sum_{x \in A} \mathbb{E}[Z_x \mid \text{offset}] - \mu_Z \right| \leq C \|Z\|_{\infty} \frac{b_k}{a}.$$

Counting pairs with nonzero conditional covariance also gives

$$\text{Var} \left(\frac{1}{|A|} \sum_{x \in A} Z_x \mid \text{offset} \right) \leq C \|Z\|_{\infty}^2 \frac{\rho_{\text{loc}}^2}{a^2}.$$

Both bounds hold for every square position, including one chosen from the offset. The same bounded-range counting applies on the torus, with periodic distances. Average Equation (8.17) over A , take the square and torus sizes large, and then send ε to zero. This proves the asserted conditional uniform convergence in mean and hence in probability. Notice that a conditional point mean need not equal d_n ; it is the period average that removes its dependence on the offset pattern. Finally, the expected difference between the full and truncated spatial averages is at most $\omega_n(D)$, even conditional on the offset. Markov's inequality, followed by Equation (8.15), proves the bounded-jump assertion. $\square$

8.5 Compact profiles and convergence of affine energies

Proposition 8.6 (Compact-profile estimate). *Let $\varphi : \mathbb{R}^2 \rightarrow \mathbb{R}^q$ be compactly supported and Lipschitz, piecewise smooth with bounded gradients and piecewise regular interfaces. Let its center c be prescribed from the offset and external independent randomness. At full stage n ,*

$$\limsup_{\substack{R \rightarrow \infty, \tau \rightarrow 0 \\ b_N/R \rightarrow \infty}} \mathbb{E} Q_{\text{start}}(\varphi((\cdot - c)/R)) \leq C d_n \int_{\mathbb{R}^2} |\nabla \varphi|^2 \, dx, \quad (8.18)$$

where C is absolute. Here n and the profile are fixed; the thresholds may depend arbitrarily on them. The estimate is uniform over the allowed masks and centers, and remains valid with the expectation conditioned on the offset.

More generally, let f be supported in a square of side at most AR whose position is prescribed from the offset and external independent randomness, and suppose that f has global Lipschitz constant at most B/R . Then

$$\mathbb{E} \mathcal{Q}_{\text{start}}(f) \leq C_{A,B}(d_n + o(1)) \quad (8.19)$$

in the same fixed- n regime, uniformly over all such data and prescriptions. Thus each assertion holds to any fixed additive tolerance for all sufficiently large spatial and torus scales and all sufficiently small τ ; no rate uniform in n is asserted.

Proof. Fix a large jump cutoff D . If a profile difference is nonzero, at least one endpoint belongs to its support. By symmetry of S , the contribution of jumps longer than D is bounded by

$$\frac{C \text{Lip}(\varphi)^2}{R^2} \sum_{x \in \text{supp } \varphi_R} d_{n,N}^{>D}(x), \quad \varphi_R(x) = \varphi((x - c)/R).$$

There are $O(R^2)$ sites in this support. Its expected value tends uniformly to zero as $D \rightarrow \infty$, by Equation (8.13).

For jumps of length at most D , the squared profile difference is bounded by R^{-2} times the squared jump length times a local supremum of $|\nabla \varphi|^2$. Away from the interfaces, these suprema converge to $|\nabla \varphi|^2$ in the Riemann-sum limit. The sites whose rescaled D -neighborhood meets an interface have vanishing relative area, and the global Lipschitz bound controls their contribution. Thus the short-jump energy is bounded by

$$\frac{C}{R^2} \sum_x w_R(x) d_{n,N}^{\leq D}(x),$$

where w_R is uniformly bounded, supported in an $O(R)$ square, and has Riemann-sum limit $\int |\nabla \varphi|^2$.

To justify averaging these weights also conditional on the offset, use the fixed local approximants from Equation (8.17). Their bounded periodic conditional means have period average arbitrarily close to d_n . On complete periods, the slowly varying Riemann-sum weights may be replaced by a representative value with error tending to zero; the incomplete boundary periods contribute vanishingly. The uniform conditional expected approximation error contributes at most $C\epsilon R^{-2} \sum_x w_R(x)$. This proves the bound for the short jumps after sending the spatial and torus scales to infinity and τ to zero. Then let the local approximation error and the jump-tail error tend to zero. All finite cutoffs and local neighborhoods were chosen at fixed n before taking these limits.

For the more general data, use the global Lipschitz bound directly. Symmetry charges every nonzero difference to the supporting square, so

$$\mathcal{Q}_{\text{start}}(f) \leq \frac{CB^2}{R^2} \sum_{x \text{ in the supporting square}} d_{n,N}(x).$$

The uniform spatial mean estimate in Proposition 8.5 now gives Equation (8.19). This argument only uses support size and the Lipschitz bound, so it is uniform for varying profiles with the stated bounds. $\square$

To approximate affine coordinate data on a torus without choosing a discontinuous coordinate branch, put

$$f_N(x) = \frac{b_N}{2\pi} (\cos(2\pi x_i/b_N), \sin(2\pi x_i/b_N))_{i=1,2} \in \mathbb{R}^4. \quad (8.20)$$

Its chord distances satisfy

$$\frac{2}{\pi} \text{dist}_2(x, y) \leq |f_N(x) - f_N(y)| \leq \text{dist}_2(x, y), \quad (8.21)$$

and their ratio to $\text{dist}_2(x, y)$ tends to one uniformly for bounded nonzero jumps as $N \rightarrow \infty$. Indeed, for shortest coordinate displacements δ_i , the squared chord distance is $\sum_i (b_N \sin(\pi \delta_i / b_N) / \pi)^2$. Truncate the jumps, use this local limit, and then use Equation (8.13) for the tails. Translation after offset averaging and Equation (8.14) give

$$b_N^{-2} \mathbb{E} Q_{S_{n,N}}(f_N) \rightarrow d_n. \quad (8.22)$$

There are no artificial probes in this identity.

Proposition 8.7 (Limit of the affine energy densities). *The fixed-stage limits satisfy*

$$d_n \rightarrow d \quad \text{for some } 0 \leq d < \infty. \quad (8.23)$$

More precisely, a deterministic $\varepsilon_n \rightarrow 0$ can be chosen such that

$$d_m \leq (1 + \varepsilon_n) d_n + \varepsilon_n \quad (m > n) \quad (8.24)$$

for all sufficiently large n , uniformly in m .

Proof. Apply Theorem 7.3 without extras to f_N in the full transition from stage n to stage m . The data have a uniformly bounded Lipschitz constant by Equation (8.21). The area-normalized charge estimate of Lemma 7.5 has a vanishing n -tail independent of m, N , as does the multiplicative error in the comparison. Divide by b_N^2 , fix $n < m$, and send N to infinity using Equation (8.22) for both stages. This proves Equation (8.24) with errors uniform in $m > n$; no limit uniform in the pair (n, m) was taken.

Fixing one sufficiently large n_0 in that inequality first bounds d_m for all $m > n_0$. Let d be the finite nonnegative lower limit of this bounded sequence and choose $n_k \rightarrow \infty$ with $d_{n_k} \rightarrow d$. For each k , Equation (8.24) gives

$$\limsup_{m \rightarrow \infty} d_m \leq (1 + \varepsilon_{n_k}) d_{n_k} + \varepsilon_{n_k}.$$

Sending $k \rightarrow \infty$ proves that the upper limit is at most d . This proves convergence. $\square$

The order of parameters in this section will be used explicitly below. One fixes n before choosing observation scales, quantiles, jump cutoffs, or local approximation radii. Their possibly very large values impose only fixed thresholds on subsequent spatial scales. With fixed additional parameters, one can then take $m > n$ so large that scales of order $\exp(b_m^2)$ exceed every such threshold. A probe strength $\tau = C_0/R$ at those scales tends to zero and eventually satisfies $\tau \leq \exp(-c_* b_m)$, as required by Theorem 7.3. Finally N is increased after these choices. Neither this procedure nor Equation (8.23) supplies or uses a rate of convergence uniform in n .

9 Vanishing of the limiting energy in two dimensions

Throughout this section, the disorder strength and the interior energy are fixed. All auxiliary constructions are those of the preceding sections. In particular, Proposition 8.7 provides finite numbers d_n with $d_n \rightarrow d \geq 0$. No convergence rate uniform in n is assumed.

Proposition 9.1. *The limiting affine energy vanishes: $d = 0$.*

Suppose for contradiction that $d > 0$. We first thin the gaps between retained inner squares and grid walls. A localized change of the affine data disappears after the complementary thinning pass. Since the expected initial and final affine energy densities approach the same limit, choosing the sign of this change forces its local first variation to be small in expectation.

We then place weak probes in a square annulus between one inner square and its wall. A logarithmic voltage profile makes total transmission to the probes small, so the cell-to-cell scattering block is nearly unitary. Its surviving internal jump energy gives an analytic trace a nonzero central value. Closing the probes identifies the boundary modulus with the preceding first variation evaluated after a support-preserving shift of the true potentials. A bounded cost for changing the law transfers smallness to boundary probabilities, and dimension-independent moment bounds turn this into a small boundary integral, contradicting submean.

9.1 Geometry and order of scales

Fix a number $M_0 > 20$, to be enlarged below. After choosing n , choose $m > n$ and a power of three L_* such that

$$e^{b_m^2} \leq L_* < 3e^{b_m^2}, \quad R = \frac{L_*}{M_0^2}. \quad (9.1)$$

The torus side b_N will be much larger than L_* . Since L_* is a power of three, the torus admits an aligned partition into grid tiles D of side L_* . The alignment is specified by the hierarchy offset. Write c_D for a tile center and use max-norm radii about this center.

Let I_D be the union of level- m tiles whose centers have radius at most R , and let O_D be the union of level- m tiles whose centers have radius at least $L_*/3$. Thus the inner square has radius $R + O(b_m)$, and the outer walls begin at radius $L_*/3 + O(b_m)$. The remaining level- m tiles form the gap. Thin levels $n, \dots, m-1$ in all gaps, leaving the inner squares and walls at stage n . The result is called the intermediate system. A complementary pass on the retained regions finishes the transition to uniform stage m . Both pass regions are unions of whole level- m tiles, as required by [Theorem 7.3](#).

Define the selector event

$$\mathcal{P}_D = \{\text{no selected tile of level at least } m \text{ meets } D\}. \quad (9.2)$$

This event involves selectors only. Conditional on the offset,

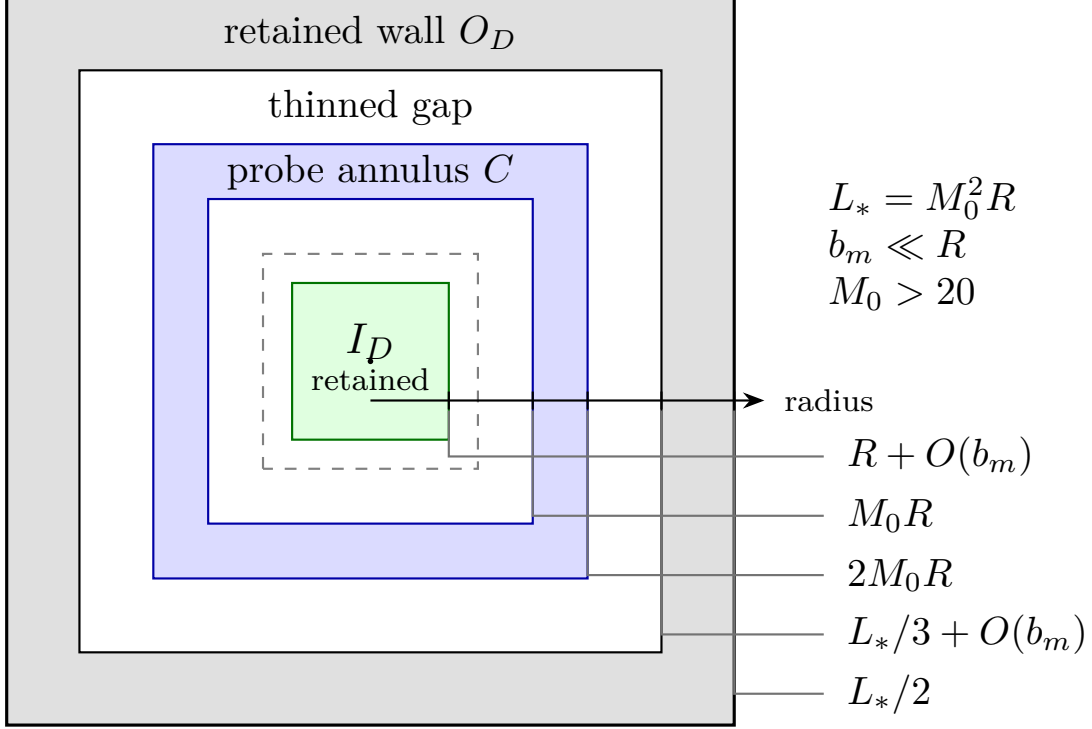
$$\mathbb{P}(\mathcal{P}_D^c \mid \text{offset}) \leq C \sum_{l \geq m} (1 + L_*/b_l)^2 q_l = o(1). \quad (9.3)$$

Indeed $q_l = e^{-b_l^6}$ and $L_* < 3e^{b_m^2}$, so the right side tends to zero faster than any fixed negative power of R . On $\mathcal{P}_D$, the intermediate cell ports in D lie only in $I_D \cup O_D$; after the complementary pass there are no cell ports in D .

Independently of the potentials, selectors, and acceptance variables, choose a uniformly random grid tile, conditional on the offset. For this test tile write $D, c, I, O, \mathcal{P}$ instead of $D, c_D, I_D, O_D, \mathcal{P}_D$. All expectations below include this choice. It is a permitted independent prescription for the screening and comparison estimates. The retained regions and the probe annulus used below are shown in [Figure 2](#).

9.2 An adaptive adjustment of the affine data

For the moment use no artificial probes. Let S' be the intermediate scattering matrix and let $f = f_N$ be the four-component torus embedding used to define the affine energy in [Proposition 8.7](#). Its chord



Dashed square: adjustment support ends at radius $2R$.

Figure 2: One grid tile in the intermediate system. The annulus is a square annulus, because all indicated radii use the max norm. The picture is schematic, not to scale: in the proof M_0 is large, so $R \ll M_0 R < 2M_0 R \ll L_*/3$. Inner and wall boundaries are rounded to unions of level- m tiles; the probe annulus is not so rounded. On $\mathcal{P}_D$, the gap has no cell ports, and the added probes occur only in the displayed annulus.

distances are globally comparable to shortest Euclidean torus distances, and its nearest-neighbor increments are bounded. Define four-component data on the intermediate cell ports by

$$F = \frac{f - f(c)}{R}, \quad F_I = \mathbf{1}_I F, \quad H^0 = \sum_{\nu=1}^4 \text{tr}[F_{I,\nu}(F_\nu - S' F_\nu S'^*)]. \quad (9.4)$$

As elsewhere, real scalar data inside a matrix expression denote the corresponding diagonal matrices. The trace H^0 is real, since every diagonal entry of $S' F_\nu S'^*$ is real. It is the local first variation of the affine energy: for fixed S' ,

$$\left. \frac{d}{dt} \mathcal{Q}_{S'}(f - tRF_I) \right|_{t=0} = -2R^2 H^0.$$

Here the perturbation is data on the intermediate ports. The proof below extends it by a cutoff to site data with bounded increments and chooses its sign separately in each grid tile. On the selector event, the affected ports disappear in the complementary pass.

Lemma 9.2. *Fix M_0 . For every $a > 0$, one can arrange*

$$\mathbb{E}[\mathbf{1}_{\mathcal{P}} |H^0|] \leq a \quad (9.5)$$

by taking n , then $m > n$, then N sufficiently large. More precisely, the estimate holds for all sufficiently large n , all sufficiently large m depending on n , and all sufficiently large N depending on the fixed pair (n, m) .

Proof. For each grid tile choose a cutoff η_D that is one on I_D , vanishes at radius $2R$, and has Lipschitz constant at most C/R . Such cutoffs exist once b_m/R is small. Define H_D^0 by Equation (9.4) with D as the test tile, and set

$$\varphi(x) = - \sum_D \mathbf{1}_{\mathcal{P}_D} \operatorname{sgn}(H_D^0) \eta_D(x) (f(x) - f(c_D)), \quad \operatorname{sgn}(0) = 0.$$

The supports are disjoint. On each support, $|f - f(c_D)| \leq CR$, so the product has uniformly bounded nearest-neighbor increments. Across support boundaries the same bound holds because the cutoff vanishes there. Using a shortest lattice path and global comparability of chord distances gives

$$|\varphi(x) - \varphi(y)| \leq C|f(x) - f(y)| \quad \text{for all torus sites } x, y. \quad (9.6)$$

This random adjustment is admissible at the start of the complementary pass in Theorem 7.3. The selector events may be used as pre-information. Also S' uses the calibrated reference value at every still-open root, not the hidden true value at that root. By Lemma 7.2, the disjoint eligible cells and their recorded statuses do not expose any of those hidden values. Thus H_D^0 and its sign are measurable with respect to the allowed pre-information. The adjustment does not use the subsequent thinning marks.

On $\mathcal{P}_D$, all intermediate ports meeting $\operatorname{supp} \eta_D$ lie in I_D , where $\eta_D = 1$. No final port meets this support. Hence $g = f + t\varphi$, $0 < t < 1$, agrees with f on the final stage- m ports. By symmetry and unitarity of S' , the quadratic energy expands as

$$\mathcal{Q}_{S'}(g) = \mathcal{Q}_{S'}(f) + 2t \sum_{\nu} \operatorname{tr}[\varphi_{\nu}(f_{\nu} - S'f_{\nu}S'^*)] + t^2 \mathcal{Q}_{S'}(\varphi).$$

One may subtract $f_{\nu}(c_D)$ in the trace belonging to a fixed tile, because S' is unitary. Its linear contribution is therefore $-2tR^2 \mathbf{1}_{\mathcal{P}_D} |H_D^0|$. There are b_N^2/L_*^2 grid tiles. Averaging the uniformly selected test tile and dividing by the torus area yields

$$\frac{\mathbb{E} \mathcal{Q}_{S'}(g)}{b_N^2} = \frac{\mathbb{E} \mathcal{Q}_{S'}(f)}{b_N^2} - 2tM_0^{-4} \mathbb{E}[\mathbf{1}_{\mathcal{P}} |H^0|] + t^2 \frac{\mathbb{E} \mathcal{Q}_{S'}(\varphi)}{b_N^2}. \quad (9.7)$$

By Equation (9.6), the last energy is at most a constant times $\mathcal{Q}_{S'}(f)$.

Apply Theorem 7.3 and Lemma 7.5 to the first pass with data f . Then apply the comparison to the complementary pass with data g , using the pointwise increment bound for g obtained from Equation (9.6). The allowed increment constant is uniform for $0 < t < 1$. The fixed-stage affine limits give

$$\begin{aligned} b_N^{-2} \mathbb{E} \mathcal{Q}_{S'}(f) &\leq d_n + \varepsilon_n + o_{n,m;N}(1), \\ b_N^{-2} \mathbb{E} \mathcal{Q}_{S'}(g) &\geq d_m - \varepsilon_n - o_{n,m;N}(1), \end{aligned}$$

where $\varepsilon_n \rightarrow 0$ uniformly in $m > n$. Here and below, $o_{n,m;N}(1)$ tends to zero as $N \rightarrow \infty$ with n, m fixed. Boundedness of d_n now gives, from Equation (9.7),

$$2tM_0^{-4} \mathbb{E}[\mathbf{1}_{\mathcal{P}} |H^0|] \leq Ct^2 + d_n - d_m + 2\varepsilon_n + o_{n,m;N}(1). \quad (9.8)$$

First take t sufficiently small that the quadratic contribution, after division by $2tM_0^{-4}$, is less than $a/2$. Since d_n is convergent, the remaining terms are less than the required tolerance for sufficiently large n, m, N in the stated order. This proves Equation (9.5). $\square$

9.3 An annulus of artificial probes

Add probes at all sites of

$$C = \{x : M_0 R \leq |x - c|_\infty \leq 2M_0 R\},$$

and nowhere else, using strength $\tau\vartheta(v_x)$, where

$$\tau = \frac{C_0}{R}. \quad (9.9)$$

The constant C_0 will be fixed before n, m, N are chosen. For sufficiently large m , Equation (9.1) implies $\tau \leq e^{-c_* b_m}$, so this mask is allowed throughout the first comparison pass. Use the same underlying hierarchy choices as before. The resulting intermediate scattering matrix has block form

$$S = \begin{pmatrix} A & T \\ T^\top & B \end{pmatrix}. \quad (9.10)$$

The first block consists of all cell ports, and the second consists of the new probes. We also use C for their index set. In this section I denotes the cell ports in the retained inner square and J denotes all other cell ports; these symbols are local to this section. On $\mathcal{P}$, every port in J is outside the probe annulus toward the exterior. Set

$$K = \|T\|_{\text{HS}}^2.$$

The data F, F_I still refer to Equation (9.4), now restricted to the cell ports of Equation (9.10).

We first give the deterministic use of a good annular barrier. A site is called good for this purpose when $\vartheta(v_x) \geq d_0$, where $d_0 > 0$ is fixed sufficiently small. Count only edges with both endpoints good and lying in a buffered square band strictly inside C , for example with both endpoint radii in $[1.2M_0 R, 1.8M_0 R]$. Give such an edge cost one, and all other edges cost zero. Denote by $\ell(x)$ the resulting cost distance from the region $|x - c|_\infty \geq 2M_0 R$. Let $\mathcal{B}$ be the event

$$\ell(x) \geq \gamma M_0 R \quad \text{whenever } |x - c|_\infty \leq M_0 R, \quad (9.11)$$

where $\gamma > 0$ is a sufficiently small fixed constant.

Lemma 9.3. *For suitable fixed $d_0, \gamma > 0$, $\mathbb{P}(\mathcal{B}^c) \leq C e^{-c M_0 R}$. On $\mathcal{B} \cap \mathcal{P}$, for every compatible matrix D' on the source columns indexed by J ,*

$$\|A_{IJ} D'\|_{\text{HS}} \leq \frac{C}{M_0 R \tau d_0} \|S_{CJ} D'\|_{\text{HS}}. \quad (9.12)$$

The constant includes the fixed factor γ^{-1} . The same cutoff argument controls every absorbed output supported where the cutoff equals one; it requires no norm bound on the exterior resolvent.

Proof. Since ϑ is positive in the interior of the single-site support, the iid probability $p_0 = \mathbb{P}\{\vartheta(v_x) < d_0\}$ tends to zero with d_0 . If Equation (9.11) fails, trim a cheap crossing to the buffered band and erase its loops. The resulting simple path has length $j \geq c M_0 R$. Choosing γ sufficiently small, a fixed positive fraction of its edges must touch bad sites. Since the path is simple, a fixed positive fraction of its distinct vertices are bad. There are at most $C(M_0 R)^2 4^j$ candidate paths of length j , and at most 2^{j+1} ways to designate a subset of their vertices. Independence therefore bounds the failure probability by

$$C(M_0 R)^2 \sum_{j \geq c M_0 R} 4^j 2^{j+1} p_0^{c' j}.$$

For sufficiently small fixed d_0 , this is exponentially small.

For the deterministic estimate, use the physical inverse G of the intermediate system with the probes inserted, and set $u = GY_J D'$. Define

$$\chi(x) = \min\{1, \ell(x)/(\gamma M_0 R)\}.$$

It is zero at the J sources and one at I . An edge across which χ changes is a counted edge; on that edge $|\chi(x) - \chi(y)| \leq (\gamma M_0 R)^{-1}$. Since diagonal terms commute with χ , the equation for χu has only the residual

$$q = [G^{-1}, \chi]u$$

and no original source. This residual is supported on the probe sites at endpoints of changing edges. If $\sigma_x = \tau \vartheta(v_x)$, write $q = Y_C a$ there. In each matrix entry of the map $Y_C^\top u \mapsto a$, the conversion introduces $\sigma_x^{-1/2} \sigma_y^{-1/2} \leq (\tau d_0)^{-1}$. The vertex degree is bounded, and hence

$$\|a\|_{\text{HS}} \leq \frac{C}{M_0 R \tau d_0} \|Y_C^\top u\|_{\text{HS}}. \quad (9.13)$$

By [Lemma 3.1](#), the norm of propagation from probe sources to all absorbed outputs is at most one. Since $\chi = 1$ at I ,

$$\|Y_I^\top u\|_{\text{HS}} = \|Y_I^\top G Y_C a\|_{\text{HS}} \leq \|a\|_{\text{HS}}.$$

The blocks in [Equation \(9.12\)](#) are off diagonal, so replacing the physical fields by their scattering blocks multiplies both sides by the same factor 2. This proves the assertion and also the stated all-output version of the cutoff argument. $\square$

The next event combines small total transmission with positive internal jump energy and a bound on transmission weighted by F . The weighted bound is needed because F grows on distant cell ports: small K alone does not control those coordinate factors in the trace.

Lemma 9.4. *Assume $d > 0$. There is a constant $c_1 > 0$ with the following property. For every sufficiently small fixed $\kappa > 0$, one can choose M_0 sufficiently large and finite constants C_1, C_F , before choosing C_0, n, m, N , so that the following holds. For any fixed sufficiently large C_0 , and then n, m, N sufficiently large successively, an event $\mathcal{G}$ of probability at least $1/2$ satisfies*

$$\mathcal{G} \subset \mathcal{P} \cap \mathcal{B}, \quad K \leq \kappa, \quad (9.14)$$

$$c_1 d \leq \sum_{i,j \in I} |F(i) - F(j)|^2 |A_{ij}|^2 \leq C_1, \quad (9.15)$$

$$\|F_\nu T\|_{\text{HS}} \leq C_F \quad (1 \leq \nu \leq 4). \quad (9.16)$$

In particular, on this event,

$$\|A_{IJ}\|_{\text{HS}} + \sum_{\nu=1}^4 \|A_{IJ} F_\nu|_J\|_{\text{HS}} \leq \frac{C(\sqrt{\kappa} + C_F)}{M_0 C_0 d_0}. \quad (9.17)$$

The right side tends to zero as C_0 increases, with all constants chosen before it kept fixed.

Proof. Fix small exception budgets for each of the following assertions, with their sum less than $1/2$. All probability bounds are unconditional; in particular, none of the screening or spatial-average estimates is conditioned on $\mathcal{P}$. Let D_* be a finite upper bound for all d_n after increasing the starting level if necessary.

Small total transmission to the annulus. Choose voltage one on C , zero within radius $2R$ and at radii at least $L_*/4$, with logarithmic ramps in the intervening regions. For example, as a function of $r = |x - c|_\infty$, the inner ramp is $\log(r/(2R))/\log(M_0/2)$ and the outer ramp is $\log((L_*/4)/r)/\log(M_0/8)$. It has a compact profile at scale R and

$$\int_{\mathbb{R}^2} |\nabla \varphi|^2 \, dx \leq \frac{C}{\log M_0}. \quad (9.18)$$

Indeed the area element for square shells is a constant times $r \, dr$, while the gradient on either ramp is at most $C/(r \log M_0)$. On $\mathcal{P}$, all cell-port voltages are zero, and all probe voltages are one. The intermediate energy is therefore exactly K : each cell-to-probe pair is counted once by the symmetric energy.

Use [Proposition 8.6](#) for the starting stage- n energy and [Theorem 7.3](#) and [Lemma 7.5](#) for the first pass. For fixed M_0, C_0, n , with m, N subsequently large, this gives

$$\mathbb{E}[\mathbf{1}_{\mathcal{P}} K] \leq \frac{CD_*}{\log M_0} + o_{n \rightarrow \infty}(1) + o_{m, N \rightarrow \infty; n, M_0, C_0}(1).$$

Choose M_0 large enough for the first term to meet the assigned Markov bound at threshold κ , then take n, m, N large in order. Together with [Equation \(9.3\)](#), this proves [Equation \(9.14\)](#) outside the assigned exception.

Internal jump energy. Inside I , the intermediate system has exactly the same data as the full stage- n system without artificial probes. Consider a deep square of radius $R/4$ and its bounded-jump neighborhoods. A shield at fixed relative radii inside I has intact stage- n tests. By [Propositions 8.2](#) and [8.3](#), comparison across that shield changes the relevant scattering block by at most $C_n e^{-c_n R^\alpha}$ in operator norm outside an event of vanishing probability. The block ranks grow only polynomially in R .

Fix n with $d_n \geq d/2$. By [Proposition 8.5](#), choose a finite jump cutoff capturing the spatial average on this deep square up to, say, a quarter of d_n . For sufficiently large m, N , that truncated average exceeds $d_n/2$ with the prescribed high probability. Every such jump lies in I . Its squared chord length under F is its squared physical length divided by R^2 , up to factors tending to one as $b_N/R \rightarrow \infty$. The deep square has area comparable to R^2 . The preceding operator-norm comparison incurs a vanishing error in the resulting weighted sum, because the rank is polynomial and the shield error is exponential. This gives the lower bound in [Equation \(9.15\)](#), with $c_1 > 0$ independent of the later scales.

For the upper bound, extend $F|_I$ to data vanishing by radius $2R$, with Lipschitz constant C/R . Apply the coarse profile bound in [Proposition 8.6](#) and then the first-pass comparison. The expected intermediate energy is at most $CD_* + o(1)$. The sum of internal jump terms is at most twice this energy. Markov's inequality therefore gives a fixed upper cutoff C_1 , chosen before C_0 , for the assigned exception budget.

Weighted transmission. For cell ports within a sufficiently large absolute multiple of L_* from c , chord bounds give $|F_\nu| \leq CM_0^2$. Their contribution to $\|F_\nu T\|_{\text{HS}}$ is at most $CM_0^2 \sqrt{K}$. Beyond this range, retain only the good-edge costs on the thick grid walls O_D . Those costs are still available in the intermediate system, including their buffered tests. The wall estimate in [Proposition 8.4](#), followed by [Proposition 8.2](#), gives for a dyadic distance range of order D_1 the expectation bound

$$C_{n, M_0} R^2 (D_1/R)^2 \left(e^{-c_n D_1^\alpha} + D_1^{-30} \right). \quad (9.19)$$

To see the rank factor, there are $O_{M_0}(R^2)$ input probes; the operator norm of transmission across a successful wall shield is exponentially small, and it is always at most one by unitarity. The chord

weight is at most CD_1/R . At torus-size distances one uses a separating band with radius a sufficiently small fixed multiple of D_1 , so the same estimates apply. The dyadic sum of Equation (9.19) tends to zero as m increases at fixed n , uniformly for sufficiently large N : its polynomial part is a sum of D_1^{-28} . Thus one may choose, for example, $C_F = CM_0^2\sqrt{\kappa} + 1$, with a sufficiently large absolute constant and harmless extra slack for all four components. This choice is independent of n, C_0 ; only the scale required to make the far error small depends on them.

Finally include $\mathcal{B}$ using Lemma 9.3. The total exception probability is less than $1/2$. This proves the event assertion. Apply Equation (9.12) with $D' = 1$ and with $D' = F_\nu|_J$. Since $S_{CJ} = (T^\top)_{CJ}$, its unweighted Hilbert-Schmidt norm is at most $\sqrt{K}$, and its weighted norm is at most $\|F_\nu T\|_{\text{HS}}$. Substituting Equation (9.9) proves Equation (9.17). $\square$

9.4 An analytic trace with nonzero central value

Close all inserted probes with the common phase z . For real s , the corresponding physical shifts and the phase are

$$v_x \mapsto v_x + s\vartheta(v_x) \quad (x \in C), \quad z = \frac{s + i\tau}{s - i\tau}. \quad (9.20)$$

By Lemma 3.2, the scattering matrix on cell ports is

$$S_z = A + \Delta(z), \quad \Delta(z) = Tz(1 - zB)^{-1}T^\top. \quad (9.21)$$

Define $H(z)$ on the unit circle by replacing S' with S_z in Equation (9.4). Its matrix expression contains the adjoint S_z^* . To remove that obstruction to analyticity, replace the adjoint by the adjugate and multiply the constant term by the determinant:

$$\mathfrak{h}(z) = \sum_\nu \text{tr}[F_{I,\nu}\{F_\nu \det S_z - S_z F_\nu \text{adj } S_z\}] \quad (9.22)$$

This function is analytic in the disk. Indeed B is a contraction, and the scattering entries in Equation (9.21) are rational analytic functions there. Possible boundary poles are removable: away from finitely many points their boundary values are unitary, which excludes a genuine pole of a rational entry on the circle. Thus $\mathfrak{h}$ extends continuously to the boundary. There,

$$\mathfrak{h}(z) = \det(S_z)H(z), \quad |\mathfrak{h}(z)| = |H(z)|, \quad (9.23)$$

because $\text{adj } S_z = (\det S_z)S_z^*$.

Lemma 9.5. *Choose κ sufficiently small depending on d , then the constants in Lemma 9.4, and then C_0 sufficiently large. There are constants $c_2 > 0$ and $B_p < \infty$, independent of the later choices n, m, N , such that on $\mathcal{G}$*

$$|\mathfrak{h}(0)| \geq c_2, \quad (9.24)$$

and, for every fixed $0 < p < 1/2$,

$$\int_{\mathbb{T}} |H(z)|^p d\gamma_{\mathbb{T}}(z) \leq B_p. \quad (9.25)$$

Here $\gamma_{\mathbb{T}}$ is normalized Haar measure. Neither constant depends on the number of cell ports or probes.

Proof. Unitarity in Equation (9.10) gives $AA^* = 1 - TT^*$. If the eigenvalues of TT^* are $\delta_j \geq 0$, then $\sum_j \delta_j = K \leq \kappa$. Thus A is invertible and

$$|\det A|^2 = \prod_j (1 - \delta_j) \geq 1 - K, \quad \|A^{-1} - A^*\|_1 = \|A^{-1}TT^*\|_1 \leq \frac{K}{\sqrt{1-K}} \leq 2K \quad (9.26)$$

for sufficiently small κ . At zero,

$$\frac{\mathfrak{h}(0)}{\det A} = \sum_\nu \text{tr} [F_{I,\nu}(F_\nu - AF_\nu A^{-1})]. \quad (9.27)$$

On I every F_ν is bounded by an absolute constant, because the radius of I is $R + O(b_m)$.

First retain only the I, I indices in the product term of Equation (9.27), along with the entire constant trace. Replacing the internal block of A^{-1} by that of A^* costs at most CK in trace norm, by Equation (9.26). Symmetry of A makes the resulting internal expression exactly

$$\frac{1}{2} \sum_{i,j \in I} |A_{ij}|^2 |F(i) - F(j)|^2 + \sum_{i \in I} |F(i)|^2 \left(1 - \sum_{j \in I} |A_{ij}|^2 \right). \quad (9.28)$$

The second term is nonnegative, since A is a contraction. The first is at least $c_1 d/2$ on $\mathcal{G}$.

For the external part, put

$$\delta_A = \|A_{IJ}\|_{\text{HS}} + \sum_\nu \|A_{IJ}F_\nu|J\rangle\|_{\text{HS}}.$$

Symmetry and Equation (9.26) give

$$\|(A^{-1})_{JI}\|_{\text{HS}} \leq \|A_{IJ}\|_{\text{HS}} + 2K.$$

The external trace is consequently bounded in modulus by $C\delta_A(\delta_A + K)$. Choose κ so that the internal replacement error is less than $c_1 d/8$. After the choices in Lemma 9.4, choose C_0 so large that Equation (9.17) makes the external error less than $c_1 d/8$. The real part of Equation (9.27) is then at least $c_1 d/4$. Combining with $|\det A| \geq \sqrt{1-\kappa}$ proves Equation (9.24).

For the boundary moment, expand the trace using $S_z = A + \Delta$. The part H_A with A, A^* has a bounded modulus on $\mathcal{G}$. Indeed its internal part is Equation (9.28), whose first term is bounded by $C_1/2$, and

$$\sum_{i \in I} \left(1 - \sum_{j \in I} |A_{ij}|^2 \right) \leq K + \|A_{IJ}\|_{\text{HS}}^2.$$

The external part is bounded using Equation (9.17). All these bounds involve only the fixed cutoffs in Lemma 9.4.

Consider a linear term, for one component,

$$\text{tr}(F_I \Delta F A^*) = \text{tr} [z(1 - zB)^{-1} D_F], \quad D_F = T^\top F A^* F_I T.$$

Its coefficient satisfies

$$\|D_F\|_1 \leq \|T^\top F A^*\|_{\text{HS}} \|F_I T\|_{\text{HS}} \leq \|F T\|_{\text{HS}} \|F_I T\|_{\text{HS}}.$$

The other linear term is its complex conjugate on the circle. By Lemma 3.5, these terms have weak L^1 bounds depending only on C_F and κ , since $\|F_I T\|_{\text{HS}} \leq C\sqrt{\kappa}$.

The quadratic term is bounded by

$$|\operatorname{tr}(F_I \Delta F \Delta^*)| \leq \|F_I \Delta\|_{\text{HS}} \|\Delta F\|_{\text{HS}}.$$

Again [Lemma 3.5](#), with the respective sandwiches $(F_I T, T^\top)$ and $(T, T^\top F)$, gives weak L^1 bounds for each factor. A nonnegative random variable with weak L^1 bound a has every moment $q < 1$ bounded by $C_q a^q$, by integration of its tail. Applying Cauchy–Schwarz with exponent $2p < 1$ therefore bounds the p -moment of this product. There are four components and finitely many terms. Subadditivity at exponent $p < 1$ proves [Equation \(9.25\)](#). The use of Hilbert-Schmidt products and trace norms makes the bounds independent of all matrix dimensions. $\square$

9.5 Changing the potential law under a bounded twist

The artificial profile vanishes at the support endpoints and belongs to $C^{1,1}([-h, h])$. For sufficiently small $|s|$, the map

$$T_s(v) = v + s\vartheta(v)$$

is therefore a diffeomorphism of $[-h, h]$ onto itself, with derivative at least $1/2$. If r_s is the density of its pushforward of the uniform law relative to that same law, then

$$r_s(y) = \frac{1}{1 + s\vartheta'(T_s^{-1}(y))}, \quad |r_s - 1| \leq C|s|, \quad \int r_s \, d\mu = 1.$$

Consequently

$$\int r_s^2 \, d\mu = 1 + \int (r_s - 1)^2 \, d\mu \leq 1 + Cs^2. \quad (9.29)$$

Here μ is the uniform law on $[-h, h]$.

Fix $W < \infty$ and assume $|s|/\tau \leq W$. Conditional on the offset and the test tile, change every true variable in C by T_s and leave all other true variables, selectors, and auxiliary uniforms unchanged. There are at most $C(M_0 R)^2$ changed sites. By independence and [Equation \(9.29\)](#), the resulting joint law has a density $\mathscr{D}_s$ relative to the original joint law with

$$\|\mathscr{D}_s\|_{L^2}^2 \leq \exp(CM_0^2 R^2 s^2) \leq \exp(CM_0^2 C_0^2 W^2) = D_W^2. \quad (9.30)$$

This also holds after averaging over the offset and the test tile. The bound is uniform in m, N once they are sufficiently large for the maps to be diffeomorphisms. Its constant may be large; only its independence of the later scales matters.

We next check why this change of true potentials reproduces the closed-probe system despite the potential-dependent hierarchy. This check concerns recomputed statuses, not an assumption that statuses are fixed by definition.

On $\mathcal{P}$, selected tiles of level at least m do not meet the test tile. Every tile of level below m touching C lies wholly in the thinned gap, since $b_m \ll R$ and the annulus is separated from both retained regions by a distance comparable to R . Thus a selected tile capable of carrying an intermediate cell port is disjoint from C . Its own true variables, own corner labels, local likelihood, and reference potential are unchanged by the shift. Eligibility depends only on selectors and is unchanged as well.

There is one remaining possibility: a changed coordinate may alter the validity of an ancestor label used by a retained tile. Any such ancestor contains both the retained root and a coordinate in C . Their separation is at least a fixed multiple of R , so the ancestor side is at least cR . The probability that any such large tile has an invalid own label is bounded by

$$u_R = C_{M_0} \sum_{b_l \geq cR} (1 + R/b_l)^2 b_l^{-K_0} = o(1). \quad (9.31)$$

This counts only tiles meeting C , which also covers every relevant changed triple. The original bound is uniform in the offset and torus size. After the shift the probability is at most $D_W \sqrt{u_R}$ by Equation (9.30). Exclude both events. All potentially changed ancestor validity tests are then valid before and after the shift. Ancestor box identities themselves need not be the same: the acceptance rules use those identities only through the validity gates, whereas each retained tile's own box and likelihood are unchanged.

On the remaining part of $\mathcal{P}$, all intermediate cell ports and references are therefore unchanged when the hierarchy is recomputed with the same selectors and auxiliary uniforms. Changes of acceptance inside the gap have no effect, because those ports have already been thinned, and $\mathcal{P}$ excludes the larger ones. Before closure the real potentials on C were the true ones; the only added terms there were the artificial absorbers. Hence Lemma 3.2 and Equation (9.20) imply that $H(z)$ is exactly the no-extra-probe intermediate trace H^0 evaluated at the shifted data. Almost-everywhere inverse statements suffice throughout.

Let $a_{n,m,N} = \mathbb{E}[\mathbf{1}_{\mathcal{P}}|H^0|]$ in the original law. Since $\mathcal{P}$ uses only unchanged selectors, Markov's inequality and Equation (9.30) give, for every fixed $\delta > 0$,

$$\mathbb{P}\{\mathcal{G}, |H(z)| > \delta\} \leq e_R(W) + D_W \left(\frac{a_{n,m,N}}{\delta} \right)^{1/2}, \quad |s|/\tau \leq W, \quad (9.32)$$

where one may take $e_R(W) = u_R + D_W \sqrt{u_R}$. This tends to zero as m grows at fixed W, M_0, C_0 . The estimate is uniform over bounded twists; it does not assert simultaneous invariance for all twists on a single realization. Only the probability bound for each twist will be integrated.

9.6 The contradiction

Proof of Proposition 9.1. Retain the assumption $d > 0$. Choose the constants in the following order. First fix the barrier constants d_0, γ , then choose κ small enough for Lemma 9.5. Choose M_0 and the cutoffs C_1, C_F using Lemma 9.4. Then choose C_0 sufficiently large for the same analytic lemma. All these choices precede n, m, N . They provide a fixed central lower bound $c_2 > 0$ and a fixed boundary moment bound B_p on an event $\mathcal{G}$ of probability at least $1/2$ once those scales are sufficiently large.

Fix $0 < p' < p < 1/2$. Under Haar measure, s/τ in Equation (9.20) has the standard Cauchy law. Denote its tail probability beyond W by β_W , so $\beta_W \rightarrow 0$. Apply Hölder's inequality on the product of the underlying probability space and the circle. The pointwise good-data bound Equation (9.25) gives

$$\begin{aligned} & \mathbb{E} \left[\mathbf{1}_{\mathcal{G}} \int_{\mathbb{T}} |H(z)|^{p'} d\gamma_{\mathbb{T}}(z) \right] \\ & \leq \delta^{p'} + B_p^{p'/p} \left[\beta_W^{1-p'/p} + \left\{ e_R(W) + D_W \left(\frac{a_{n,m,N}}{\delta} \right)^{1/2} \right\}^{1-p'/p} \right]. \end{aligned} \quad (9.33)$$

For completeness, the first term covers $|H| \leq \delta$. The Haar tail costs the first Hölder term. On the bounded-twist part, Equation (9.32) bounds the product measure of $\{\mathcal{G}, |H| > \delta\}$, and the same p -moment bound gives the second Hölder term. Tonelli's theorem applies to these nonnegative quantities; no realization-wise uniformity in the twist is used.

The right side can be made smaller than $c_2^{p'}/4$. To see the order without interchanging limits, first choose W so the tail term is small, then choose δ so $\delta^{p'}$ is small. With these fixed, D_W is a fixed finite constant. Choose the tolerance in Lemma 9.2 sufficiently small relative to D_W and δ . Take n large for that adjustment and the good-data requirements; then take m large for all fixed- n spatial estimates, the probe-strength constraint, and $e_R(W)$; finally take N large for the finite-torus

approximations at this fixed pair. Increasing m, N as required preserves the preceding assertions. Thus Equation (9.33) is indeed less than $c_2^{p'}/4$.

On the other hand, $|\mathfrak{h}|^{p'}$ is subharmonic in the disk. For example, one may first use the nonnegative Laplacian of $(|\mathfrak{h}|^2 + \epsilon)^{p'/2}$ and then let $\epsilon \downarrow 0$. The continuous boundary extension and Equations (9.23) and (9.24) give on every good realization

$$\int_{\mathbb{T}} |H(z)|^{p'} d\gamma_{\mathbb{T}}(z) = \int_{\mathbb{T}} |\mathfrak{h}(z)|^{p'} d\gamma_{\mathbb{T}}(z) \geq |\mathfrak{h}(0)|^{p'} \geq c_2^{p'}.$$

Taking the expectation over $\mathcal{G}$ gives a lower bound $c_2^{p'}/2$, contradicting the preceding upper bound. Therefore $d = 0$. $\square$

10 A fixed-energy fractional-moment estimate

We return to the true product law and remove the hierarchy from the operator whose decay is to be estimated. Throughout this section, $h > 0$ and $E \in (-4 - h, 4 + h)$ are fixed. All choices may depend on these two parameters. Distance means the shortest-coordinate max-norm distance on the torus $\mathbb{T}_N^2 = (\mathbb{Z}/b_N\mathbb{Z})^2$. The notation for the probe sets below is local to this section.

Theorem 10.1 (True torus Green function). *Fix $0 < s < 1/16$. There are constants $A = A(h, E, s) < \infty$, $a = a(h, E, s) > 0$, and an integer $N_0 = N_0(h, E, s)$ such that, for every $N \geq N_0$ and every $x, y \in \mathbb{T}_N^2$,*

$$\mathbb{E} \left| (H_N - E)_{xy}^{-1} \right|^{s/2} \leq A \exp\{-a \text{dist}(x, y)\}. \quad (10.1)$$

Here H_N has the true iid uniform $[-h, h]$ diagonal and periodic nearest-neighbor hopping. Its inverse at the specified real energy exists almost surely. No uniformity in E , nor any statement about time evolution, is part of this theorem.

We prove the theorem by first retaining arbitrary deterministic masks of artificial probes. At each site a mask prescribes absence, strength τ , or strength $\tau\vartheta(v_i)$, with at most one probe per site. The profile ϑ is the one fixed earlier; in particular it equals one on $[-h/2, h/2]$, is $C^{1,1}$, and vanishes quadratically only at the endpoints of the potential interval. Write η_i for a prescribed positive probe strength and let Y have columns $\sqrt{\eta_i}e_i$. In this section

$$G = (H_N - E - iYY^\top)^{-1}, \quad S = 1 + 2iY^\top GY.$$

Thus S is symmetric and unitary, and $Y^\top GY$ is a contraction by Lemma 3.1. All exceptional algebraic singularities below can be discarded under the continuous product law.

The parameter order is

$$s, d_0, C_0, n, m, N, \quad R = \lfloor \exp(b_m^2) \rfloor, \quad \tau = C_0/R. \quad (10.2)$$

The barrier threshold $d_0 > 0$ will be small; C_0 will be large. After they have been fixed, n and then $m > n$ are made sufficiently large. Finally N is sufficiently large in terms of all these choices; all ensuing estimates hold for every larger N . We always require $b_N > 100R$ and $\tau \leq 1$.

10.1 Filling and twisting an annulus

Let X, Y' be arbitrary deterministic subsets of the probe sites in the radius- R squares centered at x, y , respectively, where $\text{dist}(x, y) \geq 10R$. Let

$$C = \{i : 2R \leq \text{dist}(i, x) \leq 4R\}.$$

Fill every previously unoccupied site of C with a probe of strength $\tau\vartheta(v_i)$, keeping existing probes and their strengths. The filled mask is again deterministic. Superscripts old and fill distinguish the resulting scattering matrices. Identify the probes in C with their sites and put

$$U = \|S_{CX}^{\text{fill}}\|_{\text{HS}}, \quad V = \|S_{CY'}^{\text{fill}}\|_{\text{HS}}, \quad D = \|S_{Y'X}^{\text{fill}}\|_{\text{HS}}.$$

Lemma 10.2 (Mixture domination). *There is $C_s < \infty$, independent of the masks, centers, R, N , and $C_0 \geq 1$, such that*

$$\mathbb{E} \|S_{Y'X}^{\text{old}}\|_{\text{HS}}^s \leq C_s(1 + C_0)\mathbb{E}[D^s + V^s U^s]. \quad (10.3)$$

As everywhere in this section, the constant may depend on the fixed h, E, s and profile.

Proof. Close just the newly added probes using a common Haar phase z . The closure identity in [Lemma 3.2](#) and the sandwich estimate in [Lemma 3.5](#), followed by integration of the weak L^1 tail, give

$$\int \| (S_z)_{Y'X} \|_{\text{HS}}^s \, dz \leq C_s(D^s + V^s U^s).$$

Here dz is normalized Haar measure, and the blocks linking the new probes to X, Y' are subblocks of those defining U, V . The corresponding real closing shift at each newly occupied site is $w\vartheta(v_i)$, where w has density

$$\frac{\tau}{\pi(w^2 + \tau^2)}.$$

Only newly occupied sites are shifted, so after closing we have precisely the old mask evaluated at these shifted true potentials.

For $|w| \leq c/R$, with fixed sufficiently small $c > 0$, the map $T_w(v) = v + w\vartheta(v)$ is a diffeomorphism of $[-h, h]$ onto itself. If $u = T_w(v)$, its pushforward density relative to the uniform law is $r_w(u) = (1 + w\vartheta'(v))^{-1}$. Uniformly in u ,

$$\log r_w(u) = -w\vartheta'(u) + O(w^2).$$

This follows from the inverse Jacobian and the Lipschitz continuity of ϑ' . There are at most KR^2 shifted sites, so at each fixed output configuration the joint density satisfies

$$\log \prod_i r_w(u_i) = -w \sum_i \vartheta'(u_i) + O(R^2 w^2).$$

Choose the half interval $[0, c/R]$ or $[-c/R, 0]$ on which the linear term is nonnegative. On this entire half interval the joint density is at least a fixed positive constant. Its Cauchy probability is

$$\frac{1}{\pi} \arctan \frac{c}{C_0} \geq \frac{c_1}{1 + C_0}.$$

Consequently the full shifted mixture dominates $c_2(1 + C_0)^{-1}$ times the true product law, pointwise in the output configuration. Applying this domination to the nonnegative old-mask block power and then averaging the Haar estimate proves the claim. $\square$

We seek a bound by an arbitrarily small multiple of $\mathbb{E}V^s$. Thus [Equation \(10.3\)](#) leaves two tasks: attenuate D across the filled annulus, and combine a small mean return U with a resampling bound for $U^s V^s$. An unweighted estimate for barrier failure would not suffice when $\mathbb{E}V^s$ is small. We therefore control that failure also under the V^s -weighted law.

10.2 Two barriers under a transmission tilt

Call a site good when $\vartheta(v_i) \geq d_0$. Give cost one to an edge precisely when both endpoints are good and both radii about x belong to $[2.2R, 2.8R]$; all other edges have cost zero. Let d_{in} be the associated path-cost distance from the radius- $2R$ square. Independently of this definition, count good-good edges with both radii in $[3.2R, 3.8R]$, and let d_{out} be the resulting distance from $\{i : \text{dist}(i, x) \geq 4R\}$. Integer roundings of the four band boundaries make no difference to the constants. For a sufficiently small absolute $\gamma > 0$, define

$$\mathcal{B} = \left\{ \begin{array}{ll} d_{\text{in}}(i) \geq \gamma R & \text{if } \text{dist}(i, x) \geq 3R, \\ d_{\text{out}}(i) \geq \gamma R & \text{if } \text{dist}(i, x) \leq 3R. \end{array} \right\}. \quad (10.4)$$

These two strictly buffered bands will allow separate resampling of the inside and outside of radius $3R$.

Lemma 10.3 (Uniform one-site estimates under tilt). *Suppose $\mathbb{E}V^s > 0$, and define the probability law*

$$d\nu = \frac{V^s}{\mathbb{E}V^s} d\mathbb{P}.$$

For a site $i \in C$, conditional on all other potentials, put $W(v_i) = V^s$. Then

$$(\mathbb{E}_i W^2)^{1/2} \leq K_s \mathbb{E}_i W, \quad (10.5)$$

with a constant uniform in every frozen exterior and in the mask. For $d_0 > 0$ sufficiently small, chosen after s and before C_0 , there are $K, c > 0$ such that

$$\mathbb{P}(\mathcal{B}^c) + \nu(\mathcal{B}^c) \leq K e^{-cR}, \quad \mathbb{E}_\nu d_{\text{min}}^{-2s} \leq K R^2, \quad d_{\text{min}} = \min_{i \in C} \vartheta(v_i). \quad (10.6)$$

No independence under ν is asserted or needed.

Proof. Fix all variables except $v = v_i$, and write the probe factor at i as $a(v) \in \{1, \vartheta(v)\}$. Schur complementation at i expresses the other rows and the row at i of the block defining V in the form

$$A' + \frac{D'}{v - i\tau a(v) - w_*}, \quad \frac{D'' \sqrt{a(v)}}{v - i\tau a(v) - w_*}, \quad \Im w_* \geq 0.$$

The coefficients are fixed matrices of compatible sizes. Passivity gives the sign of $\Im w_*$. The deleted physical-site system is invertible almost surely: the imaginary part forces a kernel vector to vanish at all remaining probe sites, so it would lie in the kernel of the true real restriction, whose nonzero determinant polynomial vanishes only on a product-law null set. Writing both rows over their common denominator, their joint numerator is

$$N(v) = ((v - i\tau a(v) - w_*)A' + D', D'' \sqrt{a(v)}).$$

On the plateau $P = [-h/2, h/2]$, this is the affine vector $N_P(v) = (vA' + B, D'')$, where $B = D' - (i\tau + w_*)A'$. If $M = \max_{v \in P} \|N_P(v)\|_{\text{HS}}$, then

$$\|A'\|_{\text{HS}} \leq 2M/h, \quad \|(B, D'')\|_{\text{HS}} \leq M, \quad \sup_{[-h, h]} \|N(v)\|_{\text{HS}} \leq K_h M.$$

Moreover an interval in P of length at least $h/4$, adjacent to an endpoint attaining the maximum, has numerator norm at least $M/2$. This uses the displayed slope bound. If $M = 0$, the entire block vanishes and the claimed inequality is immediate.

Put $b(v) = v - i\tau a(v) - w_*$. Its modulus is at most $K_h(1 + |w_*|)$. For every $0 < q < 1$,

$$\mathbb{E}_i |b(v)|^{-q} \leq K_{h,q}(1 + |w_*|)^{-q}. \quad (10.7)$$

Indeed, for bounded w_* use $|b(v)| \geq |v - \Re w_*|$ and integrate the one-dimensional singularity; for $|w_*|$ sufficiently large use $|b(v)| \geq |w_*| - h - 1$. Hence

$$\mathbb{E}_i W \geq c_{h,s} \left(\frac{M}{1 + |w_*|} \right)^s, \quad \mathbb{E}_i W^2 \leq K_{h,s} \left(\frac{M}{1 + |w_*|} \right)^{2s}.$$

This proves Equation (10.5), uniformly even for arbitrarily large exterior impedance.

For any measurable single-site set A , Cauchy–Schwarz now gives

$$\nu(v_i \in A \mid (v_j)_{j \neq i}) \leq K_s \mathbb{P}(v_i \in A)^{1/2}$$

whenever the conditional normalizing mean is positive. The remaining conditional configurations have zero ν -measure. In particular the right side for $A = \{\vartheta < d_0\}$ tends to zero with d_0 , uniformly in the conditioning. Iterating this estimate over any fixed set of distinct sites bounds the probability that all of them are bad by β^k , where β can be made arbitrarily small.

A failure of either barrier gives, after trimming to intermediate radii and erasing loops, a simple path in the corresponding band of length $j \geq c_3 R$ and cost less than γR . For γ small, a fixed positive fraction of its edges are not good-good. Since the path is simple, this forces at least $c_4 j$ distinct bad sites. There are at most $K R^2 4^j$ candidate paths of length j , and at most 2^{j+1} subsets of their vertices. Choose d_0 so that $8\beta^{c_4} < 1$. Summing the resulting geometric series for $j \geq c_3 R$ proves the tilted exponential bound. The same argument with the original independent site probabilities proves the untilted bound.

Finally, the same conditional Cauchy–Schwarz estimate gives

$$\mathbb{E}_\nu [\vartheta(v_i)^{-2s} \mid (v_j)_{j \neq i}] \leq K_s (\mathbb{E}_i \vartheta(v_i)^{-4s})^{1/2} \leq K.$$

The integral is finite because ϑ vanishes quadratically and $8s < 1$. Since $d_{\min}^{-2s} \leq \sum_{i \in C} \vartheta(v_i)^{-2s}$ and $|C| \leq K R^2$, this proves the last assertion. $\square$

10.3 Cutoffs and the small return estimate

We record explicitly the consequence of Lemma 9.3 that will be used several times. Suppose $\Psi = GY_J$ is a matrix of fields sourced at a probe set J , and a scalar cutoff χ changes only across edges whose two endpoints belong to C and carry probes of strengths at least τd . If $|\chi_i - \chi_j| \leq L$, then the hopping commutator can be written as a probe source:

$$[H_N, \chi] \Psi = Y_C B_\chi, \quad \|B_\chi\|_{\text{HS}} \leq \frac{KL}{\tau d} \|Y_C^\top \Psi\|_{\text{HS}}. \quad (10.8)$$

Only endpoints of changing edges need be included on the right. Indeed, in the coefficient at an endpoint one divides by its square root strength, and replaces each neighboring field by its absorbed field divided by that neighbor's square root strength. Bounded lattice degree gives the stated operator bound. Ward then yields

$$\|Y^\top G[H_N, \chi] \Psi\|_{\text{HS}} \leq \|B_\chi\|_{\text{HS}}. \quad (10.9)$$

The output here includes *all* probes. These estimates also compare two systems agreeing on the cutoff support and its changing edges: cut off a field of the first system and correct its commutator residual in the second. No estimate on the exterior resolvent is involved.

Lemma 10.4 (Small return to a filled annulus). *For each fixed sufficiently large C_0 and each $\epsilon > 0$, one can choose n , then $m > n$, then a threshold for N , so that*

$$\mathbb{E}U^2 < \epsilon$$

for every subsequent N , every deterministic mask, every center, and every permitted subset X .

Proof. Carry the filled artificial mask through the auxiliary hierarchy. Take a scalar voltage f_R equal to one on the radius- R square, zero at radii at least $2R$, and Lipschitz with constant K/R . By [Proposition 8.6](#), at fixed n ,

$$\mathbb{E}\mathcal{Q}_{n,N}(f_R) \leq Kd_n + o_{m,N;n,C_0}(1).$$

Apply [Theorem 7.3](#) to thin to uniform stage m . The profile is supported in the radius- $2R$ square and has Lipschitz constant K/R , while $b_m \ll R$. Thus [Lemma 7.5\(ii\)](#) bounds its unnormalized charge by Kr_n , where $r_n = \sum_{l \geq n} q_l p_l \rightarrow 0$, independently of R . Writing U_m for the same artificial-probe block at stage m , we obtain

$$\mathbb{E}U_m^2 \leq Kd_n + o_{n \rightarrow \infty}(1) + o_{m,N;n,C_0}(1). \quad (10.10)$$

Each probe pair contributing to U_m^2 has unit squared voltage difference and is counted in both directions in $\mathcal{Q}$, canceling its factor $1/2$. The artificial probes are allowed in the comparison because $C_0/R \leq \exp(-c_* b_m)$ for sufficiently large m .

Let $\mathcal{A}$ be the event that no selected tile of level at least m meets the radius- $5R$ square. Tile counting gives

$$\mathbb{P}(\mathcal{A}^c) \leq K \sum_{l \geq m} (1 + R/b_l)^2 e^{-b_l^6} = o(R^{-4}).$$

For example each fixed polynomial loss in R is dominated by $\exp(-b_m^6)$, since $\log R \sim b_m^2$. On $\mathcal{A}$, the stage- m and true filled coefficients and probes coincide throughout this square. On $\mathcal{B}$ use

$$\chi = (1 - d_{\text{in}}/(\gamma R))_+.$$

It equals one on X , vanishes outside radius $3R$, and changes only on shared good-good edges. Cut off stage- m fields sourced at X , and correct in the true system using [Equations \(10.8\) and \(10.9\)](#). Since $R\tau = C_0$,

$$U \leq \left(1 + \frac{K}{C_0 d_0}\right) U_m \quad \text{on } \mathcal{A} \cap \mathcal{B}.$$

Always $U^2 \leq |X| \leq KR^2$, by unitarity. The exceptional contribution is therefore $o(1)$, by the preceding hazard estimate and the product-law path count in the proof of [Lemma 10.3](#). That count does not require $\mathbb{E}V^s > 0$. Now $d_n \rightarrow 0$ by [Proposition 9.1](#); choosing n, m, N in order in [Equation \(10.10\)](#) proves the assertion. The error notation means that at fixed n, C_0 , any desired tolerance holds for all sufficiently large m , and then for all N above a threshold depending on m . It asserts no rate uniform in n . $\square$

10.4 Direct transmission and independent resampling

We next prove two pointwise estimates:

$$D \leq Kd_{\min}^{-1}V \quad \text{almost surely,} \quad D \leq Ke^{-cC_0}V \quad \text{on } \mathcal{B}. \quad (10.11)$$

For the first, use fields sourced at Y' , symmetry of S , and a cutoff equal to one inside radius $2R$, zero outside radius $4R$, with gradient at most K/R . Its source term vanishes, while both endpoints of every changing edge have strength at least $\tau d_{\min}$. Equations (10.8) and (10.9) give $D \leq K(R\tau d_{\min})^{-1}V$; here $C_0 \geq 1$. In particular, if $\mathbb{E}V^s = 0$, then $D = 0$ almost surely, and no tilted law is required.

For the improved estimate choose an integer

$$\ell = \left\lceil \frac{K_1}{\tau d_0} \right\rceil$$

with K_1 sufficiently large. For $j \geq 1$ use the cutoff

$$\chi_j(i) = \min \left\{ 1, \frac{(d_{\text{out}}(i) - (j-1)\ell)_+}{\ell} \right\}.$$

All original sources have $d_{\text{out}} = 0$. Nonzero gradients have both endpoint distances at least $(j-1)\ell$, because cost distances are integer valued and change by at most one across an edge. Gradients only occur on counted good-good edges in the outer band. For the field $\Psi = G^{\text{fill}}Y_{Y'}$, put

$$A_j = \left\| \mathbf{1}_{\{d_{\text{out}} \geq j\ell\}} Y^\top \Psi \right\|_{\text{HS}}, \quad B_j = \left\| \mathbf{1}_{C \cap \{d_{\text{out}} \geq j\ell\}} Y^\top \Psi \right\|_{\text{HS}}.$$

The cutoff equations and Ward give

$$A_j \leq \frac{K}{\ell \tau d_0} B_{j-1} \leq \frac{1}{2} B_{j-1}, \quad B_j \leq A_j.$$

Start with $B_0 = V/2$. On $\mathcal{B}$, the set X has $d_{\text{out}} \geq \gamma R$, so it remains beyond every one of the $\lfloor \gamma R/\ell \rfloor$ cutoffs. For sufficiently large fixed C_0 , followed by sufficiently large R , this number is at least $c(d_0)C_0$. The iteration proves the second estimate in Equation (10.11). Notice that A_j includes every absorbed output, including X ; the proof does not require $X \subset C$.

To separate the product $V^s U^s$, enlarge the probability space with two independent fresh families of true potentials. Let $A = \{i : \text{dist}(i, x) < 3R\}$. Form one comparison configuration by retaining the original potentials on A^c and replacing those on A by the first fresh family. Form a second by retaining the original potentials on A and replacing those on A^c by the second fresh family. Keep the filled deterministic mask in both configurations, and recompute its strengths from the corresponding potentials. Denote the first configuration's transmission to Y' by V^{ext} , and the second's transmission from X by U^{int} .

On $\mathcal{B}$, the cutoff $(1 - d_{\text{out}}/(\gamma R))_+$ is supported in the shared exterior A^c , equals one on Y' , and has all changing edges in its buffered outer band. Applying it to the fresh-interior fields and correcting in the original system proves

$$V \leq \left(1 + \frac{K}{C_0 d_0} \right) V^{\text{ext}}.$$

Likewise $(1 - d_{\text{in}}/(\gamma R))_+$ is supported in the shared interior A and gives

$$U \leq \left(1 + \frac{K}{C_0 d_0} \right) U^{\text{int}}.$$

Take C_0 large enough that both factors are at most two. The two comparison configurations are independent *unconditionally*: they use disjoint parts of the original product sample and independent

fresh families. Each has the original true product distribution. Dropping $\mathbf{1}_{\mathcal{B}}$ before factoring therefore yields

$$\mathbb{E}[\mathbf{1}_{\mathcal{B}} V^s U^s] \leq 2^{2s} \mathbb{E} V^s \mathbb{E} U^s \leq 2^{2s} \mathbb{E} V^s (\mathbb{E} U^2)^{s/2}. \quad (10.12)$$

This factorization makes no assertion of independence conditional on the original barrier event.

The exceptional event is controlled under the tilted law from [Lemma 10.3](#). Unitarity gives $U \leq KR$, and [Equations \(10.6\)](#) and [\(10.11\)](#) give, respectively,

$$\begin{aligned} \frac{\mathbb{E}[\mathbf{1}_{\mathcal{B}^c} V^s U^s]}{\mathbb{E} V^s} &\leq KR^s \nu(\mathcal{B}^c), \\ \frac{\mathbb{E}[\mathbf{1}_{\mathcal{B}^c} D^s]}{\mathbb{E} V^s} &\leq K \mathbb{E}_\nu[\mathbf{1}_{\mathcal{B}^c} d_{\min}^{-s}] \\ &\leq K (\mathbb{E}_\nu d_{\min}^{-2s})^{1/2} \nu(\mathcal{B}^c)^{1/2} \leq KR e^{-cR/2}. \end{aligned}$$

Both quantities tend to zero uniformly over all the prescriptions. Combining these estimates with [Lemma 10.2](#), we obtain

$$\mathbb{E} \left\| S_{Y'X}^{\text{old}} \right\|_{\text{HS}}^s \leq K_s (1 + C_0) \left[e^{-scC_0} + (\mathbb{E} U^2)^{s/2} + \epsilon_R \right] \mathbb{E} V^s, \quad \epsilon_R \rightarrow 0. \quad (10.13)$$

The constants $K_s, c > 0$ are independent of large C_0 . First choose C_0 to make its exponential term small, then use [Lemma 10.4](#) with n, m, N in order to make the return term small, and enlarge m, N as necessary for ϵ_R . Thus the entire multiplier can be made arbitrarily small. These are uniform expectation bounds for each choice of mask and centers; no simultaneous good event over all choices is needed.

10.5 Contraction over all masks

The passage from finite-scale smallness to exponential fractional-moment decay parallels the finite-volume criteria of [\[4\]](#). We prove the contraction here for the probe transmission moments and deterministic masks.

Proposition 10.5 (Uniform mask contraction). *For fixed h, E, s , parameters through R and an integer N_0 can be chosen so that, for every $N \geq N_0$,*

$$F_N(r) \leq \frac{1}{2} F_N(r - 4R), \quad r \geq 10R.$$

Here $F_N(r)$ is the supremum of $\mathbb{E} \|S_{Y'X}\|_{\text{HS}}^s$ over all deterministic artificial masks, all subsets X, Y' lying in radius- R squares, and all centers at distance at least r . A supremum over an empty family is zero. Consequently there are constants $K, a_1 > 0$, independent of $N \geq N_0$, such that

$$F_N(r) \leq KR^s e^{-a_1 r/R}, \quad r \geq 0. \quad (10.14)$$

Proof. Partition C into at most J sets C_k , each contained in a radius- R square with center in C ; J is an absolute constant. Then

$$V^s = \left(\sum_{k=1}^J \|S_{C_k Y'}^{\text{fill}}\|_{\text{HS}}^2 \right)^{s/2} \leq \sum_{k=1}^J \|S_{C_k Y'}^{\text{fill}}\|_{\text{HS}}^s.$$

Each new center is at distance at least $r - 4R$ from y . The filled model is included in the same mask class. Choose the multiplier in [Equation \(10.13\)](#) at most $1/(2J)$ and take suprema, using symmetry to interchange the two block roles when necessary. This proves the contraction. Always $F_N(r) \leq KR^s$ by unitarity and the rank of a radius- R input set. Iterating in steps of $4R$ until the argument falls below $10R$ proves [Equation \(10.14\)](#). All thresholds were fixed before taking the supremum and are valid on every sufficiently large torus. $\square$

10.6 Removing the two endpoint probes

Choose a mask consisting only of two constant-strength probes τ at distinct sites x, y . Let $G^0 = (H_N - E)^{-1}$, and let g be its two-by-two compression to $\{x, y\}$. Resolvent subtraction gives the damped compression

$$g^{\text{damp}} = (1 - i\tau g)^{-1} g.$$

For a symmetric two-by-two matrix, direct multiplication by the adjugate shows

$$(g^{\text{damp}})_{xy} = \frac{g_{xy}}{\det(1 - i\tau g)}.$$

Since $S_{xy} = 2i\tau(g^{\text{damp}})_{xy}$, the exact identity is

$$G_{xy}^0 = \frac{S_{xy}}{2i\tau} \det(1 - i\tau g). \quad (10.15)$$

For completeness, the moments needed to remove the determinant are uniform in the exterior and the volume. Conditional on all other potentials, Schur complementation gives

$$g = (A + \text{diag}(v_x, v_y))^{-1}$$

with A a fixed real symmetric matrix. Put $t = (v_x + v_y)/2$, $u = (v_x - v_y)/2$. The integration square becomes $|u| \leq h$, $|t| \leq h - |u|$, with Jacobian two. For each fixed u , let $\lambda_1(u), \lambda_2(u)$ be the real eigenvalues of $A + \text{diag}(u, -u)$. If $0 < q < 1$,

$$\int_{-(h-|u|)}^{h-|u|} \left\| (A + \text{diag}(u, -u) + t1)^{-1} \right\|^q dt \leq \sum_{j=1}^2 \int_{-h}^h |t + \lambda_j(u)|^{-q} dt \leq K_q h^{1-q}.$$

Integration in u , with the normalized joint uniform density, therefore proves

$$\mathbb{E} \|g\|^q \leq K_{h,q}, \quad 0 < q < 1. \quad (10.16)$$

The bound is uniform over all real symmetric A , so no exterior norm enters. The deleted system is generically invertible; its exceptional set has zero product measure. A one-site Schur complement and the same scalar integral also give $\mathbb{E} |G_{xx}^0|^q \leq K_{h,q}$.

Since $|\det(1 - i\tau g)| \leq (1 + \tau \|g\|)^2$, Equation (10.16) with $q = 2s < 1$ gives a uniform bound on $\mathbb{E} |\det(1 - i\tau g)|^s$. Apply Cauchy-Schwarz to Equation (10.15):

$$\mathbb{E} |G_{xy}^0|^{s/2} \leq (2\tau)^{-s/2} (\mathbb{E} |S_{xy}|^s)^{1/2} (\mathbb{E} |\det(1 - i\tau g)|^s)^{1/2}.$$

The first factor of expectations decays exponentially by Equation (10.14), applied to singleton subsets. The diagonal is covered by the one-site bound. The now fixed R, τ and all constants are absorbed into $A(h, E, s)$ and $a(h, E, s) > 0$, proving Theorem 10.1. This conclusion remains a fixed-real-energy finite-volume estimate; the spectral passage uses it separately at each energy.

11 Infinite-volume spectral type

We now remove the fixed-energy restriction from the spectral conclusion. The probabilistic input is the true finite-torus estimate Theorem 10.1. We use the bounded self-adjoint spectral theorem and its Borel functional calculus in the form of [16, Theorems 3.1 and 3.7]. We include the spectral-averaging step to make the quantifiers explicit; it is the rank-one mechanism of [15, Theorem 2', Proposition 7 and Theorem 8(i)].

Lemma 11.1. *For every fixed $h > 0$, every fixed $E \in (-4 - h, 4 + h)$ and every site $x \in \mathbb{Z}^2$, almost surely there is $u \in \ell^2(\mathbb{Z}^2)$ such that*

$$(H_v - E)u = \delta_x.$$

If μ_x is the scalar spectral measure of δ_x , then almost surely

$$\Gamma_x(E) := \int_{\mathbb{R}} \frac{\mu_x(\mathrm{d}e)}{(e - E)^2} < \infty. \quad (11.1)$$

The integrand is interpreted as $+\infty$ at $e = E$.

Proof. Couple the torus of side b_N to the infinite sample by taking the true potentials in the square of that side centered at x . Let $G_N(E)$ be its real Green matrix. By [Theorem 10.1](#), there are fixed $q, c, C > 0$, with $q < 1$, such that for all sufficiently large N ,

$$\mathbb{E} |G_N(E)_{yx}|^q \leq C \exp(-c|y - x|_\infty).$$

The torus distance from x equals the physical coordinate distance in this centered square. Hence for $0 < a < c$,

$$\sup_N \mathbb{E} \sum_{y \text{ in the square}} e^{a|y-x|_\infty} |G_N(E)_{yx}|^q < \infty,$$

where the supremum may omit the finitely many smaller tori. Fatou implies that the liminf of these nonnegative sums is finite almost surely. Choose a subsequence on which they are bounded. Its coordinates are bounded, and a diagonal subsequence converges at every lattice site. Fatou for the counting measure gives a limit u with finite weighted ℓ^q sum. In particular $u \in \ell^2$: an ℓ^q sequence is bounded, and $\sum_y |u_y|^2 \leq \|u\|_\infty^{2-q} \sum_y |u_y|^q$. Every fixed nearest-neighbor equation eventually avoids the torus boundary, so the limit solves $(H_v - E)u = \delta_x$.

The bounded self-adjoint spectral theorem gives

$$\mu_x(\mathrm{d}e) = (e - E)^2 \mu_u(\mathrm{d}e),$$

where μ_u is the scalar spectral measure of u . Thus $\mu_x(\{E\}) = 0$ and the integral in [Equation \(11.1\)](#) is at most $\|u\|^2$. $\square$

Lemma 11.2 (Rank-one spectral averaging). *Let H_0 be bounded self-adjoint and let ξ be a unit vector. For $a \in \mathbb{R}$, let μ_a be the scalar spectral measure of ξ for $H_0 + a\langle \xi, \cdot \rangle \xi$. Then, as measures on $\mathbb{R}$,*

$$\int_{\mathbb{R}} \mu_a \mathrm{d}a \leq \text{Leb}. \quad (11.2)$$

Suppose on a Borel set B the base measure μ_0 satisfies $\int (e - E)^{-2} \mu_0(\mathrm{d}e) < \infty$ for Lebesgue-almost every $E \in B$. For almost every a , the restriction of μ_a to B is concentrated on eigenvalues of the perturbed operator.

Proof. For $\text{Im } z > 0$ write

$$F(z) = \int \frac{\mu_0(\mathrm{d}e)}{e - z}, \quad F_a(z) = \frac{F(z)}{1 + aF(z)}.$$

The second identity follows from rank-one resolvent inversion. Since $\text{Im } F(z) > 0$, direct integration in a gives $\int_{\mathbb{R}} \text{Im } F_a(z) \mathrm{d}a = \pi$. For a nonnegative continuous compactly supported function f ,

integrate against $f(\operatorname{Re} z)$, use Tonelli, and let $\operatorname{Im} z \downarrow 0$. The normalized Poisson kernels are an approximate identity, and Fatou in a gives

$$\int_{\mathbb{R}} \int_{\mathbb{R}} f(e) \mu_a(\mathrm{d}e) \mathrm{d}a \leq \int_{\mathbb{R}} f(e) \mathrm{d}e.$$

This proves Equation (11.2).

Let B_0 be the subset of B where the base inverse-square integral is finite. It is Borel, and $B \setminus B_0$ is Lebesgue null. Equation (11.2) shows that it has zero μ_a -measure for almost every a . At $E \in B_0$, Cauchy–Schwarz and dominated convergence show that $F(E + i0)$ exists and is real. If $1 + aF(E + i0) \neq 0$, then

$$\operatorname{Im} F_a(E + i\eta) \longrightarrow 0, \quad \frac{\mu_a([E - \eta, E + \eta])}{\eta} \leq 2 \operatorname{Im} F_a(E + i\eta) \longrightarrow 0.$$

For any finite Borel measure μ , the set of points where this last ratio tends to zero has μ -measure zero. To see this directly on a bounded interval, fix $\varepsilon > 0$ and restrict first to points where the ratios are at most ε at every dyadic radius below a common threshold. At a finer dyadic mesh, each interval meeting this set lies in a centered interval of that radius at one of its points, so its measure is at most ε times the mesh length. Summing gives at most a constant times ε times the length of the bounded ambient interval. Increase the threshold sets to their union, then let $\varepsilon \downarrow 0$, and finally exhaust the real line.

It remains to consider $E \in B_0$ with $1 + aF(E + i0) = 0$. For each positive integer k , apply the bounded Borel multiplier $\mathbf{1}_{\{|e-E| \geq 1/k\}}(e - E)^{-1}$ for H_0 to ξ . The inverse-square integral makes these vectors Cauchy in norm. Their limit u satisfies

$$(H_0 - E)u = \xi, \quad \langle \xi, u \rangle = F(E + i0).$$

Indeed, the base spectral measure has no atom at E , so the truncated equations converge to $(H_0 - E)u = \xi$ by boundedness of H_0 . Cauchy–Schwarz gives the asserted inner product. Consequently

$$(H_0 + a\langle \xi, \cdot \rangle \xi - E)u = (1 + aF(E + i0))\xi = 0.$$

The vector is nonzero because its base equation has a nonzero right side. Hence E is an eigenvalue. All these eigenvectors belong to the closed cyclic subspace generated by ξ under H_0 . This subspace is separable and invariant under the perturbed operator. Eigenvectors for distinct eigenvalues are orthogonal, so the energies obtained in this way form a countable set. The remaining scalar spectral mass is therefore pure point, as asserted. $\square$

Proposition 11.3. *For each fixed $h > 0$, almost surely every site spectral measure of H_v is pure point. Consequently H_v has pure-point spectral type and a complete orthonormal eigenbasis.*

Proof. The property in Equation (11.1) is jointly measurable in the true potential configuration and E . Indeed $\|H_v\| \leq 4 + h$, and polynomial functional calculus followed by uniform approximation of continuous functions makes $v \mapsto \mu_x$ a Borel measure kernel. Integration of the nonnegative Borel kernel $(e - E)^{-2}$ then gives the asserted joint measurability. This property no longer mentions any auxiliary port construction or its energy-dependent parameters.

By Lemma 11.1 and Fubini, for almost every setting of the potentials away from x , almost every base value $v_x^0 \in [-h, h]$ has finite $\Gamma_x(E)$ for Lebesgue-almost every $E \in (-4 - h, 4 + h)$. Fix such an exterior and one such base value. Varying v_x corresponds to the rank-one parameter $a = v_x - v_x^0$. By Lemma 11.2, for almost every allowed a the site measure is pure point on that open interval.

The same averaging inequality removes the two endpoint energies. For every allowed a the norm bound confines the spectrum to the closed interval, so the entire site measure is pure point.

This conditional conclusion integrates to an almost-sure assertion. For completeness, purity of a scalar measure kernel is measurable: $(v, e) \mapsto \mu_x(v)(\{e\})$ is Borel by continuous approximations to the diagonal indicator, and its zero set has measurable $\mu_x(v)$ -mass. Purity is exactly the vanishing of that mass. Intersect now over the countable set of sites x . Each δ_x lies in the closed pure-point subspace; their span is dense, so that subspace is the entire Hilbert space. Choosing orthonormal bases in the mutually orthogonal eigenspaces yields the required eigenbasis. We have not assumed that one site vector is cyclic for the full two-dimensional operator. $\square$

Proposition 11.4. *For each fixed $h > 0$, almost surely $\sigma(H_v) = [-4 - h, 4 + h]$. Under [Proposition 11.3](#), the eigenvalues are dense in this interval.*

Proof. The norm bound gives the inclusion into the displayed interval. For each prescribed constant in a countable dense subset of $[-h, h]$, each positive rational accuracy, and each integer box size, infinitely many independent disjoint-box trials have positive probability that every potential in the box is within that accuracy of the constant. Almost surely such boxes therefore occur; take the countable intersection over these choices.

The free adjacency has plane waves with energies $2 \cos k_1 + 2 \cos k_2$, whose range is $[-4, 4]$. Normalize a plane wave cut off to one of the large nearly constant boxes. The hopping error is supported within bounded distance of its boundary and has norm tending to zero because boundary size divided by box volume tends to zero. The diagonal error is at most the prescribed accuracy. Let size tend to infinity and accuracy to zero, using countable dense choices of the wave energies and the constants. The approximate eigenvector criterion puts a dense subset of $[-4 - h, 4 + h]$ in the spectrum, and its closedness gives the entire interval.

For an operator with a complete orthonormal eigenbasis, the spectrum is the closure of its eigenvalues: outside that closure the reciprocal of the diagonal eigenvalue differences defines a bounded inverse. The last assertion follows. Together with [Proposition 11.3](#), this proves [Theorem 1.1](#). $\square$

References

- [1] E. Abrahams, P. W. Anderson, D. C. Licciardello, and T. V. Ramakrishnan, *Scaling theory of localization: Absence of quantum diffusion in two dimensions*, Physical Review Letters **42** (1979), no. 10, 673–676. doi:10.1103/PhysRevLett.42.673.
- [2] M. Aizenman, A. Elgart, S. Naboko, J. H. Schenker, and G. Stolz, *Moment analysis for localization in random Schrödinger operators*, Inventiones Mathematicae **163** (2006), 343–413. doi:10.1007/s00222-005-0463-y.
- [3] M. Aizenman and S. Molchanov, *Localization at large disorder and at extreme energies: An elementary derivation*, Communications in Mathematical Physics **157** (1993), no. 2, 245–278. doi:10.1007/BF02099760.
- [4] M. Aizenman, J. H. Schenker, R. M. Friedrich, and D. Hundertmark, *Finite-volume fractional-moment criteria for Anderson localization*, Communications in Mathematical Physics **224** (2001), 219–253. doi:10.1007/s002200100441.
- [5] P. W. Anderson, *Absence of diffusion in certain random lattices*, Physical Review **109** (1958), no. 5, 1492–1505. doi:10.1103/PhysRev.109.1492.

- [6] A. Benedek, A. P. Calderón, and R. Panzone, *Convolution operators on Banach space valued functions*, Proceedings of the National Academy of Sciences of the United States of America **48** (1962), no. 3, 356–365. doi:10.1073/pnas.48.3.356.
- [7] A. Black, R. Drogin, and F. Hernández, *Self-consistent equations and quantum diffusion for the Anderson model*, arXiv:2506.06468v2 (2025). Version 2, November 6, 2025.
- [8] J. M. Combes and L. Thomas, *Asymptotic behaviour of eigenfunctions for multiparticle Schrödinger operators*, Communications in Mathematical Physics **34** (1973), no. 4, 251–270. doi:10.1007/BF01646473.
- [9] F. Delyon, Y. Lévy, and B. Souillard, *Anderson localization for multi-dimensional systems at large disorder or large energy*, Communications in Mathematical Physics **100** (1985), no. 4, 463–470. doi:10.1007/BF01217724.
- [10] J. Ding and C. K. Smart, *Localization near the edge for the Anderson Bernoulli model on the two dimensional lattice*, Inventiones Mathematicae **219** (2020), 467–506. arXiv:1809.09041v3.
- [11] J. Fröhlich, F. Martinelli, E. Scoppola, and T. Spencer, *Constructive proof of localization in the Anderson tight binding model*, Communications in Mathematical Physics **101** (1985), no. 1, 21–46. doi:10.1007/BF01212355.
- [12] J. Fröhlich and T. Spencer, *Absence of diffusion in the Anderson tight binding model for large disorder or low energy*, Communications in Mathematical Physics **88** (1983), no. 2, 151–184. doi:10.1007/BF01209475.
- [13] O. Hurtado, *Localization and unique continuation for non-stationary Schrödinger operators on the 2D lattice*, Communications in Mathematical Physics **407** (2026), article 64. doi:10.1007/s00220-026-05559-8.
- [14] B. Simon, *Schrödinger operators in the twenty-first century*, in *Mathematical Physics 2000*, A. Fokas, A. Grigoryan, T. Kibble, and B. Zegarlinski (eds.), Imperial College Press, London, 2000, pp. 283–288. Author-hosted published version.
- [15] B. Simon and T. Wolff, *Singular continuous spectrum under rank one perturbations and localization for random Hamiltonians*, Communications on Pure and Applied Mathematics **39** (1986), no. 1, 75–90. doi:10.1002/cpa.3160390105.
- [16] G. Teschl, *Mathematical Methods in Quantum Mechanics: With Applications to Schrödinger Operators*, Graduate Studies in Mathematics **99**, American Mathematical Society, Providence, RI, 2009. Author-hosted first edition.